
Modern Physics

basics

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This material is part of the **basics-books project**. It is also available as a .pdf document.

Please check out the Github repo of the project, [basics-book project](#).

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Part I

Special Relativity

SPECIAL RELATIVITY

Principles. (1) invariant nature of physical laws, (2) c as the maximum speed of propagation of information in space. Every inertial observer measures the same speed of light.

Lorentz's transformation as a special coordinate transformations for space-time description as seen by two inertial observers in relative motion with uniform velocity. Physical laws governing a physical process have the same expressions if seen by two inertial observers. In a homogeneous time-space the expression of equations of physics doesn't depend on translation of origin of time-space coordinates; in an isotropic space the expression of equations of physics doesn't depend on rotations space rotation; as two inertial observers may write the same expressions of the physical equations using their own coordinates, the expression of equations of physics doesn't depend on Lorentz's transformations.

- standard configuration: derivation of Lorentz's transformations from principles and symmetry considerations
- composition of transformations and general Lorentz's transformation

Some consequences and examples:

- **Finite speed of propagation and loss of simultaneity.**
- **Speed of light and causality.** Causality follows from the principle that c is the maximum speed of information: all the observers perceive causes before consequences.
- **Length contraction and time dilation**

Mechanics. Point kinematics: 4-velocity and 4-acceleration; dynamics of free particle: 4-momentum, energy-momentum relation, rest energy; dynamics of particles in external force field: which field? For particles in EM field, Lorentz's force; Lagrangian approach

Electromagnetism. Special relativity and equations of electromagnetism. From classical electromagnetism to electromagnetism in special relativity theory; EM potentials, gauge condition, 4-current and EM field tensor; EM field equations; motion of a point charge in an EM field; energy balance equations. Relativity of electromagnetism: low-speed relativity for classical electromagnetism. Lagrangian approach for the equations of motion of charges in an EM field, and for the EM field equations.

SPECIAL RELATIVITY - NOTES

An event is determined by spatio-temporal information together, $t, \vec{r}$. Absolute nature of physics needs vector algebra and calculus formalism

$$\mathbf{X} = ct\mathbf{E}_0 + \vec{r} = ct\mathbf{E}_0 + x^1\mathbf{E}_1 + x^2\mathbf{E}_2 + x^3\mathbf{E}_3 = X^\alpha\mathbf{E}_\alpha,$$

having used Cartesian coordinates for the space coordinate.

Geometry of time-space in special relativity. Minkowski metric reads

$$g_{\alpha\beta} = \mathbf{E}_\alpha \cdot \mathbf{E}_\beta = \text{diag}\{1, -1, -1, -1\}$$

Reciprocal basis

The reciprocal basis reads $\mathbf{E}_\alpha \cdot \mathbf{E}^\beta = \delta_\alpha^\beta$, $\mathbf{E}_\alpha = g_{\alpha\beta}\mathbf{E}^\beta$, s.t. the elementary interval between two events can be written as

$$d\mathbf{X} = dX^\alpha \mathbf{E}_\alpha = \underbrace{dX^\alpha g_{\alpha\beta}}_{=dX_\beta} \mathbf{E}^\beta = dX_\beta \mathbf{E}^\beta,$$

having used Cartesian coordinates,

$$\begin{array}{cccc} X^0 = ct & X^1 = x & X^2 = y & X^3 = z \\ X_0 = ct & X_1 = -x & X_2 = -y & X_3 = -z \end{array}$$

Elementary interval between two events

$$ds^2 = d\mathbf{X} \cdot d\mathbf{X} = (dX_\alpha \mathbf{E}^\alpha) \cdot (dX^\beta \mathbf{E}_\beta) = c^2 dt^2 - (dx^1)^2 - (dx^2)^2 - (dx^3)^2 = c^2 dt^2 - |d\vec{r}|^2$$

2.1 Kinematics of point

Proper time. For a co-moving observer, $d\vec{r}' = \vec{0}$, and t' is commonly indicated with τ , and its differential is invariant itself, being the product of a constant (c is a universal constant in special relativity) and an invariant quantity.

$$ds^2 = c^2 dt'^2 - |d\vec{r}'|^2 = c^2 d\tau^2.$$

Given the invariant nature of ds ,

$$ds^2 = c^2 d\tau^2 = c^2 dt^2 - |d\vec{r}|^2 = c^2 dt^2 \left[1 - \frac{1}{c^2} \frac{|d\vec{r}|^2}{dt^2} \right] = c^2 dt^2 \left[1 - \frac{|\vec{v}|^2}{c^2} \right]$$

and thus

$$ds = c d\tau = \gamma^{-1}(v/c) c dt ,$$

with $\gamma(w) = \frac{1}{\sqrt{1-w^2}}$.

Note: ds is invariant **todo** prove it. And/or add a section about the role of invariance.

4-Velocity. Given the parametric representation of an event in space-time as a function of its proper time, $\mathbf{X}(\tau)$ or coordinate s , $\mathbf{X}(s)$ the derivative w.r.t. this parameter is defined as the 4-velocity of the event in space time. Using Cartesian coordinates inducing constant and uniform basis $\mathbf{E}_\alpha$, as a function of the observer time t , ct , $x^i(t)$, and the transformation of coordinates $t(\tau)$, with differential $dt = \frac{1}{\gamma} d\tau$

$$\mathbf{U}(\tau) := \mathbf{X}'(\tau) = \frac{d}{d\tau} (X^\alpha(\tau) \mathbf{E}_\alpha) = \frac{dt}{d\tau} (c \mathbf{E}_0 + x^i(t) \mathbf{E}_i) = \gamma(v/c) (c \mathbf{E}_0 + \dot{x}^i(t) \mathbf{E}_i) = \gamma(v/c) (c \mathbf{E}_0 + \vec{v})$$

or

$$\mathbf{U}(s) := \mathbf{X}'(s) = \frac{dt}{ds} \frac{d}{dt} \mathbf{X}(t) = \dots = \gamma(v/c) \left(\mathbf{E}_0 + \frac{\vec{v}}{c} \right) .$$

Note: Using s as the parameter, $\mathbf{U}$ is non-dimensional, and has pseudo-norm = 1,

$$\mathbf{U}(s) \cdot \mathbf{U}(s) = \gamma^2 \underbrace{\left(1 - \frac{|\vec{v}|^2}{c^2} \right)}_{=\gamma^{-2}} = 1 .$$

Using τ as the parameter, $\mathbf{U}$ has physical dimension of a velocity and pseudo-norm = c .

4-acceleration $\mathbf{X}''(\tau)$ or $\mathbf{X}''(s)$, **todo**

2.2 Dynamics of a point mass

4-momentum

$$\mathbf{P} = m\mathbf{U}$$

Using Cartesian coordinates and τ as independent variable,

$$\mathbf{P} = m\mathbf{U} = m \frac{d\mathbf{X}}{d\tau} = m\gamma(c, \vec{v}) .$$

Low-speed limit - classical mechanics

The spatial component is γ times the 3-dimensional momentum $\vec{p} = m\vec{v}$; the time component reads

$$P^0 = m\gamma(w)c ,$$

and for small ratio $w := \frac{v}{c}$ it can be expanded in Taylor series around $w = 0$ as

$$\gamma(w) \sim \gamma(0) + w\gamma'(0) + \frac{1}{2} w^2 \gamma''(0) + o(w^2) ,$$

with

$$\begin{aligned}\gamma(w)|_{w=0} &= \frac{1}{\sqrt{1-w^2}} \Big|_{w=0} = 1 \\ \gamma'(w)|_{w=0} &= -\frac{1}{2}(1-w^2)^{-\frac{3}{2}}(-2w) \Big|_{w=0} = w(1-w^2)^{-\frac{3}{2}} = 0 \\ \gamma''(w)|_{w=0} &= \left((1-w^2)^{-\frac{3}{2}} + w \left(-\frac{3}{2} \right) (1-w^2)^{-\frac{5}{2}}(-2w) \right) \Big|_{w=0} = \\ &= \left((1-w^2)^{-\frac{3}{2}} + 3w^2(1-w^2)^{-\frac{5}{2}} \right) \Big|_{w=0} = 1\end{aligned}$$

and thus

$$\gamma(w) = 1 + \frac{1}{2}w^2 + o(w^2)$$

and

$$\gamma(v/c) m c \sim m c \left(1 + \frac{v^2}{c^2} \right) = \frac{1}{c} \left(mc^2 + \frac{1}{2} m |\vec{v}|^2 \right)$$

Energy-momentum relation

Recognizing energy ($E = \gamma mc^2$) and 3-momentum ($\vec{p} = m_3 \vec{v}$, with $m_3 := \gamma m$)¹, the 4-momentum can be written as

$$\mathbf{P} = m\mathbf{U} = \gamma m \left(1, \frac{\vec{v}}{c} \right) =: \frac{1}{c} \left(\frac{E}{c}, \vec{p} \right)$$

Its pseudo-norm reads

$$m^2 = \mathbf{P} \cdot \mathbf{P} = \frac{1}{c^4} (E^2 - c^2 |\vec{p}|^2)$$

and thus the relation between E , $\vec{p}$, m and c ,

$$E^2 = m^2 c^4 + c^2 |\vec{p}|^2,$$

from which, for $\vec{v} = \vec{0} \rightarrow \vec{p} = \vec{0}$,

$$E^2 = m^2 c^4,$$

and keeping only the solution with positive energy (**todo** reference to Dirac's equation and anti-matter?)

$$E = mc^2.$$

4-force and dynamical equation for a point mass. The dynamical equation of a point mass subject to a 4-force $\mathbf{f}$ reads

$$\frac{d\mathbf{P}}{d\tau} = \mathbf{f}. \quad (2.1)$$

Examples:

- **Free particle** is subject to no net force, $\mathbf{f}$,
- Point charge q in EM field is subject to **Lorentz force**, $\mathbf{f} = q\mathbf{F} \cdot \mathbf{U}$, being $\mathbf{F}$ the EM field tensor, (2.4),

$$\mathbf{F} = \nabla \mathbf{A} - (\nabla \mathbf{A})^T.$$

¹ This definitions naturally arise in Lagrange approach.

2.2.1 Lagrangian approach

Here variational approach to the motion of point particle in a force field $\mathbf{f}$ is derived from the weak form of the dynamical equation (2.1). In particular, the derivation is performed for a particle moving in a known EM field. The motion of a free particle immediately follows if the EM field is zero.

$$m \frac{d\mathbf{U}}{d\tau} = q\mathbf{F} \cdot \mathbf{U}$$

The equation of motion can be manipulated to get Lagrange equations

$$\begin{aligned} 0 &= m \frac{d\mathbf{U}}{d\tau} - q\nabla\mathbf{A} \cdot \mathbf{U} + q(\nabla\mathbf{A})^T \cdot \mathbf{U} = \\ &= \frac{d}{d\tau} [m\mathbf{U} + q\mathbf{A}(\mathbf{X}(\tau))] - \nabla(q\mathbf{A} \cdot \mathbf{U}) = (1) \\ &= \frac{d}{d\tau} \left[\mathbf{E}^\alpha \frac{\partial}{\partial U^\alpha} (mc\sqrt{\mathbf{U} \cdot \mathbf{U}} + q\mathbf{A}(\mathbf{X}(\tau)) \cdot \mathbf{U}) \right] - \mathbf{E}^\alpha \frac{\partial}{\partial X^\alpha} (mc\sqrt{\mathbf{U} \cdot \mathbf{U}} + q\mathbf{A} \cdot \mathbf{U}) = (2) \\ &= \mathbf{E}^\alpha \left[\frac{d}{d\tau} \frac{\partial \mathcal{L}}{\partial X'^\alpha} - \frac{\partial \mathcal{L}}{\partial X^\alpha} \right] = \\ &= \frac{d}{d\tau} \frac{\partial \mathcal{L}}{\partial \mathbf{X}'} - \frac{\partial \mathcal{L}}{\partial \mathbf{X}} \end{aligned} \quad (2.2)$$

since (1) $\mathbf{U} = \mathbf{E}^\alpha U_\alpha = c\mathbf{E}^\alpha \frac{\partial}{\partial U^\alpha} \sqrt{U^\beta U_\beta}$, see (2.3), and $\sqrt{\mathbf{U} \cdot \mathbf{U}}$ independent from $\mathbf{X}$, and equal to c^2 and (2) with the definition of the **Lagrangian function**

$$\mathcal{L}(\mathbf{X}'(\tau), \mathbf{X}(\tau), \tau) := -mc\sqrt{\mathbf{X}'(\tau) \cdot \mathbf{X}'(\tau)} - q\mathbf{A}(\mathbf{X}(\tau)) \cdot \mathbf{X}'(\tau)$$

Proof of the relation used in (1)

$$\begin{aligned} \mathbf{E}^\alpha \frac{\partial}{\partial U^\alpha} \sqrt{U^\beta U_\beta} &= \mathbf{E}^\alpha \frac{\partial}{\partial U^\alpha} \sqrt{g_{\beta\gamma} U^\beta U^\gamma} = \\ &= \mathbf{E}^\alpha \frac{1}{2\sqrt{U^\beta U_\beta}} (g_{\beta\gamma} \delta_\alpha^\beta U^\gamma + g_{\beta\gamma} U^\beta \delta_\alpha^\gamma) = \\ &= \mathbf{E}^\alpha \frac{1}{2\sqrt{U^\beta U_\beta}} (2U_\alpha) = \\ &= \mathbf{E}^\alpha \frac{U_\alpha}{\sqrt{U^\beta U_\beta}} = \\ &= \frac{\mathbf{U}}{\sqrt{\mathbf{U} \cdot \mathbf{U}}} = \frac{\mathbf{U}}{c}. \end{aligned} \quad (2.3)$$

since $\mathbf{U} \cdot \mathbf{U} = c^2$ and the components of the metric tensor are independent from the velocity.

Weak form of the problem is derived from the Lagrange equation (2.2), with dot product with arbitrary 4-vector $\mathbf{W}$,

$$0 = \mathbf{W} \cdot \left[\frac{d}{d\tau} \frac{\partial \mathcal{L}}{\partial \mathbf{X}'} - \frac{\partial \mathcal{L}}{\partial \mathbf{X}} \right].$$

and integrating over an interval of proper time and using integration by parts,

$$\begin{aligned} 0 &= \int_{\tau=\tau_0}^{\tau_1} \mathbf{W} \cdot \left[\frac{d}{d\tau} \frac{\partial \mathcal{L}}{\partial \mathbf{X}'} - \frac{\partial \mathcal{L}}{\partial \mathbf{X}} \right] d\tau = \\ &= \left[\mathbf{W}(\tau) \cdot \frac{\partial \mathcal{L}}{\partial \mathbf{X}'}(\mathbf{X}'(\tau), \mathbf{X}(\tau), \tau) \right] \Big|_{\tau_0}^{\tau_1} - \int_{\tau=\tau_0}^{\tau_1} \left[\mathbf{W}' \cdot \frac{\partial \mathcal{L}}{\partial \mathbf{X}'} + \mathbf{W} \cdot \frac{\partial \mathcal{L}}{\partial \mathbf{X}} \right] d\tau. \end{aligned}$$

With the choice of the test function $\mathbf{W}(\tau)$ as the variation of the position of the particle, $\mathbf{W} = \delta\mathbf{X}$ just to remember its variational nature, with given values at τ_0, τ_1 and thus $\mathbf{W}(\tau_0) = \mathbf{W}(\tau_1) = 0$, the weak form of the dynamical equations is thus recast as the stationary condition of an action functional in a **variational principle**,

$$0 = -\delta S = -\delta \int_{\tau=\tau_0}^{\tau_1} \mathcal{L}(\mathbf{X}'(\tau), \mathbf{X}(\tau), \tau) d\tau.$$

todo Repeat the process using generalized coordinates $q^k(\tau)$, $\mathbf{X}(q^k(\tau), \tau) \dots$

Lagrange equations and variational principle with generalized coordinates

Choosing the test function as it's usually done in deriving Lagrange formulation of mechanics from the strong formulation, after writing the position and its velocity of the particle as a function of generalized coordinates q^k ,

$$\begin{aligned} \mathbf{X}(q^k(\tau), \tau) \\ \mathbf{U}(q^k(\tau), \tau) = \frac{d\mathbf{X}}{d\tau} = q^{k'}(\tau) \frac{\partial \mathbf{X}}{\partial q^k} + \frac{\partial \mathbf{X}}{\partial \tau} \\ = \frac{\partial \mathbf{X}'}{\partial q^{k'}} \end{aligned}$$

With the particular choice of the test function, $\mathbf{W} = w^k(\tau) \frac{\partial \mathbf{X}}{\partial q^k}(q^l(\tau), \tau)$

Variational principle as a function of time. Using the relation between proper time τ and the observer time t , $d\tau = \gamma^{-1} dt = \sqrt{1 - \frac{|\vec{v}|^2}{c^2}} dt$,

$$\begin{aligned} S &= \int_{\tau=\tau_0}^{\tau_1} \mathcal{L}(\mathbf{X}'(\tau), \mathbf{X}(\tau), \tau) d\tau = \\ &= \int_{t=t_0}^{t_1} \mathcal{L}(\mathbf{X}'(\tau), \mathbf{X}(\tau), \tau) \gamma^{-1} dt = \\ &= \int_{t=t_0}^{t_1} \sqrt{1 - \frac{|\vec{v}|^2}{c^2}} (-mc^2 - \gamma q\phi(\vec{r}, t) + \gamma q\vec{a}(\vec{r}, t) \cdot \vec{v}(t)) dt = \\ &= \int_{t=t_0}^{t_1} \left(-\sqrt{1 - \frac{|\vec{v}|^2}{c^2}} mc^2 - q\phi(\vec{r}, t) + q\vec{a}(\vec{r}, t) \cdot \vec{v}(t) \right) dt = \\ &= \int_{t=t_0}^{t_1} L(\vec{v}(t), \vec{r}(t), t) dt \end{aligned}$$

Three-dimensional momentum can be defined as the partial derivative of the Lagrangian function L w.r.t. the velocity $\vec{v}$,

$$\vec{p} := \frac{\partial L}{\partial \vec{v}} = \gamma \frac{\vec{v}}{c^2} mc^2 + q\vec{a} = \gamma m\vec{v} + q\vec{a}$$

The Hamiltonian function reads

$$\begin{aligned} H &= \vec{v} \cdot \frac{\partial L}{\partial \vec{v}} - L = \\ &= \gamma m|\vec{v}|^2 + q\vec{a} \cdot \vec{v} + \gamma^{-1} mc^2 + q\phi - q\vec{a} \cdot \vec{v} = \\ &= \gamma m|\vec{v}|^2 \left[1 + \left(1 - \frac{|\vec{v}|^2}{c^2} \right) \frac{c^2}{|\vec{v}|^2} \right] + q\phi = \\ &= \gamma mc^2 + q\phi. \end{aligned}$$

The expressions of the Hamiltonian function (divided by the light speed) and the three-dimensional momentum can be recognized to be respectively the time and space components of 4-momentum,

$$\mathbf{P} = m\mathbf{U} + q\mathbf{A} ,$$

remembering the expression of the contravariant components of the 4-velocity and the 4-EM potential,

$$[\mathbf{P}]^\alpha = \left(\gamma mc + q \frac{\phi}{c}, \gamma m \vec{v} + q \vec{a} \right) = \left(\frac{H}{c}, \vec{p} \right)$$

2.3 Electromagnetism

2.3.1 Classical electromagnetic theory

Maxwell equations

Maxwell equations read

$$\begin{cases} \nabla \cdot \vec{d} = \rho_f \\ \nabla \times \vec{e} + \partial_t \vec{b} = \vec{0} \\ \nabla \cdot \vec{b} = 0 \\ \nabla \times \vec{h} - \partial_t \vec{d} = \vec{j}_f \end{cases}$$

or in vacuum, with $\rho_f = \rho$, $\vec{j} = \vec{j}_f$, $\vec{d} = \varepsilon_0 \vec{e}$, $\vec{b} = \mu_0 \vec{h}$

$$\begin{cases} \nabla \cdot \vec{e} = \frac{\rho}{\varepsilon_0} \\ \nabla \times \vec{e} + \partial_t \vec{b} = \vec{0} \\ \nabla \cdot \vec{b} = 0 \\ \nabla \times \vec{b} - \mu_0 \varepsilon_0 \partial_t \vec{e} = \mu_0 \vec{j} \end{cases}$$

Electromagnetic potentials

The electromagnetic field can be written in terms of the electromagnetic potentials

$$\begin{cases} \vec{b} = \nabla \times \vec{a} \\ \vec{e} = -\partial_t \vec{a} - \nabla \varphi \end{cases}$$

Lorentz force

A particle in motion in a electromagnetic field is subject to Lorentz force. In classical electromagnetism, the expression of Lorentz force reads

$$\vec{F} = q \left(\vec{e} - \vec{b} \times \vec{v} \right) ,$$

whose power is

$$\vec{v} \cdot \vec{F} = \vec{v} \cdot q \left(\vec{e} - \vec{b} \times \vec{v} \right) = q \vec{v} \cdot \vec{e} .$$

2.3.2 Electromagnetic potential

$$\begin{cases} \vec{b} = \nabla \times \vec{a} \\ \vec{e} = -\partial_t \vec{a} - \nabla \varphi \end{cases}$$

$$\mathbf{A} = \mathbf{E}_\alpha A^\alpha = \frac{\varphi}{c} \mathbf{E}_0 + \vec{a}$$

$$\nabla \mathbf{A} = \left(\mathbf{E}^\alpha \frac{\partial}{\partial X^\alpha} \right) (A^\beta \mathbf{E}_\beta) = \frac{\partial A^\beta}{\partial X^\alpha} \mathbf{E}^\alpha \otimes \mathbf{E}_\beta = g^{\alpha\gamma} \frac{\partial A^\beta}{\partial X^\alpha} \mathbf{E}_\gamma \otimes \mathbf{E}_\beta.$$

whose components may be collected in a 2-dimensional array (first index for rows, second index for columns),

$$(\nabla \mathbf{A})_\alpha{}^\beta = \frac{\partial A^\beta}{\partial X^\alpha} = \begin{bmatrix} c^{-2} \partial_t \varphi & c^{-1} \partial_x \varphi & c^{-1} \partial_y \varphi & c^{-1} \partial_z \varphi \\ c^{-1} \partial_t a_x & \partial_x a_x & \partial_y a_x & \partial_z a_x \\ c^{-1} \partial_t a_y & \partial_x a_y & \partial_y a_y & \partial_z a_y \\ c^{-1} \partial_t a_z & \partial_x a_z & \partial_y a_z & \partial_z a_z \end{bmatrix}$$

or covariant-covariant coomponents,

$$(\nabla \mathbf{A})_{\alpha\beta} = \frac{\partial A_\beta}{\partial X^\alpha} = \frac{\partial A^\gamma}{\partial X^\alpha} g_{\gamma\beta} = \begin{bmatrix} c^{-2} \partial_t \varphi & -c^{-1} \partial_x \varphi & -c^{-1} \partial_y \varphi & -c^{-1} \partial_z \varphi \\ c^{-1} \partial_t a_x & -\partial_x a_x & -\partial_y a_x & -\partial_z a_x \\ c^{-1} \partial_t a_y & -\partial_x a_y & -\partial_y a_y & -\partial_z a_y \\ c^{-1} \partial_t a_z & -\partial_x a_z & -\partial_y a_z & -\partial_z a_z \end{bmatrix}$$

or contravariant-contravariant coomponents,

$$(\nabla \mathbf{A})^{\alpha\beta} = \frac{\partial A^\beta}{\partial X_\alpha} = g_{\alpha\gamma} \frac{\partial A^\beta}{\partial X^\gamma} = \begin{bmatrix} c^{-2} \partial_t \varphi & c^{-1} \partial_x \varphi & c^{-1} \partial_y \varphi & c^{-1} \partial_z \varphi \\ -c^{-1} \partial_t a_x & -\partial_x a_x & -\partial_y a_x & -\partial_z a_x \\ -c^{-1} \partial_t a_y & -\partial_x a_y & -\partial_y a_y & -\partial_z a_y \\ -c^{-1} \partial_t a_z & -\partial_x a_z & -\partial_y a_z & -\partial_z a_z \end{bmatrix}$$

The electromagnetic field tensor is defined as the anti-symmetric part of the gradient of the 4-electromagnetic potential,

$$\mathbf{F} = [\nabla \mathbf{A} - (\nabla \mathbf{A})^T] \quad (2.4)$$

whose components may be collected in a 2-dimensional array (first index for rows, second index for columns),

$$\begin{aligned} F^{\alpha\beta} &= \begin{bmatrix} 0 & -\frac{\underline{e}^T}{c} \\ \frac{\underline{e}}{c} & \underline{b}_\times \end{bmatrix}, & F_{\alpha\beta} &= \begin{bmatrix} 0 & \frac{\underline{e}^T}{c} \\ -\frac{\underline{e}}{c} & \underline{b}_\times \end{bmatrix} \\ F_\alpha{}^\beta &= \begin{bmatrix} 0 & -\frac{\underline{e}^T}{c} \\ -\frac{\underline{e}}{c} & -\underline{b}_\times \end{bmatrix}, & F^\alpha{}_\beta &= \begin{bmatrix} 0 & \frac{\underline{e}^T}{c} \\ \frac{\underline{e}}{c} & -\underline{b}_\times \end{bmatrix} \end{aligned}$$

2.3.3 Electromagnetic field and electromagnetic field equations

The pair of Maxwell equations

$$\begin{cases} \rho_f = \nabla \cdot \vec{d} \\ \vec{j}_f = -\partial_t \vec{d} + \nabla \times \vec{h} \end{cases}$$

can be re-written in 4-formalism, using 4-gradient in Cartesian coordinates

$$\nabla = \mathbf{E}^\alpha \frac{\partial}{\partial X^\alpha} = \mathbf{E}_0 \frac{\partial}{c \partial t} + \mathbf{E}_i \frac{\partial}{\partial x^i} = \mathbf{E}_0 \frac{\partial}{c \partial t} + \nabla,$$

and the definition of the 4-current density vector

$$\mathbf{J} = J^\alpha \mathbf{E}_\alpha = c\rho \mathbf{E}_0 + \vec{j}$$

so that

$$c\rho \mathbf{E}_0 + \vec{j} = \nabla \cdot \mathbf{F} = \nabla \cdot [(0 \mathbf{E}_0 + c\vec{d}) \otimes \mathbf{E}_0 + (-\mathbf{E}_0 c\vec{d} + \vec{h}_\times)],$$

with the displacement field tensor,

$$\mathbf{D} = D^{\alpha\beta} \mathbf{E}_\alpha \mathbf{E}_\beta,$$

with components (rows for the first index, columns for the second index)

$$D^{\alpha\beta} = \begin{bmatrix} 0 & -cd_x & -cd_y & -cd_z \\ cd_x & 0 & -h_z & h_y \\ cd_y & h_z & 0 & -h_x \\ cd_z & -h_y & h_x & 0 \end{bmatrix} = \begin{bmatrix} 0 & -c\vec{d}^T \\ c\vec{d} & \vec{h}_\times \end{bmatrix}.$$

The pair of Maxwell equations

$$\begin{cases} \nabla \cdot \vec{b} = 0 \\ \partial_t \vec{b} + \nabla \times \vec{e} = \vec{0} \end{cases}$$

can be re-written in 4-formalism as

$$0 = \partial_\mu F_{\eta\xi} + \partial_\eta F_{\xi\mu} + \partial_\xi F_{\mu\eta}$$

Among these $64 = 4^3$ equations, there are only 4 independent equations.

- If 2 indices are the same, the corresponding equation is the identity $0 = 0$. As an example, if $\mu = \eta$

$$0 = \partial_\mu F_{\mu\xi} + \partial_\mu \underbrace{F_{\xi\mu}}_{-F_{\mu\xi}} + \partial_\xi \underbrace{F_{\mu\mu}}_{=0} = 0,$$

thus only combinations with different indices may provide some information.

- Given the ordered set of indices (μ, η, ξ) , switching a pair of indices provides the same equation. As an example, switching μ and η

$$\begin{aligned} 0 &= \partial_\eta F_{\mu\xi} + \partial_\mu F_{\xi\eta} + \partial_\xi F_{\eta\mu} = \\ &= \partial_\eta (-F_{\xi\mu}) + \partial_\mu (-F_{\eta\xi}) + \partial_\xi (-F_{\mu\eta}). \end{aligned}$$

- Thus, only 4 combination of different indices, without taking order into account, provide independent information

$$\begin{aligned} (1, 2, 3) : \quad 0 &= \partial_1 F_{23} + \partial_2 F_{31} + \partial_3 F_{12} = \partial_x (-b_x) + \partial_y (-b_y) + \partial_z (-b_z) \\ (2, 3, 0) : \quad 0 &= \partial_2 F_{30} + \partial_3 F_{02} + \partial_0 F_{23} = \partial_y \left(-\frac{e_z}{c}\right) + \partial_z \left(\frac{e_y}{c}\right) + \partial_{ct} (-b_x) \\ (3, 0, 1) : \quad 0 &= \partial_3 F_{01} + \partial_0 F_{13} + \partial_1 F_{30} = \partial_z \left(\frac{e_x}{c}\right) + \partial_{ct} (-b_y) + \partial_x \left(-\frac{e_z}{c}\right) \\ (0, 1, 2) : \quad 0 &= \partial_0 F_{12} + \partial_1 F_{20} + \partial_2 F_{01} = \partial_{ct} (-b_z) + \partial_x \left(-\frac{e_y}{c}\right) + \partial_y \left(\frac{e_x}{c}\right) \end{aligned}$$

i.e.

$$\begin{aligned} (1, 2, 3) : \quad 0 &= -\nabla \cdot \vec{b} \\ (2, 3, 0) : \quad 0 &= -\frac{1}{c} [\partial_t b_x + (\partial_y e_z - \partial_z e_y)] \\ (3, 0, 1) : \quad 0 &= -\frac{1}{c} [\partial_t b_y + (\partial_z e_x - \partial_x e_z)] \\ (0, 1, 2) : \quad 0 &= -\frac{1}{c} [\partial_t b_z + (\partial_x e_y - \partial_y e_x)] \end{aligned}$$

i.e.

$$\begin{cases} 0 = \nabla \cdot \vec{b} \\ \vec{0} = \partial_t \vec{b} + \nabla \times \vec{e} \end{cases}$$

2.3.4 Point particle in electromagnetic field

Lorentz 4-force acting on a point charge of electric charge charge q reads

$$\mathbf{f} = \mathbf{F} \cdot \mathbf{J} = q \mathbf{F} \cdot \mathbf{U}.$$

so that the dynamical equation reads

$$m \mathbf{X}'' = q \mathbf{F} \cdot \mathbf{X}'$$

2.3.5 Energy balance

Continuity

$$\nabla \cdot \mathbf{J} = 0$$

Maxwell's equations

$$\begin{aligned} \nabla \cdot \mathbf{D} &= \mathbf{J} \\ \nabla \cdot (\epsilon : \mathbf{F}) &= \mathbf{0} \end{aligned}$$

Energy and momentum balance equation of the EM field using 3-dimensional formalism,

$$\begin{aligned} \frac{\partial u}{\partial t} + \nabla \cdot \vec{s} &= -\vec{e} \cdot \vec{j}^f \\ \frac{\partial \vec{s}}{\partial t} + \nabla \cdot \left\{ c^2 \left[\frac{1}{2} (\vec{d} \cdot \vec{e} + \vec{h} \cdot \vec{b}) \mathbb{I} - (\vec{d} \otimes \vec{e} + \vec{h} \otimes \vec{b}) \right] \right\} &= -c^2 (\rho^f \vec{e} - \vec{b} \times \vec{j}^f) \\ \frac{1}{c} \frac{\partial u}{\partial t} + \nabla \cdot \left(\frac{\vec{s}}{c} \right) &= -\frac{\vec{e}}{c} \cdot \vec{j}^f \\ \frac{1}{c} \frac{\partial \vec{s}}{\partial t} + \nabla \cdot \left[\frac{1}{2} (\vec{d} \cdot \vec{e} + \vec{h} \cdot \vec{b}) \mathbb{I} - (\vec{d} \otimes \vec{e} + \vec{h} \otimes \vec{b}) \right] &= -\rho^f c \frac{\vec{e}}{c} + \vec{b} \times \vec{j}^f \end{aligned} \quad (2.5)$$

The governing equations of energy and momentum of the EM field can be recast in 4-dimensional formalism,

$$\nabla \cdot \mathbf{T} = -\mathbf{F} \cdot \mathbf{J}$$

with the **energy-momentum-stress tensor**, $\mathbf{T}$,

$$\begin{aligned} T^{\alpha\beta} &= \frac{1}{4} g^{\alpha\beta} F_{\mu\nu} D^{\mu\nu} - F^{\alpha\nu} D^\beta{}_\nu = \\ &= \frac{1}{4} g^{\alpha\beta} F_{\mu\nu} D^{\mu\nu} - F^{\alpha\nu} g^{\beta\mu} D_{\mu\nu} = \\ &= \frac{1}{4} g^{\alpha\beta} F_{\mu\nu} D^{\mu\nu} + F^{\alpha\nu} D_{\nu\mu} g^{\mu\beta} \end{aligned} \quad (2.6)$$

Term $\mathbf{g} \cdot \mathbf{F} : \mathbf{D}$

$$\mathbf{F} : \mathbf{D} = F_{\mu\nu} D^{\mu\nu} = \begin{bmatrix} 0 & \frac{\underline{e}^T}{c} \\ -\frac{\underline{e}}{c} & \underline{b}_{\times} \end{bmatrix} : \begin{bmatrix} 0 & -c\underline{d}^T \\ c\underline{d} & \underline{h}_{\times} \end{bmatrix} = -2\underline{e}^T \underline{d} + 2\underline{b}^T \underline{h} = 2(-\vec{e} \cdot \vec{d} + \vec{b} \cdot \vec{h}) . \quad (2.7)$$

Term $F^{\alpha\nu} D^{\beta}_{\nu}$

$$F^{\alpha\nu} D^{\beta}_{\nu} = \begin{bmatrix} 0 & -\frac{\underline{e}^T}{c} \\ \frac{\underline{e}}{c} & \underline{b}_{\times} \end{bmatrix} \begin{bmatrix} 0 & c\underline{d}^T \\ c\underline{d} & -\underline{h}_{\times} \end{bmatrix} = \begin{bmatrix} -\underline{e}^T \underline{d} & \frac{(\underline{h} \times \underline{e})^T}{c} \\ c\underline{b} \times \underline{d} & \underline{e} \underline{d}^T + \underline{b}_{\times} \underline{h}_{\times} \end{bmatrix} \quad (2.8)$$

being

$$\underline{e}^T \underline{h}_{\times} = e_i \varepsilon_{kji} h_j = (\vec{h} \times \vec{e})_k$$

$$[\underline{b}_{\times} \underline{h}_{\times}]_{im} = \varepsilon_{ijk} b_j \varepsilon_{klm} h_l = (\delta_{il} \delta_{jm} - \delta_{im} \delta_{jl}) b_j h_l = h_i b_m - b_l h_l \delta_{im} = \underline{h} \underline{b}^T - \underline{b}^T \underline{h} I_{\equiv 3} .$$

Energy-stress tensor

Putting together the expression (2.7) of term $\frac{1}{4} g^{\alpha\beta} F_{\mu\nu} D^{\mu\nu}$ and (2.8) of term $F^{\alpha\nu} D^{\beta}_{\nu}$, with $\underline{d} = \varepsilon \underline{e}$, $\underline{b} = \mu \underline{h}$ (todo do energy balance for all charges and currents and spearating contributions of free and bound charges and currents)

$$\begin{aligned} T^{\alpha\beta} &= \frac{1}{4} \begin{bmatrix} 2(-\underline{e}^T \underline{d} + \underline{b}^T \underline{h}) & 0^T \\ \underline{0} & -2(-\underline{e}^T \underline{d} + \underline{b}^T \underline{h}) I_{\equiv 3} \end{bmatrix} - \begin{bmatrix} -\underline{e}^T \underline{d} & \frac{(\underline{e} \times \underline{h})^T}{c} \\ c\underline{b} \times \underline{d} & \underline{e} \underline{d}^T + \underline{b}_{\times} \underline{h}_{\times} \end{bmatrix} = \\ &= \begin{bmatrix} \frac{1}{2} (\underline{e}^T \underline{d} + \underline{b}^T \underline{h}) & \frac{(\underline{e} \times \underline{h})^T}{c} \\ \frac{\underline{e} \times \underline{h}}{c} & \frac{1}{2} (\underline{e}^T \underline{d} + \underline{b}^T \underline{h}) I_{\equiv 3} - \underline{e} \underline{d}^T - \underline{b} \underline{h}^T \end{bmatrix} = \\ &= \begin{bmatrix} u & \frac{\vec{s}}{c} \\ \frac{\vec{s}}{c} & u \mathbb{I} - \vec{e} \otimes \vec{d} - \vec{b} \otimes \vec{h} \end{bmatrix} \end{aligned}$$

Energy-stress equation in 4-dimensional formalism

Energy and momentum balance equations (2.5) can be recast using 4-dimensional formalism,

$$\nabla \cdot \mathbf{T} = -\mathbf{F} \cdot \mathbf{J}$$

$$\frac{\partial T^{\nu\mu}}{\partial X^{\nu}} = -F^{\mu\nu} J_{\nu}$$

as

$$\begin{aligned} F^{\mu\nu} J_{\nu} &= \begin{bmatrix} 0 & -\frac{\underline{e}^T}{c} \\ \frac{\underline{e}}{c} & \underline{b}_{\times} \end{bmatrix} \begin{bmatrix} \rho c \\ -\underline{j} \end{bmatrix} = \begin{bmatrix} \frac{1}{c} \underline{e}^T \underline{j} \\ \rho \underline{e} - \underline{b}_{\times} \underline{j} \end{bmatrix} \\ 0 : \quad \frac{\partial T^{\nu 0}}{\partial X^{\nu}} &= -F^{0\nu} J_{\nu} \quad \frac{1}{c} \frac{\partial}{\partial t} u + \frac{\partial}{\partial x_k} \frac{s_k}{c} = -\frac{1}{c} e_k j_k \\ 1 : 3 : \quad \frac{\partial T^{\nu i}}{\partial X^{\nu}} &= -F^{i\nu} J_{\nu} \quad \frac{1}{c} \frac{\partial}{\partial t} \frac{s_i}{c} + \frac{\partial \sigma_{ki}}{\partial x_k} = -\rho e_i + \varepsilon_{ijk} b_j j_k \end{aligned}$$

Relativity under Lorentz's transformation. Components of the description of two inertial observers in relative motion, with aligned Cartesian space coordinates, are related by Lorentz's transformation

Energy-momentum-stress tensor

todo Complete

Contravariant (Cartesian in space) components of the energy-momentum-stress tensor can be collected in a symmetric matrix

$$\mathbf{T}' = \mathbf{\Lambda} \mathbf{T} \mathbf{\Lambda}^T$$

$$\mathbf{T} = \begin{bmatrix} u & \frac{\mathbf{s}^T}{c} \\ \frac{\mathbf{s}}{c} & \boldsymbol{\sigma} \end{bmatrix}$$

$$\mathbf{\Lambda} = \begin{bmatrix} \gamma & -\gamma \mathbf{v}^T \\ -\gamma \mathbf{v} & \mathbf{I}_3 + (\gamma - 1) \hat{\mathbf{v}} \hat{\mathbf{v}}^T \end{bmatrix} = \gamma \begin{bmatrix} 1 & -\mathbf{v}^T \\ -\mathbf{v} & \hat{\mathbf{v}} \hat{\mathbf{v}}^T \end{bmatrix} + \begin{bmatrix} 0 & \mathbf{0}^T \\ \mathbf{0} & \mathbf{I} - \hat{\mathbf{v}} \hat{\mathbf{v}}^T \end{bmatrix} = \begin{bmatrix} \gamma & -\gamma \mathbf{v}^T \\ -\gamma \mathbf{v} & \mathbf{P} \end{bmatrix}$$

$$\begin{aligned} \mathbf{T}' &= \begin{bmatrix} \gamma & -\gamma \mathbf{v}^T \\ -\gamma \mathbf{v} & \mathbf{P} \end{bmatrix} \begin{bmatrix} u & \frac{\mathbf{s}^T}{c} \\ \frac{\mathbf{s}}{c} & \boldsymbol{\sigma} \end{bmatrix} \begin{bmatrix} \gamma & -\gamma \mathbf{v}^T \\ -\gamma \mathbf{v} & \mathbf{P} \end{bmatrix} = \\ &= \begin{bmatrix} \gamma u - \frac{\gamma}{c} \mathbf{s}^T \mathbf{v} & \frac{\gamma}{c} \mathbf{s}^T - \gamma \mathbf{v}^T \boldsymbol{\sigma} \\ -\gamma u \mathbf{v} + \mathbf{P} \frac{\mathbf{s}}{c} & -\frac{\gamma}{c} \mathbf{v} \mathbf{s}^T + \mathbf{P} \boldsymbol{\sigma} \end{bmatrix} \begin{bmatrix} \gamma & -\gamma \mathbf{v}^T \\ -\gamma \mathbf{v} & \mathbf{P} \end{bmatrix} = \\ &= \begin{bmatrix} \gamma^2 u - 2 \frac{\gamma^2}{c} \mathbf{s}^T \mathbf{v} + \gamma^2 \mathbf{v}^T \boldsymbol{\sigma} \mathbf{v} & -\gamma^2 u \mathbf{v}^T + \frac{\gamma^2}{c} \mathbf{s}^T \mathbf{v} \mathbf{v}^T + \frac{\gamma}{c} \mathbf{s}^T \mathbf{P} - \gamma \mathbf{v}^T \boldsymbol{\sigma} \mathbf{P} \\ -\gamma^2 u \mathbf{v} + \frac{\gamma}{c} \mathbf{P} \mathbf{s} + \frac{\gamma^2}{c} \mathbf{v} \mathbf{s}^T - \gamma \mathbf{P} \boldsymbol{\sigma} \mathbf{v} & \gamma^2 u \mathbf{v} \mathbf{v}^T - \frac{\gamma}{c} (\mathbf{P} \mathbf{s} \mathbf{v}^T + \mathbf{v} \mathbf{s}^T \mathbf{P}) + \mathbf{P} \boldsymbol{\sigma} \mathbf{P} \end{bmatrix} \end{aligned}$$

INERTIAL REFERENCE FRAMES AND LORENTZ'S TRANSFORMATIONS

Equations of physics as seen by inertial reference frames (**todo** but what's an inertial reference frame?) have the same expressions. Change of observer can be related to change of coordinates (and choice of basis vectors, in special relativity) used to describe a physical phenomenon, that is invariant to this change for its very nature: reality doesn't change if observed from different point of view (if observations don't interact with reality itself, at least). Proper mathematical tools for dealing with invariance are vectors and tensors in general.

Using 2 different sets of coordinates associated with constant and uniform basis vectors $\mathbf{E}_\alpha$, $\mathbf{E}'_\beta$ a vector can be written as

$$\mathbf{V} = V^\alpha \mathbf{E}_\alpha = V'^\beta \mathbf{E}'_\beta .$$

Lorentz transformation describe the change of description between two inertial reference frames.

3.1 Lorentz's transformations

3.1.1 Lorentz's transformations in standard configuration

Standard configuration is defined for two observers with Cartesian bases for the space components, with the axes aligned, and with relative motion along x , x' axes.

...**todo** derivation...see material for high school: *Special Relativity:Relativity and Lorentz's transformations*

$$\begin{aligned} ct' &= \gamma(ct - vx) \\ x' &= \gamma(-vct + x) \\ y' &= y \\ z' &= z \end{aligned} ,$$

with $\gamma := \frac{1}{\sqrt{1-v^2}}$, so that the coordinates transformations can be written using matrix of change of coordinates

$$\begin{bmatrix} ct' \\ x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} \gamma & -\gamma v & 0 & 0 \\ -\gamma v & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} ct \\ x \\ y \\ z \end{bmatrix}$$

$$X'^\alpha = \Lambda^\alpha_\beta X^\beta .$$

This is the same transformation of the components of all the vectors of the 4-dimensional space as seen by two different inertial observers

3.1.2 Lorentz's transformation in general configuration

Beside change of origin of the coordinates, general transformation can be derived composing Lorentz's transformations in standard configuration (the only one derived so far), and rotations of the space coordinates.

As an example, the general expression of Lorentz's transformation between two inertial reference frames with general relative (space) velocity $\vec{v}$ - and aligned axes - can be derived introducing two intermediate inertial reference frames: reference $Otxyz$ and $O't'x'y'z'$ are the two reference frames we'd like to link with a Lorentz transformation; $O_at_ax_ay_az_a$ is a reference frame at rest w.r.t. $Otxyz$ with the x_a axis with the same direction of the relative (space) velocity $\vec{v}$, $O_bt_bx_by_az_b$ is in standard configuration with O_a . The transformation of coordinates reads

$$\begin{aligned} \begin{bmatrix} x'_0 \\ \mathbf{r}' \end{bmatrix} &= \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & \mathbf{R}^T \end{bmatrix} \begin{bmatrix} x_{b,0} \\ \mathbf{x}_b \end{bmatrix} = \\ &= \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & \mathbf{R}^T \end{bmatrix} \begin{bmatrix} \gamma & -\gamma v & & \\ -\gamma v & \gamma & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \begin{bmatrix} x_{a,0} \\ \mathbf{x}_a \end{bmatrix} = \\ &= \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & \mathbf{R}^T \end{bmatrix} \left(\begin{bmatrix} \gamma & -\gamma v & & \\ -\gamma v & \gamma & & \\ & & 0 & \\ & & & 0 \end{bmatrix} + \begin{bmatrix} 0 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{bmatrix} \right) \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & \mathbf{R} \end{bmatrix} \begin{bmatrix} x_0 \\ \mathbf{x} \end{bmatrix}, \end{aligned}$$

and the two contributions give

$$\begin{aligned} \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & \mathbf{R}^T \end{bmatrix} \begin{bmatrix} \gamma & -\gamma v & & \\ -\gamma v & \gamma & & \\ & & 0 & \\ & & & 0 \end{bmatrix} \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & \mathbf{R} \end{bmatrix} &= \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & \mathbf{R}^T \end{bmatrix} \begin{bmatrix} \gamma & -\gamma v \mathbf{R}_{1,:} \\ -\gamma v & (\gamma - 1) \mathbf{R}_{1,:} \\ \mathbf{0}_{2,1} & \mathbf{0}_{2,4} \end{bmatrix} = \\ &= \begin{bmatrix} \gamma & -\gamma v \mathbf{R}_{1,:} \\ -\gamma v \mathbf{R}_{1,:}^T & (\gamma - 1) \mathbf{R}_{1,:}^T \mathbf{R}_{1,:} \end{bmatrix} = \\ &= \begin{bmatrix} \gamma & -\gamma R_{11}v & -\gamma R_{12}v & -\gamma R_{13}v \\ -\gamma R_{11}v & (\gamma - 1) R_{11}R_{11} & (\gamma - 1) R_{11}R_{12} & (\gamma - 1) R_{11}R_{13} \\ -\gamma R_{12}v & (\gamma - 1) R_{12}R_{11} & (\gamma - 1) R_{12}R_{12} & (\gamma - 1) R_{12}R_{13} \\ -\gamma R_{13}v & (\gamma - 1) R_{13}R_{11} & (\gamma - 1) R_{13}R_{12} & (\gamma - 1) R_{13}R_{13} \end{bmatrix}. \\ \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & \mathbf{R}^T \end{bmatrix} \begin{bmatrix} 0 \\ \mathbf{I}_3 \end{bmatrix} \begin{bmatrix} 1 & \mathbf{0}^T \\ \mathbf{0} & \mathbf{R} \end{bmatrix} &= \begin{bmatrix} 0 \\ \mathbf{I}_3 \end{bmatrix} \end{aligned}$$

Here, components $(R_{11}v, R_{12}v, R_{13}v)$ are the components of velocity $\vec{v}$ as seen by observer O , and thus the matrix for coordinate transformation reads

$$\begin{aligned} \Lambda &= \begin{bmatrix} 0 & \\ & \mathbf{I}_3 \end{bmatrix} + \begin{bmatrix} \gamma & -\gamma \mathbf{v}^T \\ -\gamma \mathbf{v} & (\gamma - 1) \hat{\mathbf{v}} \hat{\mathbf{v}}^T \end{bmatrix} = \\ &= \begin{bmatrix} \gamma & -\gamma \mathbf{v}^T \\ -\gamma \mathbf{v} & \mathbf{I}_3 + (\gamma - 1) \hat{\mathbf{v}} \hat{\mathbf{v}}^T \end{bmatrix} \end{aligned} \quad (3.1)$$

and the transformation of coordinates can be written as

$$\begin{cases} ct' &= \gamma(ct - \vec{v} \cdot \vec{r}) \\ \vec{r}' &= -\gamma ct \vec{v} + (\gamma - 1) \hat{v} \otimes \hat{v} \cdot \vec{r} + \vec{r} = \\ &= -\gamma ct \vec{v} + \underbrace{[\mathbb{I} - \hat{v} \otimes \hat{v}]}_{\mathbb{P}_{\perp \hat{v}}} \cdot \vec{r} + \gamma \underbrace{\hat{v} \otimes \hat{v}}_{\mathbb{P}_{\parallel \hat{v}}} \cdot \vec{r}, \end{cases} \quad (3.2)$$

having introduced the two orthogonal projectors in directions parallel and orthogonal to the direction of the velocity, $\hat{v}$.

Example (Standard configuration, as a special case.)

If $\vec{v} = v\hat{x}$, Lorentz's transformation for two inertial observers in standard configuration is retrieved from the general expression of Lorentz's transformation

$$\begin{cases} ct' &= \gamma(ct - vx) \\ \vec{r}' &= -\gamma vct\hat{x} + y\hat{y} + z\hat{z} + \gamma vx\hat{x} \end{cases}$$

3.2 Transformation of vector components and vectors of the bases

$$\mathbf{V} = V^\alpha \mathbf{E}_\alpha = V'^\beta \mathbf{E}'_\beta$$

$$V'^\beta = \Lambda^\beta_\alpha V^\alpha,$$

being α and β the indice of columns and rows of matrix Λ of change of coordinates of the general Lorentz's transformation (3.1). In order to keep invariance, vectors of the bases transform with the transpose transformation, namely

$$\mathbf{V} = V^\alpha \mathbf{E}_\alpha = V'^\beta \mathbf{E}'_\beta = V^\alpha \underbrace{\Lambda^\beta_\alpha \mathbf{E}'_\beta}_{\mathbf{E}_\alpha}$$

$$\mathbf{E}_\alpha = \Lambda^\beta_\alpha \mathbf{E}'_\beta. \quad (3.3)$$

$$[\Lambda^{-1}]^\alpha_\phi \mathbf{E}_\alpha = [\Lambda^{-1}]^\alpha_\phi \underbrace{\Lambda^\beta_\alpha}_{=\delta^\beta_\phi} \mathbf{E}'_\beta$$

3.3 Transformation of tensor components

$$\mathbf{D} = D^{\alpha\beta} \mathbf{E}_\alpha \mathbf{E}_\beta = D'^{\phi\eta} \mathbf{E}'_\phi \mathbf{E}'_\eta =$$

and using the rule of transformation of vectors of the bases (3.3)

$$D'^{\phi\eta} = D^{\alpha\beta} \Lambda^\phi_\alpha \Lambda^\eta_\beta,$$

or using matrix notation (**todo** avoid abuse of notation! Use underline for arrays of coomponents, bold for vectors and tensors)

$$\mathbf{D}' = \Lambda \mathbf{D} \Lambda^T$$

Example (Electromagnetic field tensor)

Contravariant components of the electromagnetic field tensor

$$\begin{aligned} \begin{bmatrix} 0 & -c\mathbf{d}'^T \\ c\mathbf{d}' & \mathbf{h}'_\times \end{bmatrix} &= \begin{bmatrix} \gamma & -\gamma\mathbf{v}^T \\ -\gamma\mathbf{v} & \mathbf{I}_3 + (\gamma-1)\hat{\mathbf{v}}\hat{\mathbf{v}}^T \end{bmatrix} \begin{bmatrix} 0 & -c\mathbf{d}^T \\ c\mathbf{d} & \mathbf{h}_\times \end{bmatrix} \begin{bmatrix} \gamma & -\gamma\mathbf{v}^T \\ -\gamma\mathbf{v} & \mathbf{I}_3 + (\gamma-1)\hat{\mathbf{v}}\hat{\mathbf{v}}^T \end{bmatrix}^T = \\ &= \begin{bmatrix} \gamma & -\gamma\mathbf{v}^T \\ -\gamma\mathbf{v} & \mathbf{I}_3 + (\gamma-1)\hat{\mathbf{v}}\hat{\mathbf{v}}^T \end{bmatrix} \begin{bmatrix} \gamma cvd_v & -c\mathbf{d}^T - (\gamma-1)cd_v\hat{\mathbf{v}}^T \\ \gamma c\mathbf{d} - \gamma\mathbf{h}_\times\mathbf{v} & -\gamma c\mathbf{d}\mathbf{v}^T + \mathbf{h}_\times + (\gamma-1)\mathbf{h}_\times\hat{\mathbf{v}}\hat{\mathbf{v}}^T \end{bmatrix} = \\ &= \begin{bmatrix} 0 & \text{anti-sym} \\ c\mathbf{d}' & \mathbf{h}'_\times \end{bmatrix} = \end{aligned}$$

with

$$\begin{aligned}
 c\mathbf{d}' &= -\gamma^2 cv^2 \hat{\mathbf{v}} d_v + \gamma c\mathbf{d} - \gamma \mathbf{h}_\times \mathbf{v} + \gamma(\gamma - 1) c \hat{\mathbf{v}} d_v = \\
 &= c\gamma^2 \hat{\mathbf{v}} d_v \underbrace{(1 - v^2)}_{=\gamma^{-2}} + \gamma c\mathbf{d} - \gamma \mathbf{h}_\times \mathbf{v} - \gamma c \hat{\mathbf{v}} d_v = \\
 &= \gamma c\mathbf{d} - \gamma \mathbf{h}_\times \mathbf{v} + (1 - \gamma) c \hat{\mathbf{v}} \hat{\mathbf{v}} \cdot \mathbf{d} ,
 \end{aligned}$$

and

$$\mathbf{h}'_\times = \quad (1)$$

$$\begin{aligned}
 &= \gamma cv \hat{\mathbf{v}} \mathbf{d}^T + \gamma(\gamma - 1) cv d_v \hat{\mathbf{v}} \hat{\mathbf{v}}^T + \\
 &- \gamma cv \mathbf{d} \hat{\mathbf{v}} + \mathbf{h}_\times + (\gamma - 1)(\mathbf{h} \times \hat{\mathbf{v}}) \hat{\mathbf{v}}^T - \gamma(\gamma - 1) cv d_v \hat{\mathbf{v}} \hat{\mathbf{v}}^T + (\gamma - 1) \hat{\mathbf{v}} \mathbf{v}^T \mathbf{h}_\times + \mathbf{0} = \quad (2)
 \end{aligned}$$

$$\begin{aligned}
 &= \gamma cv (\hat{\mathbf{v}} \mathbf{d}^T - \mathbf{d} \hat{\mathbf{v}}^T) + \mathbf{h}_\times + (\gamma - 1) ((\mathbf{h}_\times \hat{\mathbf{v}}) \hat{\mathbf{v}}^T - \hat{\mathbf{v}} (\mathbf{h}_\times \hat{\mathbf{v}})^T) = \quad (3) \\
 &= \gamma cv (\mathbf{d} \times \hat{\mathbf{v}})_\times + \mathbf{h}_\times + (\gamma - 1) (\hat{\mathbf{v}} \times (\mathbf{h} \times \hat{\mathbf{v}}))_\times
 \end{aligned}$$

having (1) recognized that $\hat{\mathbf{v}} \hat{\mathbf{v}}^T \mathbf{h}_\times \hat{\mathbf{v}} \hat{\mathbf{v}}^T = \hat{\mathbf{v}} (\hat{\mathbf{v}} \cdot (\mathbf{h} \times \hat{\mathbf{v}})) \hat{\mathbf{v}} = \mathbf{0}$, (2) $\{\mathbf{v}^T \mathbf{h}_\times\}_k = v_i \varepsilon_{ijk} h_j = -\{\mathbf{h} \times \hat{\mathbf{v}}\}_k$ and (3)

$$\begin{aligned}
 [(\mathbf{a} \times \mathbf{b})_\times]_{ij} &= \varepsilon_{ikj} \varepsilon_{klm} a_l b_m = \\
 &= (\delta_{jl} \delta_{im} - \delta_{jm} \delta_{il}) a_l b_m = \\
 &= a_j b_i - a_i b_j = [\mathbf{b} \mathbf{a}^T - \mathbf{a} \mathbf{b}^T]_{ij} .
 \end{aligned}$$

Thus

$$\begin{aligned}
 \mathbf{h}' &= \mathbf{h} - \gamma c \mathbf{v} \times \mathbf{d} + (1 - \gamma)(\mathbf{h} \times \hat{\mathbf{v}}) \times \hat{\mathbf{v}} = \\
 &= \mathbf{h} - \gamma c \mathbf{v} \times \mathbf{d} + (1 - \gamma) (\hat{\mathbf{v}} \hat{\mathbf{v}} \cdot \mathbf{h} - \mathbf{h}) = \\
 &= \gamma \mathbf{h} - \gamma c \mathbf{v} \times \mathbf{d} + (1 - \gamma) \hat{\mathbf{v}} \hat{\mathbf{v}} \cdot \mathbf{h} .
 \end{aligned}$$

Going back from non dimensional velocity to dimensional velocity $c\mathbf{v} \rightarrow \mathbf{v}$,

$$\begin{aligned}
 \mathbf{d}' &= \gamma \left(\mathbf{d} - \frac{\mathbf{h} \times \mathbf{v}}{c^2} \right) + (1 - \gamma) \hat{\mathbf{v}} \hat{\mathbf{v}} \cdot \mathbf{d} \\
 \mathbf{h}' &= \gamma (\mathbf{h} - \mathbf{v} \times \mathbf{d}) + (1 - \gamma) \hat{\mathbf{v}} \hat{\mathbf{v}} \cdot \mathbf{h}
 \end{aligned}$$

Repeating the same process for the electromagnetic field tensor

$$[\mathbf{F}]^{\alpha\beta} = \begin{bmatrix} 0 & -\frac{\mathbf{e}^T}{c} \\ \frac{\mathbf{e}}{c} & \mathbf{b}_\times \end{bmatrix}$$

$$\mathbf{e}' = \gamma (\mathbf{e} - \mathbf{b} \times \mathbf{v}) + (1 - \gamma) \hat{\mathbf{v}} \hat{\mathbf{v}} \cdot \mathbf{e}$$

$$\mathbf{b}' = \gamma \left(\mathbf{b} - \frac{\mathbf{v} \times \mathbf{e}}{c^2} \right) + (1 - \gamma) \hat{\mathbf{v}} \hat{\mathbf{v}} \cdot \mathbf{b}$$

Example (4-Current)

$$\mathbf{J}' = \Lambda \mathbf{J}$$

$$c\rho' = \gamma c\rho - \gamma v \mathbf{j}$$

$$\mathbf{j}' = -\gamma v c\rho + \gamma \mathbf{j}$$

Example (Constitutive equations)

Linear isotropic medium

$$\begin{aligned}\mathbf{d} &= \varepsilon_0 \mathbf{e} + \mathbf{p} \\ \mathbf{b} &= \mu_0 \mathbf{h} + \mu_0 \mathbf{m}\end{aligned}$$

with $c^2 = \frac{1}{\varepsilon_0 \mu_0}$

$$\begin{aligned}c\mathbf{d} &= \varepsilon_0 c^2 \frac{\mathbf{e}}{c} + c\mathbf{p} = \frac{1}{\mu_0} \frac{\mathbf{e}}{c} + c\mathbf{p} \\ \mathbf{h} &= \frac{1}{\mu_0} \mathbf{b} - \mathbf{m}\end{aligned}$$

the constitutive equations can be written in 4-dimensional form as

$$\begin{aligned}\mathbf{D} &= \frac{1}{\mu_0} \mathbf{F} + \mathbf{P} \\ \begin{bmatrix} 0 & -c\mathbf{d}^T \\ c\mathbf{d} & \mathbf{h}_\times \end{bmatrix} &= \frac{1}{\mu_0} \begin{bmatrix} 0 & -\mathbf{e}^T/c \\ \mathbf{e}/c & \mathbf{b}_\times \end{bmatrix} + \begin{bmatrix} 0 & -c\mathbf{p}^T \\ c\mathbf{p} & -\mathbf{m}_\times \end{bmatrix}\end{aligned}$$

ELECTROMAGNETISM

From 3-dimensional to 4-dimensional formalism. In this section the 4-dimensional formalism is introduced, and the 4-dimensional version of governing equations of electromagnetism that naturally suits special relativity theory is derived starting from the governing equations of classical electromagnetism.

Definition of physical quantities

- 4-potential vector, $\mathbf{A} = \frac{\phi}{c}\mathbf{E}_0 + a^i\mathbf{E}_i$
- 4-current vector, $\mathbf{J} = c\rho\mathbf{E}_0 + j^i\mathbf{E}_i$
- EM field tensor,

$$\begin{aligned}\mathbf{F} &= -\mathbf{E}_0 \otimes \mathbf{E}_i \frac{e_i}{c} + \frac{e_i}{c}\mathbf{E}_i \otimes \mathbf{E}_0 + \varepsilon_{ijk}b_j\mathbf{E}_i \otimes \mathbf{E}_k \\ \mathbf{D} &= -\mathbf{E}_0 \otimes \mathbf{E}_i cd_i + d_i\mathbf{E}_i \otimes \mathbf{E}_0 + \varepsilon_{ijk}h_j\mathbf{E}_i \otimes \mathbf{E}_k\end{aligned}$$

or

$$[F^{\alpha\beta}] = \begin{bmatrix} 0 & -\mathbf{e}^T/c \\ \mathbf{e}/c & \mathbf{b}_{\times} \end{bmatrix}, \quad [D^{\alpha\beta}] = \begin{bmatrix} 0 & -c\mathbf{d}^T \\ c\mathbf{d} & \mathbf{h}_{\times} \end{bmatrix}$$

- 4-momentum-energy density tensor, $\mathbf{T}$

Definitions, relations and equations

- Lorentz's gauge, $\nabla \cdot \mathbf{A} = 0$
- EM field tensor, $\mathbf{F} = \nabla\mathbf{A} - (\nabla\mathbf{A})^T$
- 4-current continuity equation, $\nabla \cdot \mathbf{J} = 0$
- Maxwell's equations

$$\begin{aligned}\nabla \cdot \mathbf{D} &= \mathbf{J}^f \\ \nabla \cdot (\epsilon : \mathbf{F}) &= \mathbf{0}\end{aligned}$$

- Constitutive equations

$$\mathbf{D} = \frac{1}{\mu_0}\mathbf{F} + \mathbf{P}$$

- Wave equation for the potential (what about wave equations for the EM field?)

$$\nabla \cdot \nabla\mathbf{A} = \mu_0\mathbf{J}$$

- Energy balance equation

$$\nabla \cdot \mathbf{T} = -\mathbf{F} \cdot \mathbf{J}$$

- Equation of motion of a point charge in a EM field,

$$m\mathbf{X}'' = q\mathbf{F} \cdot \mathbf{X}'$$

Einstein's special relativity and Lorentz's transformations for the EM quantities. After the equations are derived, Lorentz's transformations are used to discuss special relativity. Low-speed relations, used in classical electromagnetism for systems with characteristic speed $v \ll c$, are derived.

Lagrangian approach to electromagnetism in special relativity. Weak form of the equations are derived, and the Lagrangian approach to electromagnetism in special relativity is discussed: both field equations and dynamical equations of charges moving in an electromagnetic field are re-derived with a Lagrangian approach.

Part II

General Relativity

GENERAL RELATIVITY

GENERAL RELATIVITY - NOTES

- *Differential geometry*: metrics, derivatives, curvatures
- *Einstein's equation*

6.1 Differential Geometry

References

- Math basics: **todo** check and complete the chapters about **tensor calculus** and **differential geometry**

Basics

Let $\mathbf{X}(q^k)$ a parametrization of points in a n -dimensional space, $k = 1 : n$.

Natural basis, $\mathbf{b}_k := \frac{\partial \mathbf{X}}{\partial q^k}$. Vectors of the natural basis can be used to write vectors and tensor fields as a linear combination of them, or in their components w.r.t. that basis

$$\mathbf{v} = v^k \mathbf{b}_k \quad , \quad \mathbf{A} = A^{kl} \mathbf{b}_k \otimes \mathbf{b}_l .$$

Metric tensor, The covariant components are $\mathbf{b}_k \cdot \mathbf{b}_l =: g_{kl}$. Some properties: $g_k^l = \delta_k^l$, $g_{ab} g^{bc} = \delta_a^c$.

Contravariant basis, $\{\mathbf{b}^k\}$, s.t. $\mathbf{b}_k \cdot \mathbf{b}^l = \delta_k^l$. It's easy to prove that $\mathbf{b}_k = g_{kl} \mathbf{b}^l$ (just scalar product with $\mathbf{b}^j \dots$), and the law for raising or lower indices holds, $A^{ij} = g^{ik} A_k^j, \dots$

Derivatives of the basis vectors. The components of the derivative $\frac{\partial \mathbf{b}_i}{\partial q^k} = \Gamma_{ik}^l \mathbf{b}_l$ are defined as the Christoffel symbols of the second type. As these are second-order derivatives, for Schwartz theorem about mixed partial derivatives, the symmetry $\Gamma_{ij}^k = \Gamma_{ji}^k$ immediately follows. The derivative of the vectors of the contravariant basis follows from

$$\begin{aligned} 0 &= \partial_k (\mathbf{b}_l \cdot \mathbf{b}^m) = \\ &= \partial_k \mathbf{b}_l \cdot \mathbf{b}^m + \mathbf{b}_l \cdot \partial_k \mathbf{b}^m = \\ &= \Gamma_{kl}^n \underbrace{\mathbf{b}_n \cdot \mathbf{b}^m}_{=\delta_n^m} + \mathbf{b}_l \cdot \partial_k \mathbf{b}^m , \end{aligned}$$

and thus $\partial_k \mathbf{b}^m = -\Gamma_{kl}^m \mathbf{b}^l$.

Derivatives of the metric tensor. Using the definition of the covariant components of the metric tensor and the derivatives of the vectors of the natural basis, it's easy to prove

$$\partial_c g_{ab} = \Gamma_{ac}^d g_{db} + \Gamma_{bc}^d g_{ad}$$

Using the property $g_{ab}g^{bc} = \delta_a^c$, from its derivatives ∂_d ,

$$\partial_d g_{ab} g^{bc} + g_{ab} \partial_d g^{bc} = 0 ,$$

it follows that $\partial_d g^{ec} = -g^{ea} \partial_d g_{ab} g^{bc}$.

Relations between the Christoffel symbols and the derivatives of the metric tensor.

$$\partial_c g_{ab} = \Gamma_{ac}^d g_{db} + \Gamma_{bc}^d g_{ad}$$

$$\partial_a g_{bc} = \Gamma_{ba}^d g_{dc} + \Gamma_{ca}^d g_{bd}$$

$$\partial_b g_{ca} = \Gamma_{cb}^d g_{da} + \Gamma_{ab}^d g_{cd}$$

so that, summing the first 2 equations and subtracting the last one,

$$\Gamma_{ac}^b = \frac{1}{2} g^{bd} (\partial_c g_{ad} + \partial_a g_{dc} - \partial_d g_{ac}) \quad (6.1)$$

6.1.1 Gradient, covariant derivative and directional derivative

Gradient of a scalar field

$$\nabla f = \mathbf{b}^k \partial_k f .$$

Gradient of a vector field

$$\nabla \mathbf{v} = \mathbf{b}^k \mathbf{b}_i (\partial_k v^i + \Gamma_{kl}^i v^l) .$$

Proof

$$\begin{aligned} \nabla \mathbf{v} &= \mathbf{b}^k \frac{\partial}{\partial q^k} (v^i \mathbf{b}_i) = \\ &= \mathbf{b}^k \mathbf{b}_i \partial_k v^i + \mathbf{b}^k \mathbf{b}_l \Gamma_{ik}^l v^i = \\ &= \mathbf{b}^k \mathbf{b}_i (\partial_k v^i + \Gamma_{kl}^i v^l) . \end{aligned}$$

Gradient of a 2-nd order tensor field

$$\nabla \mathbf{A} = \mathbf{b}^k \mathbf{b}_i \mathbf{b}_j (\partial_k A^{ij} + \Gamma_{kl}^i A^{lj} + \Gamma_{kl}^j A^{il}) = \mathbf{b}^k \mathbf{b}_i \mathbf{b}_j \nabla_k A^{ij} .$$

Proof

...

6.1.2 Curvature Tensor

todo. Meaning of this definition

...

Definition through the action on an arbitrary vector field, whose components are

$$\{\mathbf{v} \cdot \mathbf{R}\}_{\sigma\eta\nu} = v_\xi R^\xi_{\sigma\eta\nu} = (\nabla_\eta \nabla_\nu - \nabla_\nu \nabla_\eta) v_\sigma$$

First term

$$\begin{aligned}
 \nabla \nabla \mathbf{v} &= \mathbf{b}^\eta \partial_\eta [\mathbf{b}^\nu \partial_\nu (v_\sigma \mathbf{b}^\sigma)] = \\
 &= \mathbf{b}^\eta \partial_\eta [\mathbf{b}^\nu \mathbf{b}^\sigma (\partial_\nu v_\sigma - \Gamma_{\nu\sigma}^\xi v_\xi)] = \\
 &= \mathbf{b}^\eta \mathbf{b}^\nu \mathbf{b}^\sigma [-\Gamma_{\eta\nu}^\mu (\dots)_{\mu\sigma} - \Gamma_{\eta\sigma}^\mu (\dots)_{\nu\mu} + \partial_{\eta\nu} v_\sigma - \partial_\eta (\Gamma_{\nu\sigma}^\xi v_\xi)] = \\
 &= \mathbf{b}^\eta \mathbf{b}^\nu \mathbf{b}^\sigma [-\Gamma_{\eta\nu}^\mu (\partial_\mu v_\sigma - \Gamma_{\mu\sigma}^\xi v_\xi) - \Gamma_{\eta\sigma}^\mu (\partial_\mu v_\nu - \Gamma_{\mu\nu}^\xi v_\xi) + \partial_{\eta\nu} v_\sigma - \partial_\eta \Gamma_{\nu\sigma}^\xi v_\xi - \Gamma_{\nu\sigma}^\xi \partial_\eta v_\xi] = \\
 &= \mathbf{b}^\eta \mathbf{b}^\nu \mathbf{b}^\sigma \nabla_{\eta\nu} v_\sigma .
 \end{aligned}$$

$$\nabla_{\sigma\eta} v_\nu - \nabla_{\eta\sigma} v_\nu$$

Exploiting symmetry properties of Christoffel symbols,

todo!!! change indices to match the definition of the curvature tensor

$$\begin{aligned}
 \nabla_{\sigma\eta} v_\nu - \nabla_{\eta\sigma} v_\nu &= -\underbrace{\Gamma_{\sigma\eta}^\xi \partial_\xi v_\nu}_1 + \underbrace{\Gamma_{\sigma\eta}^\mu \Gamma_{\mu\nu}^\xi v_\xi}_2 - \underbrace{\Gamma_{\sigma\nu}^\xi \partial_\eta v_\xi}_3 + \underbrace{\Gamma_{\sigma\nu}^\mu \Gamma_{\mu\eta}^\xi v_\xi}_4 + \underbrace{\partial_{\sigma\eta} v_\nu}_4 - \underbrace{\partial_\sigma \Gamma_{\eta\nu}^\xi v_\xi}_5 - \underbrace{\Gamma_{\eta\nu}^\xi \partial_\sigma v_\xi}_5 + \\
 &- \left[-\underbrace{\Gamma_{\eta\sigma}^\xi \partial_\xi v_\nu}_1 + \underbrace{\Gamma_{\eta\sigma}^\mu \Gamma_{\mu\nu}^\xi v_\xi}_2 - \underbrace{\Gamma_{\eta\nu}^\xi \partial_\sigma v_\xi}_5 + \underbrace{\Gamma_{\eta\nu}^\mu \Gamma_{\mu\sigma}^\xi v_\xi}_4 + \underbrace{\partial_{\eta\sigma} v_\nu}_4 - \underbrace{\partial_\eta \Gamma_{\sigma\nu}^\xi v_\xi}_3 - \underbrace{\Gamma_{\sigma\nu}^\xi \partial_\eta v_\xi}_3 \right] = \\
 &= [\partial_\eta \Gamma_{\sigma\nu}^\xi - \partial_\sigma \Gamma_{\eta\nu}^\xi + \Gamma_{\sigma\nu}^\mu \Gamma_{\mu\eta}^\xi - \Gamma_{\eta\nu}^\mu \Gamma_{\mu\sigma}^\xi] v_\xi .
 \end{aligned}$$

Thus, it follows that

$$\begin{aligned}
 R_{\sigma\eta\nu}^\xi &= \partial_\eta \Gamma_{\sigma\nu}^\xi - \partial_\nu \Gamma_{\eta\sigma}^\xi + \Gamma_{\sigma\nu}^\mu \Gamma_{\mu\eta}^\xi - \Gamma_{\eta\sigma}^\mu \Gamma_{\mu\nu}^\xi \\
 R_{\phi\sigma\eta\nu} &= g_{\phi\xi} R_{\sigma\eta\nu}^\xi = g_{\phi\xi} (\partial_\eta \Gamma_{\sigma\nu}^\xi - \partial_\nu \Gamma_{\eta\sigma}^\xi + \Gamma_{\sigma\nu}^\mu \Gamma_{\mu\eta}^\xi - \Gamma_{\eta\sigma}^\mu \Gamma_{\mu\nu}^\xi)
 \end{aligned}$$

Ricci's tensor

todo Meaning

Ricci's tensor is defined as the contraction of the first and third indices of the curvature tensor with the metric tensor,

$$\begin{aligned}
 R_{\sigma\nu} &:= g^{\alpha\beta} R_{\alpha\sigma\beta\nu} = \underbrace{g^{\alpha\beta} g_{\alpha\gamma}}_{\delta_\gamma^\beta} R^\gamma_{\sigma\beta\nu} = R^\mu_{\sigma\mu\nu} \\
 R_{\sigma\nu} &= \partial_\xi \Gamma_{\sigma\nu}^\xi - \partial_\nu \Gamma_{\xi\sigma}^\xi + \Gamma_{\sigma\nu}^\mu \Gamma_{\mu\xi}^\xi - \Gamma_{\xi\nu}^\mu \Gamma_{\mu\sigma}^\xi
 \end{aligned}$$

Curvature scalar

todo Meaning

$$\begin{aligned}
 R &:= g^{\sigma\nu} R_{\sigma\nu} = R^\sigma_{\sigma} . \\
 R &= g^{\sigma\nu} R_{\sigma\nu} = g^{\sigma\nu} (\partial_\xi \Gamma_{\sigma\nu}^\xi - \partial_\nu \Gamma_{\xi\sigma}^\xi + \Gamma_{\sigma\nu}^\mu \Gamma_{\mu\xi}^\xi - \Gamma_{\xi\nu}^\mu \Gamma_{\mu\sigma}^\xi)
 \end{aligned}$$

Properties

- $R_{abcd} = -R_{abdc}$

Proof

By direct inspection of the expression of the components of the curvature tensor,

$$R_{\phi\sigma\eta\nu} = g_{\phi\xi} R^{\xi}_{\sigma\eta\nu} = g_{\phi\xi} \left(\partial_{\eta} \Gamma^{\xi}_{\sigma\nu} - \partial_{\nu} \Gamma^{\xi}_{\eta\sigma} + \Gamma^{\mu}_{\sigma\nu} \Gamma^{\xi}_{\mu\eta} - \Gamma^{\mu}_{\eta\sigma} \Gamma^{\xi}_{\mu\nu} \right)$$

and the symmetry of the Christoffel symbols, $\Gamma^c_{ab} = \Gamma^c_{ba}$.

- $R_{abcd} = -R_{bacd}$
- $R_{abcd} + R_{acdb} + R_{adb c} = 0$
- $R_{abcd} = R_{cdab}$
- $R_{abcd;e} + R_{abde;c} + R_{abec;d} = 0$ (Second Bianchi identity)

Proof

$$\begin{aligned} R_{\phi\sigma\eta\nu;\chi} &= (g_{\phi\xi} R^{\xi}_{\sigma\eta\nu})_{;\chi} = \\ &= g_{\phi\xi} \left(\partial_{\eta} \Gamma^{\xi}_{\sigma\nu} - \partial_{\nu} \Gamma^{\xi}_{\eta\sigma} + \Gamma^{\mu}_{\sigma\nu} \Gamma^{\xi}_{\mu\eta} - \Gamma^{\mu}_{\eta\sigma} \Gamma^{\xi}_{\mu\nu} \right)_{;\chi} = \\ &= \end{aligned}$$

todo

Divergence of Ricci's tensor

$$\nabla_{\mu} R^{\mu}_{\nu} = \frac{1}{2} \nabla_{\nu} R.$$

Proof

$$\begin{aligned} R_{\sigma\nu} &= R^{\xi}_{\sigma\xi\nu} = \partial_{\xi} \Gamma^{\xi}_{\sigma\nu} - \partial_{\sigma} \Gamma^{\xi}_{\xi\nu} + \Gamma^{\mu}_{\sigma\nu} \Gamma^{\xi}_{\mu\xi} - \Gamma^{\mu}_{\xi\nu} \Gamma^{\xi}_{\mu\sigma} \\ R^{\mu}_{\nu} &= g^{\mu\sigma} R_{\sigma\nu} \\ \nabla_{\mu} R^{\mu}_{\nu} &= \partial_{\mu} R^{\mu}_{\nu} + \Gamma^{\mu}_{\mu\sigma} R^{\sigma}_{\nu} - \Gamma^{\mu}_{\sigma\nu} R^{\sigma}_{\mu} = \\ &= \end{aligned}$$

6.2 Einstein's equation

Idea:

1. Try with **the simplest relation** between momentum-energy tensor T and a second order tensor representing the geometry of space, i.e. **proportionality** with Ricci's curvature tensor R

$$R = \kappa T .$$

Remark. This relationship doesn't agree with local energy-momentum balance $\nabla \cdot T = 0$, as $\nabla \cdot R \neq 0$,

$$\begin{aligned} \nabla_\mu R^\mu{}_\nu &= \frac{1}{2} \nabla_\nu R \\ \nabla_\mu R^{\mu\sigma} &= \nabla_\mu (g^{\sigma\nu} R^\mu{}_\nu) = g^{\sigma\nu} \nabla_\mu R^\mu{}_\nu = \frac{1}{2} g^{\sigma\nu} \nabla_\nu R = \nabla_\mu \left(\frac{1}{2} g^{\sigma\mu} R \right) , \end{aligned}$$

as the covariant derivative of the metric components of the metric tensor are identically zero. Moving the extreme terms of this equality on the same side of the equal sign, $\nabla_\mu (R^{\mu\sigma} - \frac{1}{2} g^{\mu\sigma} R) = 0$.

Covariant derivative of the metric tensor

$$\nabla_\nu g_{\mu\eta} = \partial_\nu g_{\mu\eta} - \Gamma_{\nu\mu}^\sigma g_{\sigma\eta} - \Gamma_{\nu\eta}^\sigma g_{\mu\sigma} = 0 ,$$

as $\partial_\nu g_{\mu\eta} = \Gamma_{\nu\mu}^\sigma g_{\sigma\eta} + \Gamma_{\nu\eta}^\sigma g_{\mu\sigma}$ (see the derivative of the components of the metric tensor in *Differential geometry for general relativity*).

2. Correct the relation, replacing Ricci's tensor with its divergence-free part

$$R^{\mu\sigma} - \frac{1}{2} g^{\mu\sigma} R = \kappa T^{\mu\sigma} ,$$

or

$$R - \frac{1}{2} g R = \kappa T .$$

3. Check against the classical limit, to find:
 - if Newton gravitation is the classical limit of general relativity
 - the values of the constants involved in the model, κ

Under some assumptions:

- **weak gravitation** for the linearization of the metric tensor $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ around Minkowski flat time-space (here $\eta_{\mu\nu}$ describing Minkowski metrics).
- **pressure-less mass distribution at rest**
- **mass** distribution “at rest” (w.r.t. the “quasi inertial observer”)

6.2.1 Alternative expression of Einstein's equations

Let's evaluate the trace of R , i.e. $R = R^\sigma{}_\sigma$ as a function of the trace of T .

$$\begin{aligned} R &:= R^\sigma{}_\sigma = \\ &= g_{\sigma\varphi} R^{\varphi\sigma} = \\ &= g_{\sigma\varphi} \left(\frac{1}{2} g^{\varphi\sigma} R + \kappa T^{\varphi\sigma} \right) = 2R + \kappa T \end{aligned}$$

and thus the relation $R = -\kappa T$ holds between the trace of Ricci and energy-momentum tensors (here $g_{\sigma\varphi}g^{\sigma\varphi} = \delta_{\sigma}^{\sigma} = 4$ in the 4-dimensional time-space). Thus, Einstein equation can be recast as

$$R = \kappa \left(T - \frac{1}{2} g T \right) . \quad (6.2)$$

6.2.2 Classical limit

1. Linearization of the metric tensor. $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$, and the inverse (linearized) relation gives $g^{\mu\nu} = \eta^{\mu\nu} - h^{\mu\nu}$

2. Linearization of Christoffel symbols.

$$\Gamma_{00}^i = -\frac{1}{2} \partial_i h_{00}$$

in the static limit, $\partial_0 \equiv 0$, for the spatial components, $i = 1 : 3$.

3. Linearization of the geodesics equation.

$$\ddot{q}^\mu + \Gamma_{\nu\sigma}^\mu \dot{q}^\nu \dot{q}^\sigma = 0$$

For $\mu = i = 1 : 3$, as the $\tau \sim t$ in the slow-regime limit, and $q^0 \sim ct$ and $q^i = x_i$,

$$\ddot{x}_i = -c^2 \Gamma_{00}^i = \frac{c^2}{2} \partial_i h_{00} .$$

This equation must be compared with the dynamical equation of the Newtonian mechanics, $\ddot{\vec{r}} = -\nabla\Phi$. In order to get the same equation, $\partial_i h_{00} = -\frac{2}{c^2} \partial_i \Phi$, and thus - except for an arbitrary constant (irrelevant) - $h_{00} = -\frac{2}{c^2} \Phi$.

4. Linearized Einstein equation, with energy-momentum tensor (of the pressure-less medium) $T^{\mu\nu} = \rho c^2 \delta_0^\mu \delta_0^\nu$, so that its trace reads $T = \rho c^2$. Thus the linearized 00 component of Einstein equation reads

$$R_{00} = \kappa \left(\rho c^2 - \frac{1}{2} \rho c^2 \right) = \kappa \frac{1}{2} \rho c^2$$

5. Linearized relation between curvature tensor and the linearized metrics.

$$R_{00} = \partial_i \Gamma_{00}^i = -\frac{1}{2} \partial_{ii} h_{00} = \frac{1}{c^2} \nabla_{\vec{r}}^2 \Phi$$

6. Comparison of the two expressions of R_{00} .

$$\frac{\kappa}{2} \rho c^2 = \frac{1}{c^2} \nabla_{\vec{r}}^2 \Phi ,$$

or rearranging,

$$\nabla^2 \Phi = \frac{\kappa c^4}{2} \rho .$$

By direct comparison with the Poisson equation for Newtonian gravitation, $\nabla^2 \Phi = 4\pi G \rho$, it follows that

$$\kappa = \frac{8\pi G}{c^4} .$$

6.3 Examples

- *Schwarzschild metric*
- *Orbits in Schwarzschild spacetime*

6.3.1 Schwarzschild metrics

Spatially spherically symmetric and **static vacuum** solution of the Einstein equation, representing the exterior gravitational field of a **non-rotating, uncharged** massive body.

Schwarzschild metrics gives

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2. \quad (6.3)$$

Vacuum equation

Without mass density and electric charge and current, $T = 0$. From the expression of (6.2), it immediately follows, $R = 0$.

Metric Ansatz & Symmetry Assumptions

The four coordinates used to parametrize the space-time are (t, r, θ, ϕ) . The two physical symmetry constraints in Schwarzschild solution of EFE are

1. **Spherical Symmetry:** the spatial geometry is invariant under rotation, s.t. the angular part of the metric is given by the standard round metric on S^2 , $d\Omega^2 = d\theta^2 + \sin^2 \theta d\phi^2$
2. **Staticity:** the metric coefficients are independent of the time coordinate t ($\partial_t g_{\mu\nu} = 0$); the further assumption that the line element is invariant under time reversal eliminates cross-terms such as $dt dr$.

Under these assumptions, the general line element can be parameterized using two undetermined functions of radius, $A(r)$ and $B(r)$:

$$ds^2 = -e^{2A(r)} c^2 dt^2 + e^{2B(r)} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2)$$

Remark. Using exponential parameterizations $e^{2A(r)}$ and $e^{2B(r)}$ ensures that $g_{tt} < 0$ and $g_{rr} > 0$ outside any horizon.
todo Add some remark/examples about the role of time, and its consequence on metrics

From the choosen parametrization, the non-zero covariant components of the metric tensors are

$$\begin{aligned} g_{tt} &= -e^{-2A(r)} c^2 \\ g_{rr} &= e^{2B(r)} \\ g_{\theta\theta} &= r^2 \\ g_{\phi\phi} &= r^2 \sin^2 \theta \end{aligned}$$

As the metric tensor is diagonal, it's contravariant components are just the inverse of its covariant components.

Christoffel Symbols

The Christoffel symbols of the second kind are computed via the expression (6.1),

$$\Gamma_{ac}^b = \frac{1}{2} g^{bd} (\partial_c g_{ad} + \partial_a g_{dc} - \partial_d g_{ac}) .$$

Writing the derivative w.r.t. r as $\partial_r A(r) = A'(r)$

- $\Gamma_{tr}^t = \Gamma_{rt}^t = \frac{1}{2} g^{td} (\partial_r g_{td} + \partial_t g_{dr} - \partial_d g_{tr}) = -\frac{1}{2} e^{2A} c^{-2} \cdot (-2A') e^{-2A} c^2 = A'$
- $\Gamma_{tt}^r = A' e^{2(A-B)} c^2$
- $\Gamma_{rr}^r = B'$
- $\Gamma_{\theta\theta}^r = -r e^{-2B}$
- $\Gamma_{\phi\phi}^r = -r \sin^2 \theta e^{-2B}$
- $\Gamma_{r\theta}^\theta = \Gamma_{\theta r}^\theta = \frac{1}{r}$
- $\Gamma_{\phi\phi}^\theta = -\sin \theta \cos \theta$
- $\Gamma_{r\phi}^\phi = \Gamma_{\phi r}^\phi = \frac{1}{r}$
- $\Gamma_{\theta\phi}^\phi = \Gamma_{\phi\theta}^\phi = \cot \theta$

Components of the Ricci Tensor

The Ricci tensor is defined by contracting the Riemann curvature tensor:

$$R_{\mu u} = \partial_\lambda \Gamma_{\mu u}^\lambda - \partial_u \Gamma_{\mu \lambda}^\lambda + \Gamma_{\mu u}^\lambda \Gamma_{\lambda \sigma}^\sigma - \Gamma_{\mu \lambda}^\sigma \Gamma_{u \sigma}^\lambda$$

Evaluating the non-zero independent components yields:

- **Time-Time Component (R_{tt})**

$$R_{tt} = e^{2(A-B)} c^2 \left[A'' + (A')^2 - A' B' + \frac{2A'}{r} \right]$$

- **Radial-Radial Component (R_{rr})**

$$R_{rr} = -A'' - (A')^2 + A' B' + \frac{2B'}{r}$$

- **Angular Component ($R_{\theta\theta}$)**

$$R_{\theta\theta} = 1 - e^{-2B} [1 + r(A' - B')]$$

Remarks.

- The $R_{\phi\phi}$ component yields $R_{\phi\phi} = R_{\theta\theta} \sin^2 \theta$, providing no independent equation).
- The non-diagonal components are identically zero (**todo prove it!**)

Solving the Differential Equations

Combining R_{tt} and R_{rr} (linear combination $e^{-2(A-B)} \frac{R_{tt}}{c^2} + R_{rr} = 0$) immediately gives

$$A'(r) = -B'(r) .$$

and integrating with respect to r ,

$$A(r) + B(r) = C = \text{const.}$$

Boundary conditions at infinity. To satisfy the boundary condition of asymptotic flatness—that spacetime approaches flat Minkowski space as $r \rightarrow \infty$, then

$$\begin{aligned} g_{tt} &= -e^{2A(r)} \rightarrow -1 & A(r) &\rightarrow 0 \\ g_{rr} &= e^{2B(r)} \rightarrow 1 & B(r) &\rightarrow 0 \end{aligned}$$

and thus $C = 0$, and $A(r) = -B(r)$.

Solving for $A(r)$ via $R_{\theta\theta}$. Substituting $B' = -A'$ and $e^{-2B} = e^{2A}$ into $R_{\theta\theta} = 0$:

$$\begin{aligned} 0 &= R_{\theta\theta} = \\ &= 1 - e^{2A} (1 + 2rA') = \\ &= 1 - \frac{d}{dr} (re^{2A}) , \end{aligned}$$

and thus, integrating in r ,

$$re^{2A(r)} = r + C .$$

or

$$e^{2A(r)} = 1 + \frac{C}{r}$$

where C is a constant of integration. Since $e^{2B(r)} = e^{-2A(r)}$, it follows $e^{2B(r)} = \left(1 + \frac{C}{r}\right)^{-1}$.

Determining the Integration Constant C (The Newtonian Limit)

To determine C , we examine the metric in the weak-field, low-velocity limit ($r \rightarrow \infty$). In this limit, general relativity recovers **Newtonian gravity**, as shown in [classical limit](#),

$$g_{tt} = \eta_{tt} + h_{tt} \approx -1 - \frac{2}{c^2} \Phi ,$$

where $\Phi(r) = -\frac{GM}{r}$ is the Newtonian gravitational potential of a central mass M . Equating the two expressions of the coefficient g_{tt}

$$g_{tt} = -e^{2A(r)} = -1 - \frac{C}{r} = -1 + \frac{2}{c^2} \frac{GM}{r}$$

and thus

$$C = -\frac{2GM}{c^2} .$$

Defining the **Schwarzschild radius** $r_s := \frac{2GM}{c^2}$, the expression of **Schwarzschild metric** becomes

$$ds^2 = -\left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 .$$

Special Case: Complete Vacuum ($M = 0$)

If the spacetime contains no mass ($M = 0$), the Minkowski flat spacetime follows

$$ds^2 = -c^2 dt^2 + dr^2 + r^2 d\theta^2 + r^2 \sin^2 \theta d\phi^2 = -c^2 dt^2 + |d\vec{r}|^2 ,$$

or, using the common transformation between spherical and Cartesian space coordinates,

$$\begin{cases} t = t \\ x = r \sin \theta \cos \phi \\ y = r \sin \theta \sin \phi \\ z = r \cos \theta , \end{cases}$$

$$ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2 .$$

6.3.2 Planetary orbits in Schwarzschild spacetime

Schwarzschild metrics (6.3) provides the metrics induced by a massive object in the vacuum spacetime around. The motion of a light object approximately doesn't affect the massive object generating the steady Schwarzschild metrics. The light object moves following the geodesics. Let the motion occurs in the equatorial plane, i.e. $\theta = \frac{\pi}{2}$, and $d\theta = 0$. Thus

$$-c^2 d\tau^2 = ds^2 = -\left(1 - \frac{r_s}{r}\right) c^2 dt^2 + \left(1 - \frac{r_s}{r}\right)^{-1} dr^2 + r^2 d\phi^2 .$$

Geodesics. With $A(r) = \frac{1}{2} \ln \left(1 - \frac{r_s}{r}\right)$, $A'(r) = \frac{1}{2} \frac{1}{1 - \frac{r_s}{r}} \frac{r_s}{r^2}$

$$0 = \ddot{q}^k + \Gamma_{ab}^k \dot{q}^a \dot{q}^b$$

$$\begin{aligned}
 t : \quad 0 &= \ddot{t} + \Gamma_{ab}^t \dot{q}^a \dot{q}^b = \\
 &= \ddot{t} + 2\Gamma_{tr}^t \dot{t} \dot{r} = \\
 &= \ddot{t} + \frac{1}{1 - \frac{r_s}{r}} \frac{r_s}{r^2} \dot{t} \dot{r} \\
 0 &= \left(1 - \frac{r_s}{r}\right)^{-1} \left[\left(1 - \frac{r_s}{r}\right) \ddot{t} + \frac{r_s}{r^2} \dot{t} \dot{r} \right] = \\
 0 &= \left(1 - \frac{r_s}{r}\right)^{-1} \frac{d}{d\tau} \left[\left(1 - \frac{r_s}{r}\right) \dot{t} \right] \rightarrow \left(1 - \frac{r_s}{r}\right) \dot{t} =: \frac{E}{c} = \text{const.} \\
 r : \quad 0 &= \ddot{r} + \Gamma_{ab}^r \dot{q}^a \dot{q}^b = \\
 &= \ddot{r} + \Gamma_{tt}^r \dot{t}^2 + \Gamma_{rr}^r \dot{r}^2 + \Gamma_{\theta\theta}^r \dot{\theta}^2 + \Gamma_{\phi\phi}^r \dot{\phi}^2 = \quad (\dot{\theta} = 0) \\
 &= \ddot{r} + \Gamma_{tt}^r \dot{t}^2 + \Gamma_{rr}^r \dot{r}^2 + \Gamma_{\phi\phi}^r \dot{\phi}^2 = \\
 &= \ddot{r} + A' e^{4A} c^2 \dot{t}^2 - A'(r) \dot{r}^2 - r \sin^2 \theta e^{2A} \dot{\phi}^2 = \\
 &= \ddot{r} + A' e^{4A} c^2 \dot{t}^2 - A'(r) \dot{r}^2 - r \theta e^{2A} \dot{\phi}^2 = \\
 \phi : \quad 0 &= \ddot{\phi} + \Gamma_{ab}^\phi \dot{q}^a \dot{q}^b \\
 &= \ddot{\phi} + 2\Gamma_{r\phi}^\phi \dot{r} \dot{\phi} + 2\Gamma_{\theta\phi}^\phi \dot{\theta} \dot{\phi} = \quad (\dot{\theta} = 0) \\
 &= \ddot{\phi} + 2\Gamma_{r\phi}^\phi \dot{r} \dot{\phi} = \\
 &= \ddot{\phi} + 2 \frac{1}{r} \dot{r} \dot{\phi}, \\
 0 &= \frac{1}{r^2} \frac{d}{d\tau} (r^2 \dot{\phi}) \rightarrow r^2(\tau) \dot{\phi}(\tau) =: L = \text{const.} \\
 \theta : \quad 0 &= \ddot{\theta} + \Gamma_{ab}^\theta \dot{q}^a \dot{q}^b \\
 &= \ddot{\theta} + 2\Gamma_{r\phi}^\theta \dot{r} \dot{\phi} + \Gamma_{\phi\phi}^\theta \dot{\phi}^2 = \quad (\dot{\phi} = 0) \\
 &= \ddot{\theta} - \sin \theta \cos \theta \Gamma_{\phi\phi}^\theta \dot{\phi}^2 = \\
 &= \ddot{\theta}.
 \end{aligned}$$

Replacing the expression for $\dot{t}$ and $\dot{\phi}$ into the relation $u^\mu u_\mu = -c^2$ (from $ds^2/d\tau^2$)

$$\begin{aligned}
 -c^2 &= -\left(1 - \frac{r_s}{r}\right) c^2 \dot{t}^2 + \left(1 - \frac{r_s}{r}\right)^{-1} \dot{r}^2 + r^2 \dot{\phi}^2 = \\
 &= -()^{-1} E^2 + ()^{-1} \dot{r}^2 + \frac{L^2}{r^2}
 \end{aligned}$$

and thus

$$\dot{r}^2 = E^2 - c^2 \left(1 - \frac{r_s}{r}\right) \left(1 + \frac{L^2}{r^2 c^2}\right).$$

Now, changing variable $u := \frac{1}{r}$ and using the relation $\phi(\tau)$ to write

$$\dot{r} = \frac{dr}{d\tau} = \frac{d\phi}{d\tau} \frac{d}{d\phi} = \frac{L}{r^2} \frac{dr}{d\phi} = L u^2 \left(-\frac{1}{u^2}\right) u'(\phi) = -L u'(\phi),$$

the equation becomes

$$L^2 u'^2 = E^2 - c^2 (1 - r_s u) \left(1 + \frac{L^2}{c^2} u^2\right).$$

Differentiating both sides by ϕ

$$0 = L^2 2u' u'' - c^2 r_s u' \left(1 + \frac{L^2}{c^2} u^2\right) + c^2 (1 - r_s u) 2 \frac{L^2}{c^2} u u'$$

or dividing by $2L^2u'$,

$$0 = u'' - \frac{c^2 r_s}{2L^2} - \frac{3}{2} r_s u^2 + u$$

or, with $r_s = \frac{2GM}{c^2}$

$$u'' + u = \frac{c^2 r_s}{2L^2} + \frac{3}{2} r_s u^2 = \frac{GM}{L^2} + \frac{3GM}{c^2} u^2.$$

Solution via perturbation method. As $\frac{3GM}{c^2} \rightarrow 0$, the equation becomes the equation of the Kepler orbits.

Let the solution be $u(\phi) = u_0(\phi) + u_1(\phi)$, with $u_0(\phi)$ the solution of the equations of classical mechanics.

$$u_0(\phi) = \frac{GM}{L^2} (1 + e \cos \phi).$$

As

$$0 = \underbrace{u_0'' + u_0 - \frac{GM}{L^2}}_{=0} + u_1'' + u_1 - \frac{3GM}{c^2} (u_0 + u_1)^2,$$

the perturbation $u_1(\phi)$ (of order $\frac{GM}{c^2}$ must satisfy (approximately, truncation) the equation

$$\begin{aligned} u_1'' + u_1 &\approx \frac{3GM}{c^2} u_0^2(\phi) = \\ &= \frac{3(GM)^3}{c^2 L^4} (1 + 2e \cos \phi + e^2 \cos^2 \phi) = (\cos 2x = 2 \cos^2 x - 1) = \\ &= 3 \left(\frac{(GM)^{3/2}}{cL^2} \right)^2 \left(1 + 2e \cos \phi + \frac{e^2}{2} (1 + \cos(2\phi)) \right) = \\ &= 3 \left(\frac{(GM)^{3/2}}{cL^2} \right)^2 \left(1 + \frac{e}{2} + 2e \cos \phi + \frac{e^2}{2} \cos(2\phi) \right), \end{aligned}$$

whose solution undergoes resonance, as the forcing $2e \cos \phi$ has the same frequency as the natural frequency of the system. The general solution reads

$$u_1(\phi) = A \cos \phi + B \sin \phi + 3 \left(\frac{(GM)^{3/2}}{cL^2} \right)^2 \left[\left(1 + \frac{e}{2} \right) + e \phi \sin \phi + \frac{e^2}{6} \cos 2\phi \right]$$

Only the term $\propto \phi \sin \phi$ is unbounded. Combining this term with the unperturbed solution gives

$$u(\phi) \approx \frac{GM}{L^2} \left\{ (1 + e \cos \phi) + 3 \left(\frac{GM}{cL} \right)^2 e \phi \sin \phi \right\},$$

and using small angle approximation $\cos(\alpha + x) = \cos \alpha \cos x - \sin \alpha \sin x \approx \cos x - \alpha \sin x$, for $\alpha \sim 0$,

$$\begin{aligned} u(\phi) &\approx \frac{GM}{L^2} \left\{ 1 + e \cos \phi + 3 \underbrace{\left(\frac{GM}{cL} \right)^2}_{\sim -\alpha} \phi e \sin \phi \right\} = \\ &\approx \frac{GM}{L^2} \left\{ 1 + e \cos \left(\phi - \left(\frac{GM}{cL} \right)^2 \phi \right) \right\} = \\ &= \frac{GM}{L^2} \left\{ 1 + e \cos \left[\left(\phi - 3 \left(\frac{GM}{cL} \right)^2 \phi \right) \right] \right\}. \end{aligned}$$

As $u := \frac{1}{r}$, the perihelion occurs for $u(\phi_{perih}) = \max_{\phi} u(\phi)$, i.e. for

$$\left(1 - 3 \left(\frac{GM}{cL}\right)^2\right) \phi_{perih,n} = n2\pi \quad , \quad n \in \mathbb{Z} .$$

The **precession of the perihelion** can be evaluated as the difference between the angular coordinate of two successive revolutions (adding a 2π to the first angle, mimicing an orbit with no precession) i.e.

$$\begin{aligned} \Delta\phi &= \phi_{perih,n+1} - (\phi_{perih,n} + 2\pi) = \\ &= \frac{2\pi}{1 - 3 \left(\frac{GM}{cL}\right)^2} - 2\pi = \\ &\approx 2\pi \left(1 + 3 \left(\frac{GM}{cL}\right)^2 - 1\right) = 6\pi \left(\frac{GM}{cL}\right)^2 . \end{aligned}$$

General solution

The general solution has the expression

$$u_1(\phi) = \underbrace{A \cos \phi + B \sin \phi}_{u_{1,homo}(\phi)} + \underbrace{C + D\phi \cos \phi + E\phi \sin \phi + F \cos(2\phi)}_{u_{1,part}(\phi)} .$$

The coefficients of the particular solution are evaluated by coefficient matching

$$\begin{aligned} (\phi \cos \phi)' &= \cos \phi - \phi \sin \phi \\ (\phi \cos \phi)'' &= -2 \sin \phi - \phi \cos \phi \\ (\phi \sin \phi)' &= \sin \phi + \phi \cos \phi \\ (\phi \sin \phi)'' &= 2 \cos \phi - \phi \sin \phi \\ (\cos(2\phi))'' &= -4 \cos(2\phi) \end{aligned}$$

then

$$\begin{aligned} \cos \phi : \quad 0 &= A - A - 2E + 2\alpha e \\ \sin \phi : \quad 0 &= B - B + 2D \\ \phi \cos \phi : \quad 0 &= D - D \\ \phi \sin \phi : \quad 0 &= E - E \\ 1 : \quad 0 &= -C + \alpha \left(1 + \frac{e}{2}\right) \\ \cos 2\phi : \quad 0 &= 4F - F + \alpha \frac{e^2}{2} , \end{aligned}$$

so that

$$\begin{aligned} E &= \alpha e \\ D &= 0 \\ C &= \alpha \left(1 + \frac{e}{2}\right) \\ F &= -\frac{1}{6} \alpha e^2 . \end{aligned}$$

Part III

Statistical Mechanics

STATISTICAL PHYSICS

LIOUVILLE THEOREM AND BOLTZMANN EQUATION

8.1 Liouville theorem

Theorem 8.1.1 (Liouville's theorem)

Let $f(\mathbf{r}_i, \mathbf{v}_i, t)$ the **multi-particle** probability density function. Liouville's equations reads

$$0 = \frac{\partial f}{\partial t} + \mathbf{v}_i \cdot \nabla_{\mathbf{r},i} f + \frac{\mathbf{F}_i}{m} \cdot \nabla_{\mathbf{v},i} f .$$

Let $f(\mathbf{r}_i, \mathbf{v}_i, t)$ be the N -particle distribution of a system as a function of the $2dN$ coordinates and components of the velocity. Probability is normalized to 1

$$1 = \int_{\Omega} f(\mathbf{r}_i, \mathbf{v}_i, t) d\mathbf{r}_i d\mathbf{v}_i .$$

The governing equations of each particle read

$$\begin{cases} \dot{\mathbf{r}}_i = \mathbf{v}_i \\ \dot{\mathbf{v}}_i = \frac{\mathbf{F}_i}{m} . \end{cases}$$

A governing equation for the probability P_{Ω} of the system to be in a state belonging to an arbitrary “material” volume Ω_t in the state space reads

$$\begin{aligned} 0 &= \frac{d}{dt} P_{\Omega} = \frac{d}{dt} \int_{\Omega_t} f = \\ &= \int_{\Omega_t} \frac{\partial f}{\partial t} + \oint_{\partial\Omega_t} (\hat{\mathbf{n}}_{\mathbf{r},i} \cdot \mathbf{v}_i + \hat{\mathbf{n}}_{\mathbf{v},i} \cdot \mathbf{a}_i) f = \\ &= \int_{\Omega_t} \left\{ \frac{\partial f}{\partial t} + \nabla_{\mathbf{r},i} \cdot (\mathbf{v}_i f) + \nabla_{\mathbf{v},i} \cdot (\mathbf{a}_i f) \right\} , \end{aligned}$$

having used [Reynolds' transport theorem](#) in the state space, applied the multi-dimensional version of the divergence theorem. As this balance holds for any arbitrary volume Ω_t , the differential form follows

$$0 = \frac{\partial f}{\partial t} + \nabla_{\mathbf{r},i} \cdot (\mathbf{v}_i f) + \nabla_{\mathbf{v},i} \cdot (\mathbf{a}_i f) .$$

Now, using [Hamilton equations](#) in absence of non-conservative forces, in terms of generalized coordinates and generalized momenta

$$\begin{cases} \dot{\mathbf{q}}^i = \frac{\partial \mathcal{H}}{\partial \mathbf{p}_i} \\ \dot{\mathbf{p}}_i = -\frac{\partial \mathcal{H}}{\partial \mathbf{q}^i} , \end{cases}$$

or in terms of positions and velocities

$$\begin{cases} \mathbf{v}_i = \dot{\mathbf{r}}_i = \frac{1}{m} \nabla_{\mathbf{v}_i} \mathcal{H} \\ \mathbf{a}_i = \dot{\mathbf{v}}_i = -\frac{1}{m} \nabla_{\mathbf{r}_i} \mathcal{H} \end{cases},$$

it's possible to find the “convective form” of the Liouville's theorem

$$\begin{aligned} 0 &= \frac{\partial f}{\partial t} + \nabla_{\mathbf{r},i} \cdot (\mathbf{v}_i f) + \nabla_{\mathbf{v},i} \cdot (\mathbf{a}_i f) = \\ &= \frac{\partial f}{\partial t} + f \underbrace{(\nabla_{\mathbf{r},i} \cdot \mathbf{v}_i + \nabla_{\mathbf{v},i} \cdot \mathbf{a}_i)}_{=0} + \mathbf{v}_i \cdot \nabla_{\mathbf{r},i} f + \mathbf{a}_i \cdot \nabla_{\mathbf{v},i} f, \end{aligned}$$

i.e.

$$\begin{aligned} 0 &= \frac{\partial f}{\partial t} + \mathbf{v}_i \cdot \nabla_{\mathbf{r},i} f + \mathbf{a}_i \cdot \nabla_{\mathbf{v},i} f \\ 0 &= \frac{\partial f}{\partial t} + \mathbf{v}_i \cdot \nabla_{\mathbf{r},i} f + \frac{\mathbf{F}_i}{m} \cdot \nabla_{\mathbf{v},i} f \\ 0 &= \frac{\partial f}{\partial t} + \mathbf{v}_i \cdot \nabla_{\mathbf{r},i} f + \frac{\mathbf{F}_i^{ext}}{m} \cdot \nabla_{\mathbf{v},i} f + \sum_{j \neq i} \frac{\mathbf{F}_{ij}}{m} \cdot \nabla_{\mathbf{v},i} f. \end{aligned}$$

8.2 Boltzmann equation

Indistinguishable particles. State counting...

1-particle probability distribution function. Integrating all but one particle (marginalizing over all the variables expect for $\mathbf{r}_1, \mathbf{v}_1$),

$$f_1(\mathbf{r}_1, \mathbf{v}_1, t) = \int_{\mathbf{r}_{k=2:N}} \int_{\mathbf{v}_{k=2:N}} f(\mathbf{r}_i, \mathbf{v}_i, t) d\mathbf{r}_k d\mathbf{v}_k,$$

s-particle probability distribution function.

$$f_s(\mathbf{r}_{j=1:s}, \mathbf{v}_{j=1:s}, t) = \int_{\mathbf{r}_{k=s+1:N}} \int_{\mathbf{v}_{k=s+1:N}} f(\mathbf{r}_i, \mathbf{v}_i, t) d\mathbf{r}_k d\mathbf{v}_k,$$

Time derivative.

$$\begin{aligned} \partial_t f_s &= \partial_t \int_{\mathbf{r}_{k=s+1:N}} \int_{\mathbf{v}_{k=s+1:N}} f(\mathbf{r}_i, \mathbf{v}_i, t) d\mathbf{r}_k d\mathbf{v}_k = \\ &= - \int_{\mathbf{r}_{k=s+1:N}} \int_{\mathbf{v}_{k=s+1:N}} \left\{ \mathbf{v}_i \cdot \nabla_{\mathbf{r}_i} f + \frac{\mathbf{F}_i^{ext}}{m} \cdot \nabla_{\mathbf{v}_i} f + \sum_{j \neq i} \frac{\mathbf{F}_{ij}}{m} \cdot \nabla_{\mathbf{v}_i} f \right\} d\mathbf{r}_k d\mathbf{v}_k = \end{aligned}$$

Using Hamilton's equation to get a divergence, and assuming zero probability at infinity (impossible to be at infinite distance, impossible to have infinite velocity. Sounds likely),

Details

$$\begin{aligned}
 \int_{\mathbf{r}_k} \left\{ \mathbf{v}_i \cdot \nabla_{\mathbf{r}_i} f + \frac{\mathbf{F}_i}{m} \cdot \nabla_{\mathbf{v}_i} f \right\} &= \int_{\mathbf{r}_k} \left\{ \mathbf{v}_i \cdot \nabla_{\mathbf{r}_i} f + \frac{\mathbf{F}_i}{m} \cdot \nabla_{\mathbf{v}_i} f + f \left(\nabla_{\mathbf{r}_i} \cdot \mathbf{v}_i + \nabla_{\mathbf{v}_i} \cdot \frac{\mathbf{F}_i}{m} \right) \right\} = \\
 &= \int_{\mathbf{r}_k} \left\{ \nabla_{\mathbf{r}_i} \cdot (f \mathbf{v}_i) + \nabla_{\mathbf{v}_i} \cdot \left(f \frac{\mathbf{F}_i}{m} \right) \right\} = \\
 &= \sum_{i \neq k} \int_{\mathbf{r}_k} \left\{ \nabla_{\mathbf{r}_i} \cdot (f \mathbf{v}_i) + \nabla_{\mathbf{v}_i} \cdot \left(f \frac{\mathbf{F}_i}{m} \right) \right\} = \\
 &= \sum_{i \neq k} \int_{\mathbf{r}_k} \left\{ \mathbf{v}_i \cdot \nabla_{\mathbf{r}_i} f + \frac{\mathbf{F}_i}{m} \cdot \nabla_{\mathbf{v}_i} f \right\} = \\
 &= \sum_{i \neq k} \int_{\mathbf{r}_k} \left\{ \mathbf{v}_i \cdot \nabla_{\mathbf{r}_i} f + \left(\frac{\mathbf{F}_i^{ext}}{m} + \sum_{j \in K} \frac{\mathbf{F}_{ij}}{m} + \sum_{j \notin K} \frac{\mathbf{F}_{ij}}{m} \right) \cdot \nabla_{\mathbf{v}_i} f \right\} = \\
 &= \sum_{i \neq k} \left\{ \mathbf{v}_i \cdot \nabla_{\mathbf{r}_i} \int_{\mathbf{r}_k} f + \frac{\mathbf{F}_i^{ext}}{m} \cdot \nabla_{\mathbf{v}_i} \int_{\mathbf{r}_k} f + \sum_{j \in K} \frac{\mathbf{F}_{ij}}{m} \cdot \nabla_{\mathbf{v}_i} \int_{\mathbf{r}_k} f \right\} + \sum_{i \neq k} \sum_{j \in K} \int_{\mathbf{r}_k} \frac{\mathbf{F}_{ij}}{m} \cdot \nabla_{\mathbf{v}_i} f = \\
 &= \sum_{i \neq k} \left\{ \mathbf{v}_i \cdot \nabla_{\mathbf{r}_i} f_s + \frac{\mathbf{F}_i^{ext}}{m} \cdot \nabla_{\mathbf{v}_i} f_s + \sum_{j \in K} \frac{\mathbf{F}_{ij}}{m} \cdot \nabla_{\mathbf{v}_i} f_s \right\} + \sum_{i \neq k} \sum_{j \in K} \int_{\mathbf{r}_k} \frac{\mathbf{F}_{ij}}{m} \cdot \nabla_{\mathbf{v}_i} f = \\
 0 &= \partial_t f_s + \sum_{i \notin K} \left\{ \mathbf{v}_i \cdot \nabla_{\mathbf{r}_i} f_s + \frac{\mathbf{F}_i^{ext}}{m} \cdot \nabla_{\mathbf{v}_i} f_s + \sum_{j \in K} \frac{\mathbf{F}_{ij}}{m} \cdot \nabla_{\mathbf{v}_i} f_s \right\} + \sum_{i \notin K} \sum_{j \in K} \int_{\mathbf{x}_k} \frac{\mathbf{F}_{ij}}{m} \cdot \nabla_{\mathbf{v}_i} f.
 \end{aligned}$$

with $K = s + 1 : N$.

1-particle equation. With $K = 2 : N$,

$$0 = \partial_t f_1 + \mathbf{v}_1 \cdot \nabla_{\mathbf{r}_1} f_1 + \frac{\mathbf{F}_1^{ext}}{m} \cdot \nabla_{\mathbf{v}_1} f_1 + \sum_{j=2:N} \int_{\mathbf{x}_k, k=2:N} \frac{\mathbf{F}_{1j}}{m} \cdot \nabla_{\mathbf{v}_1} f d\mathbf{r}_k d\mathbf{v}_k.$$

Details

Force $\mathbf{F}_{1j}$ only depends on variables of particle 1 and j . Therefore,

$$\begin{aligned}
 \sum_{j=2:N} \int_{\mathbf{r}_k, k=2:N} \frac{\mathbf{F}_{1j}}{m} \cdot \nabla_{\mathbf{v}_1} f d\mathbf{r}_k d\mathbf{v}_k &= \sum_{j=2:N} \int_{\mathbf{x}_j} \frac{\mathbf{F}_{1j}}{m} \cdot \int_{\mathbf{r}_k, k=2:N, k \neq j} \nabla_{\mathbf{v}_1} f d\mathbf{r}_k d\mathbf{v}_k = \\
 &= \sum_{j=2:N} \int_{\mathbf{x}_j} \frac{\mathbf{F}_{1j}}{m} \cdot \nabla_{\mathbf{v}_1} \int_{\mathbf{x}_k, k=2:N, k \neq j} f d\mathbf{r}_k d\mathbf{v}_k = \\
 &= \sum_{j=2:N} \int_{\mathbf{x}_j} \frac{\mathbf{F}_{1j}}{m} \cdot \nabla_{\mathbf{v}_1} f_2^{K/\{1,j\}}(\mathbf{r}_1, \mathbf{v}_1, \mathbf{r}_j, \mathbf{v}_j, t) d\mathbf{r}_j d\mathbf{v}_j.
 \end{aligned}$$

For identical particles, the summation becomes $(N-1) \int_{\mathbf{x}_2} \frac{\mathbf{F}_{12}}{m} \cdot \nabla_{\mathbf{v}_1} f_2(\mathbf{x}_1, \mathbf{x}_2, t)$

Collision integral. Assumptions

Details

When a collision occurs, the dynamics of a particle is not smooth anymore. Thus, is it possible to use partial derivative in time? Force is impulsive. Thus, either function in the sense of distribution and/or integration in time is required

Trash

- only 2-particle collision. Good assumption for “dilute systems”.
- a collision occurs when 2 particles are in the same position (i.e. the dimension of the particles is negligible), $\mathbf{r}_2 = \mathbf{r}_1$

$$\propto \int \Gamma_{\mathbf{p}'_1, \mathbf{p}'_2 \rightarrow \mathbf{p}_1, \mathbf{p}_2} f(\mathbf{r}_1, \mathbf{r}_2, \mathbf{p}'_1, \mathbf{p}'_2) - \Gamma_{\mathbf{p}_1, \mathbf{p}_2 \rightarrow \mathbf{p}'_1, \mathbf{p}'_2} f(\mathbf{r}_1, \mathbf{r}_2, \mathbf{p}_1, \mathbf{p}_2)$$

with

- $\Gamma_{\mathbf{p}'_1, \mathbf{p}'_2 \rightarrow \mathbf{p}_1, \mathbf{p}_2}$ representing the probability of starting from $\mathbf{p}'_i$ before the collision and ending with momenta $\mathbf{p}_i$
 - this probability is determined by the analysis of 2-particle collision dynamics
 - this probability is “scaled” by the probability of being in the state before the collision $f(\mathbf{r}_1, \mathbf{r}_2 = \mathbf{r}_1, \mathbf{p}'_1, \mathbf{p}'_2)$

Molecular chaos (Stosszahlansatz). Before and after the collision

$$f_2(\mathbf{v}_1, \mathbf{v}_2, t) \sim f_1(\mathbf{v}_1) f(\mathbf{v}_2, t)$$

8.3 H-theorem

Starting from Boltzmann equation for the 1-particle probability density function, with the molecular chaos assumption,

$$D_t f_1 = \int d\mathbf{v}_2 \int d\Omega \sigma(\Omega) |\mathbf{v}_1 - \mathbf{v}_2| (f'_1 f'_2 - f_1 f_2)$$

Functional $H(t)$ is defined as

$$H(t) = \int f(\mathbf{v}, t) \ln f(\mathbf{v}, t) d\mathbf{v}$$

Taking time derivative (integration domain is unbounded. Use divergence theorem and boundary conditions to prove that flux contributions are zero)

$$d_t H(t) = \int \partial_t (f \ln f) d\mathbf{v} = \int \partial_t f (1 + \ln f) d\mathbf{v} .$$

Details

todo include convective terms of Boltzmann equation

todo prove each step

$$\begin{aligned}
 d_t H(t) &= \int \partial_t f (1 + \ln f) d\mathbf{v} = \\
 &= \int \int_{\mathbf{v}_1} |\mathbf{v}_1 - \mathbf{v}_2| (f'_1 f'_2 - f_1 f_2) (1 + \ln f_1) \sigma_\Omega d\mathbf{v}_1 d\mathbf{v}_2 d\Omega = \\
 &= \frac{1}{2} \int \int_{\mathbf{v}_1} |\mathbf{v}_1 - \mathbf{v}_2| (f'_1 f'_2 - f_1 f_2) (\ln f_1 + \ln f_2) \sigma_\Omega d\mathbf{v}_1 d\mathbf{v}_2 d\Omega = \\
 &= \frac{1}{4} \int \int_{\mathbf{v}_1} |\mathbf{v}_1 - \mathbf{v}_2| (f'_1 f'_2 - f_1 f_2) (\ln f_1 + \ln f_2 - \ln f'_1 - \ln f'_2) \sigma_\Omega d\mathbf{v}_1 d\mathbf{v}_2 d\Omega = \\
 &= \frac{1}{4} \int \int_{\mathbf{v}_1} |\mathbf{v}_1 - \mathbf{v}_2| (f'_1 f'_2 - f_1 f_2) \ln \frac{f_1 f_2}{f'_1 f'_2} \sigma_\Omega d\mathbf{v}_1 d\mathbf{v}_2 d\Omega = \\
 &\leq 0,
 \end{aligned}$$

as

$$(a - b) \ln \frac{b}{a} \leq 0,$$

since, if $a > b$ the first factor is positive and the second negative, and viceversa, and $a = b$ implies equality.

STATISTICAL PHYSICS - NOTES

9.1 Ensembles

9.2 Microcanonical ensemble

9.3 Canonical ensemble

9.4 Macrocanonical ensemble

9.5 Statistics

Each of the N components of the system is in an **energy level** i . Energy level i has g_i sublevels with the same energy level.

- energy levels, E_i of each component
- occupation number N_i of level i
- **Central role of energy.** In a system macroscopically at rest, the energy of a system is the only macroscopic meaningful non-zero mechanical quantity, constant for closed and isolated systems
- **Principle of maximum uncertainty, maximum entropy, minimum information:** given a measurement of a macroscopic variable V , describing the macrostate of the system, the feasible un-observed/able microstates of the system are the microstates consistent with it: there's usually a sharp maximum of in the probability density of the microstates.

Given a macrostate, what's the number of ways $W(N_i; g_i)$ to get a consistent microstate? Once the expression is found, constrained optimization follows: optimization w.r.t. N_i is usually performed in the limit of $N_i \rightarrow +\infty$ (why in Fermi-Dirac distribution, obeying Pauli exclusion principle?), with the values of the macroscopic variables as constraints usually treated with Lagrange multiplier.

9.5.1 Boltzmann

Statistics of distinguishable components. Let N_i the number of elementary components in the energy level i with energy E_i . Let the number of elementary components $N = \sum_i N_i$ and the total energy of the system $E = \sum_i N_i E_i$ be given.

9.5.2 Bose-Einstein

Statistics of indistinguishable components that can be in the same (sub)level. Given the number of elementary components $\sum_i N_i = N$ and the energy $\sum_i N_i E_i = E$,

$$W_{BE,i} = \frac{(N_i + g_i - 1)!}{N_i! (g_i - 1)!} \quad , \quad W_{BE} = \prod_i W_{BE,i} . \quad (9.1)$$

Counting microstates

todo write page *Combinatorics* and add link

Most likely microstate. Instead of maximizing (9.1), the objective function is $\ln W_{BE}$, after using Stirling approximation in the limit of large N_i and g_i , $N_i! \sim \left(\frac{N_i}{e}\right)^{N_i}$. The approximate occupation number of one of the G_i sublevels of the i^{th} level of the most likely microstate is

$$n_i := \frac{N_i}{g_i} = \frac{1}{e^{\alpha + \beta E_i} - 1} .$$

Optimization

$$\begin{aligned} J(N_i, \alpha, \beta) &= \ln W_{BE} + \alpha \left(N - \sum_i N_i \right) + \beta \left(E - \sum_i N_i E_i \right) = \\ &= \sum_i \{ \ln(N_i + g_i - 1)! - \ln N_i! - \ln(g_i - 1)! \} + \alpha \left(N - \sum_i N_i \right) + \beta \left(E - \sum_i N_i E_i \right) \simeq \\ &\simeq \sum_i \{ (N_i + g_i - 1) \ln(N_i + g_i - 1) - N_i \ln N_i - (g_i - 1) \ln(g_i - 1) + N_i + g_i - 1 - N_i - (g_i - 1) \} + \alpha \left(N - \sum_i N_i \right) \\ &= \sum_i \{ (N_i + g_i - 1) \ln(N_i + g_i - 1) - N_i \ln N_i - (g_i - 1) \ln(g_i - 1) \} + \alpha \left(N - \sum_i N_i \right) + \beta \left(E - \sum_i N_i E_i \right) \end{aligned}$$

Using $\partial_n(n + a) \ln(n + a) = \ln(n + a) + 1$,

$$0 = \partial_{N_k} J \simeq \{ \ln(N_k + g_k - 1) - \ln N_k \} - \alpha - \beta E_k ,$$

and thus

$$\begin{aligned} \ln \frac{N_k + g_k - 1}{N_k} &= \alpha + \beta E_k , \\ \frac{N_k + g_k - 1}{N_k} &= e^{\alpha + \beta E_k} \\ N_k &= \frac{g_k - 1}{e^{\alpha + \beta E_k} - 1} \simeq \frac{g_k}{e^{\alpha + \beta E_k} - 1} , \end{aligned}$$

Thus, in the limit of $g_k \gg 1$, the occupation number of the k level is

$$N_k = \frac{G_k}{e^{\alpha + \beta E_k} - 1} ,$$

and the average occupation number of one of the g_k sublevels in the k level is

$$n_k := \frac{N_k}{G_k} = \frac{1}{e^{\alpha + \beta E_k} - 1}$$

Meaning of α, β

Example 9.5.1 (Black-body radiation: Planck, Wien, and Stefan-Boltzmann laws)

Planck's law. Energy density w.r.t. frequency

$$u_f(f, T) = \frac{8\pi h f^3}{c^3} \frac{1}{e^{\frac{hf}{k_B T}} - 1}$$

Planck's law in a cubic box

Planck's law uses:

- relation between pulsation and wave vector, or frequency and wave number and the speed of light c for light waves

$$c = \frac{\omega}{|\vec{k}|} = \lambda f$$

$$f = \frac{\omega}{2\pi} = \frac{c|\vec{k}|}{2\pi}$$

- Planck assumption that the minimum non-zero energy of a mode with frequency f is $E = hf$, and all the possible values of the energy of the mode is

$$E_m = mhf \quad , \quad m \in \mathbb{N} .$$

Taking a cubic box with sides $L_x = L_y = L_z = L$, the possible modes have (**todo** why? Which boundary condition? Periodic? Some physical? Just fictitious discretization?) in each direction wave-lengths $\lambda_n = \frac{L}{|\vec{n}|} = \frac{2\pi}{|\vec{k}|}$,

$$\vec{k} = \frac{2\pi}{L} \vec{n} .$$

Mode density in $\vec{n}$ -domain is 2 mode per each volume of unit length (2 polarization), and thus the number of modes dN in an elementary volume is

$$dN = 2 d^3 \vec{n} ,$$

Changing variables, it's possible to find the mode density w.r.t. wave vector $\vec{k}$,

$$dN = 2 d^3 \vec{n} = 2 \frac{L^3}{(2\pi)^3} d^3 \vec{k} ,$$

or with its absolute value, exploiting the isotropy of the density function - and writing the elementary volume using “spherical coordinates” $d^3\vec{k} = 4\pi |\vec{k}|^2 d|\vec{k}|$,

$$\begin{aligned} dN &= \frac{V}{(2\pi)^3} 8\pi |\vec{k}|^2 d|\vec{k}| = \\ &= \frac{V}{(2\pi)^3} 8\pi \frac{8\pi^3}{c^3} f^2 df = \\ &= V \frac{8\pi}{c^3} f^2 df =: V g(f) df . \end{aligned}$$

Average energy of a mode

Using Boltzmann distribution (**why?**) for the energy distribution in a single mode,

$$P(E_r) = \frac{e^{-\beta E_r}}{Z} ,$$

with $E_r = r h f$, and the partition function

$$Z = \sum_s e^{-\beta E_s} = \sum_s e^{-\beta h f s} = \frac{1}{1 - e^{-\beta h f}} .$$

The average energy of the mode reads

$$\begin{aligned} \langle E \rangle &= \sum_r E_r P(E_r) = \\ &= \sum_r r h f \frac{e^{-\beta h f r}}{Z} = \\ &= h f (1 - e^{-\beta h f}) \sum_r r e^{-\beta h f r} = \\ &= h f (1 - e^{-\beta h f}) \frac{e^{-\beta h f}}{(1 - e^{-\beta h f})^2} = \\ &= \frac{h f}{e^{\beta h f} - 1} . \end{aligned}$$

Putting together the mode number density and the average energy of a mode, the energy density per unit volume, per frequency reads

$$\begin{aligned} u(f, T) &= \langle E \rangle(f) g(f) = \\ &= \frac{h f}{e^{\beta h f} - 1} \frac{8\pi}{c^3} f^2 = \\ &= \frac{8\pi h f^3}{c^3} \frac{1}{e^{\beta h f} - 1} . \end{aligned}$$

Property of the series

$$\sum_{n=0}^{+\infty} n x^n = \frac{x}{(1-x)^2}$$

Proof. If the series is convergent (is this the required condition?)

$$\frac{d}{dx} \sum_{n=0}^{+\infty} x^n = \frac{d}{dx} \frac{1}{1-x} = \frac{1}{(1-x)^2}$$

$$\frac{d}{dx} \sum_{n=0}^{+\infty} x^n = \sum_{n=0}^{+\infty} n x^{n-1}$$

$$x \frac{d}{dx} \sum_{n=0}^{+\infty} x^n = \sum_{n=0}^{+\infty} n x^n = \frac{x}{(1-x)^2}$$

Spectral radiance, B_f , so that an infinitesimal amount of power radiated by a surface ... is $dP = B_f(f, T) \cos \theta dA d\Omega df$

$$B_f(f, T) = \frac{2hf^3}{c^2} \frac{1}{e^{\frac{hf}{k_B T}} - 1}.$$

This expression is obtained¹ assuming homogeneous radiation from a small hole cut into a wall of the box. Only half of the energy radiates through the hole - so factor $\frac{1}{2}$ in front of the energy density - through a solid angle 2π - and thus this process give the same result as a radiation of all the energy density in all the space directions, just providing the same factor $\frac{1}{4\pi}$. The flux of energy “has velocity” c and thus

$$B_f(f, T) = \frac{1}{4\pi} u_f(f, T) c.$$

Wien’s law. Wien’s law tells that the frequency f^* corresponding to the maximum of the spectral radiance of a black-body radiation described by Planck’s law is proportional to its temperature.

From direct evaluation of the derivative of the spectral radiance as a function of f ,

$$\begin{aligned} \partial_f B_f(f, T) &= \frac{2h}{c^2} \left[3f^2 \frac{1}{e^{\frac{hf}{k_B T}} - 1} + f^3 \left(-\frac{\frac{h}{k_B T} e^{\frac{hf}{k_B T}}}{\left(e^{\frac{hf}{k_B T}} - 1 \right)^2} \right) \right] = \\ &= \frac{2hf^2 e^{\frac{hf}{k_B T}}}{c^2 \left(e^{\frac{hf}{k_B T}} - 1 \right)^2} \left[3 \left(1 - e^{-\frac{hf}{k_B T}} \right) - \frac{hf}{k_B T} \right]. \end{aligned}$$

Now, if $\partial_f B_f(f, T) = 0$ the frequency is either $f = 0$, or the solution of the nonlinear algebraic equation

$$0 = 3 \left(1 - e^{-\frac{hf}{k_B T}} \right) - \frac{hf}{k_B T}.$$

Defining $x := \frac{hf}{k_B T}$, this equation becomes

$$0 = 3(1 - e^x) - x,$$

whose solution $x^* \approx 2.82$ can be easily evaluated with an iterative method (or expressed in term of the Lambert’s function W , so loved at Stanford and on Youtube: they’d probaly like to look at tabulated values, or pose). Once the solution x^* of this non-dimensional equation is found, the frequency where maximum energy density occurs reads

$$f^* = \frac{k_B T}{h} x^* \simeq 2.82 \frac{k_B}{h} T.$$

¹ Derivation of Planck’s Law.

Stefan-Boltzmann law.

$$\begin{aligned}
 \frac{P}{A} &= \int B_f(f, T) \cos \phi \, df \, d\Omega = \\
 &= \int_{f=0}^{+\infty} \int_{\phi=0}^{\frac{\pi}{2}} \int_{\theta=0}^{2\pi} B_f(f, T) \cos \phi \sin \phi \, df \, d\phi \, d\theta = \\
 &= \pi \int_{f=0}^{+\infty} B_f(f, T) \, df = \\
 &= \frac{2\pi h}{c^2} \int_{f=0}^{+\infty} \frac{f^3}{e^{\frac{hf}{k_B T}} - 1} \, df = \\
 &= \frac{2\pi h}{c^2} \left(\frac{k_B T}{h} \right)^4 \int_{u=0}^{+\infty} \frac{u^3}{e^u - 1} \, du .
 \end{aligned}$$

The value of the integral is $\frac{\pi^4}{15}$ and thus

$$\frac{P}{A} = \sigma T^4 \quad , \quad \sigma = \frac{2\pi^5 k_B^4}{15c^2 h^3} .$$

Example 9.5.2 (Energy density and radiance)

Radiance. The radiance $L_{e,\Omega}$ of a surface is the flux of energy per unit solid angle, per unit projected area of the source.

Spectral radiance in frequency is the radiance per unit frequency, $L_{e,\Omega,f} = \frac{\partial L_{e,\Omega}}{\partial f}$.

9.5.3 Fermi-Dirac

Statistics of undistinguishable components that can't be in the same (sub)level, obeying to the Pauli exclusion principle. Given the number of elementary components $\sum_i N_i = N$ and the energy $\sum_i N_i E_i = E$,

$$W_{FD,i} = \frac{G_i!}{(G_i - N_i)! N_i!} \quad , \quad W_{FD} = \prod_i W_{FD,i} . \tag{9.2}$$

Counting microstates

todo write page *Combinatorics* and add link

Most likely microstate. The approximate occupation number of the i^{th} level of the most likely microstate is

$$n_i := \frac{N_i}{G_i} = \frac{1}{1 + e^{\alpha + \beta E_i}} .$$

Optimization

$$\begin{aligned}
 J(N_i, \alpha, \beta) &= \ln W_{FD} + \alpha \left(N - \sum_i N_i \right) + \beta \left(E - \sum_i N_i E_i \right) = \\
 &= \sum_i \{ \ln G_i! - \ln(G_i - N_i)! - \ln N_i! \} + \alpha \left(N - \sum_i N_i \right) + \beta \left(E - \sum_i N_i E_i \right) = \\
 &= \sum_i \{ G_i \ln G_i - (G_i - N_i) \ln(G_i - N_i) - N_i \ln N_i \} + \alpha \left(N - \sum_i N_i \right) + \beta \left(E - \sum_i N_i E_i \right) =
 \end{aligned}$$

Using $\partial_n (n + a) \ln(n + a) = \ln(n + a) + 1$,

$$0 = \partial_{N_k} J \simeq \{ \ln(G_k - N_k) - \ln N_k \} - \alpha - \beta E_k ,$$

and thus

$$\begin{aligned}
 \ln \frac{G_k - N_k}{N_k} &= \alpha + \beta E_k , \\
 \frac{G_k}{N_k} - 1 &= e^{\alpha + \beta E_k}
 \end{aligned}$$

The occupation number of the k level is

$$N_k = \frac{G_k}{1 + e^{\alpha + \beta E_k}} .$$

The average occupation of the G_k sublevels of the k level is

$$n_k := \frac{N_k}{G_k} = \frac{1}{1 + e^{\alpha + \beta E_k}} .$$

Meaning of α, β

STATISTICAL PHYSICS - STATISTICS MISCELLANEA

Information content and Entropy

Given a discrete random variable X with probability mass function $p_X(x)$, the self-information (**todo** *what about mutual information of random variables?*) is defined as the opposite of the logarithm of the mass function $p_X(x)$,

$$I_X(x) := -\ln(p_X(x)) .$$

Information content of independent random variables is additive. Since $p_{X,Y}(x,y) = p_X(x)p_Y(y)$,

$$I_{X,Y}(x,y) = -\ln(p_{X,Y}(x,y)) = -\ln(p_X(x)p_Y(y)) = -\ln p_X(x) - \ln p_Y(y) .$$

Shannon entropy. Shannon entropy of a discrete random variable X is defined as the expected value of the information content,

$$H(X) := \mathbb{E}[I_X(X)] = \sum p_X(x) I_X(x) = -\sum p_X \ln p_X(x) .$$

Gibbs entropy. Gibbs entropy was defined by J.W.Gibbs in 1878,

$$S = -k_B \sum_i p_i \ln p_i .$$

Additivity holds for independent random variables.

Boltzmann entropy. Boltzmann entropy holds for uniform distributions over Ω possible states, $p_i = \frac{1}{\Omega}$. Gibbs' entropy of this uniform distribution becomes

$$S = -k_B \Omega \frac{1}{\Omega} \ln \frac{1}{\Omega} = k_B \ln \Omega .$$

Entropy in Quantum Mechanics. todo

Boltzmann distribution

Given a set of discrete states with probability p_i , and the average measure as “macroscopic quantity” $E = \sum_i p_i E_i$, Boltzmann distribution maximizes the entropy (**todo** *Link to min info, max uncertainty*)

$$S = -k_B \sum_i p_i \ln p_i .$$

The distribution follows from the constrained optimization

$$\tilde{S} = S - \alpha \left(\sum_i p_i - 1 \right) - \beta \left(\sum_i p_i E_i - E \right)$$

$$0 = \partial_\alpha \tilde{S} = - \sum_i p_i - 1$$

$$0 = \partial_\beta \tilde{S} = - \sum_i p_i E_i - E$$

$$0 = \partial_{p_k} \tilde{S} = -k_B (\ln p_k + 1) - \alpha - \beta E_k$$

and thus

$$p_k = e^{-1 - \frac{\alpha}{k_B} - \frac{\beta}{k_B} E_k} = e^{-\left(1 + \frac{\alpha}{k_B}\right)} e^{-\frac{\beta}{k_B} E_k} = C e^{-\frac{\beta}{k_B} E_k},$$

and the normalization constant C is determined by normalization condition

$$1 = \sum_k p_k = C \sum_k e^{-\frac{\beta E_k}{k_B}}$$

The inverse $Z = C^{-1}$ is defined as the **partition function**,

$$Z = C^{-1} = \sum_k e^{-\frac{\beta E_k}{k_B}},$$

and the probability distribution becomes

$$p_k = \frac{e^{-\frac{\beta E_k}{k_B}}}{Z} = \frac{e^{-\frac{\beta E_k}{k_B}}}{\sum_i e^{-\frac{\beta E_i}{k_B}}}.$$

Properties.

$$\frac{p_k}{p_i} = e^{-\frac{\beta}{k_B} (E_k - E_i)}.$$

Thermodynamics. Comparison of statistics and classical thermodynamics

First principle of classical thermodynamics (for a monocomponent gas with no electric charge,...) reads

$$T dS = dE + P dV$$

Entropy for Boltzmann distribution reads

$$\begin{aligned} S &= -k_B \sum_i p_i \ln p_i = \\ &= -k_B \sum_i \left[p_i \left(-\frac{\beta E_i}{k_B} - \ln Z \right) \right] = \\ &= \beta \langle E \rangle + k_B \ln Z \end{aligned}$$

From classical thermodynamics, temperature T can be defined as the partial derivative of the entropy of a system w.r.t. its internal energy keeping constant all the other independent variables,

$$\begin{aligned}
 \frac{1}{T} &= \left(\frac{\partial S}{\partial E} \right) \Big|_X = \\
 &= \frac{\partial \beta}{\partial E} E + \beta + k_B \frac{\partial \ln Z}{\partial E} = \\
 &= \frac{\partial \beta}{\partial E} E + \beta + k_B \frac{1}{Z} \frac{\partial Z}{\partial E} = \\
 &= \frac{\partial \beta}{\partial E} E + \beta + k_B \frac{1}{Z} \frac{\partial Z}{\partial \beta} \frac{\partial \beta}{\partial E} = \\
 &= \frac{\partial \beta}{\partial E} E + \beta + k_B \frac{1}{Z} \left(- \sum_i \frac{E_i}{k_B} e^{-\frac{\beta E_i}{k_B}} \right) \frac{\partial \beta}{\partial E} = \\
 &= \frac{\partial \beta}{\partial E} E + \beta - \left(\sum_i E_i p_i \right) \frac{\partial \beta}{\partial E} = \\
 &= \frac{\partial \beta}{\partial E} E + \beta - E \frac{\partial \beta}{\partial E} = \beta .
 \end{aligned}$$

todo

- write the derivative above clearly in terms of composite functions
- microscopical/statistical approach to the first principle of thermodynamics

$$dE = d \left(\sum_i p_i E_i \right) = \sum_i E_i dp_i + \sum_i p_i dE_i$$

ISING MODEL

- Introduction, and some notes about analytical results and numerical methods. Ising model on regular lattices
- **todo** Notebook implementing methods and showing some examples

References.

- <https://rajeshrinet.github.io/blog/2014/ising-model/#:~:text=Flip the spin of a,Repeat 2-4>
- <https://mattbierbaum.github.io/ising.js/>

11.1 Ising model - Notes

11.1.1 Introduction

11.1.2 Time evolution

Metropolis-Hastings

Metropolis-Hastings algorithm is designed to generate a collection of states according to a desired distribution $p(s)$, using a **Markov process** with a unique **stationary distribution** $\pi(s) = p(s)$.

Algorithm

- Initialize state s_0 with energy E_0
- while **not** terminal condition:
 - flip the spin of a randomly chosen site, to get the possible state $s_{\tilde{n}}$
 - calculate the change in energy of the state $\Delta\tilde{E}_n := \tilde{E}_n - E_{n-1}$
 - update state:
 - * if $\Delta\tilde{E}_n < 0$ accept the change, $s_n = s_{\tilde{n}}$
 - * if $\Delta\tilde{E}_n > 0$ accept the change with probability $\exp\left(-\frac{\Delta\tilde{E}_n}{T}\right)$

Starting from the **detailed balance** (sufficient but not necessary condition for the existence of a (unique?) stationary state), $P(s'|s)p(s) = P(s|s')p(s')$,

$$\frac{P(s'|s)}{P(s|s')} = \frac{p(s')}{p(s)},$$

each transition is splitted in **two sub-steps**: proposal and acceptance-rejection steps. Transition is written as $P(s'|s) = g(s'|s)A(s', s)$ being $g(s'|s)$ the proposal conditional probability, and $A(s', s)$ the acceptance probability. Acceptance probability is choosen in a way to satisfy the conditions of time-reversal and ergodicity. Metropolis choice reads

$$A(s', s) = \min \left(1, \frac{p(s')}{p(s)} \frac{g(s|s')}{g(s'|s)} \right) .$$

todo check if Metropolis choice fulfills the necessary conditions.

Example: Boltmann distribution with uniform proposal. Let a system have Boltzmann stationary distribution, $p(s) \propto \exp(-\beta E(s))$, and uniform proposal conditional probability $g(s'|s) = \frac{1}{N}$, with N the number of accessible states from the state s .

Example 11.1.1 (Ising model on a 2D lattice with 1 flip per update)

As an example, for an Ising model on a 2D lattice with dimension $m \times n$ with proposal update with 1 flip $N = mn$.

Example 11.1.2 (Ising model on a 2D lattice with k flips per update)

todo Is a probability built with independent successive flips ok? Target states at each flip have the same probability, s.t. the probability of reaching steps s_k after k flips starting from s_0 is not uniform.

Let $g(s|s') = g(s'|s)$. The acceptance probability of the transition between different states $s' \neq s$ reads

$$A(s', s) = \min \left(1, e^{-\beta(E(s')-E(s))} \right) ,$$

and the corresponding element of transition matrix is

$$P(s'|s) = \frac{1}{N} \min \left(1, e^{-\beta(E(s')-E(s))} \right) .$$

The *diagonal elements* of the probability transition matrix - representing no state transition - are evaluated as the probability that makes $\sum_{s'} P(s'|s) = P(s|s) + \sum_{s' \in T(s)} P(s'|s) = 1$, i.e.

$$P(s|s) = \frac{1}{N} \sum_{\substack{s' \in T(s) \\ E(s') > E(s)}} \left(1 - e^{-\beta(E(s')-E(s))} \right) .$$

$P(s|s)$

$$\begin{aligned} P(s|s) &= 1 - \sum_{s' \in T(s)} P(s'|s) = \\ &= 1 - \sum_{s' \in T(s)} \frac{1}{N} \min \left(1, e^{-\beta(E(s')-E(s))} \right) = \\ &= \frac{1}{N} \sum_{s' \in T(s)} \left[1 - \min \left(1, e^{-\beta(E(s')-E(s))} \right) \right] = \\ &= \frac{1}{N} \sum_{s' \in T(s)} \max \left(0, 1 - e^{-\beta(E(s')-E(s))} \right) = \\ &= \frac{1}{N} \sum_{\substack{s' \in T(s) \\ E(s') > E(s)}} \left(1 - e^{-\beta(E(s')-E(s))} \right) . \end{aligned}$$

It's easy to prove that this probability functions are related by the detailed balance equations,

$$\frac{P(s'|s)}{P(s|s')} = \frac{\min(1, e^{-\beta(E(s')-E(s))})}{\min(1, e^{-\beta(E(s)-E(s'))})} = e^{-\beta(E(s')-E(s))} = \frac{e^{-\beta E(s')}}{e^{-\beta E(s)}} = \frac{p(s')}{p(s)}.$$

11.2 Markov processes

Markov Process. Let S a finite set of state s that can be indexed with integer indices i , P the transition probability between two states at consecutive time-steps, $P_{ss'} = P(s'|s)$, and p_0 the initial probability. A Markov process is uniquely defined by the triple

$$\langle S, P, p_0 \rangle.$$

If the Markov process is **time-homogeneous**, state transition matrix P is independent from the time-step.

11.2.1 State transition matrix

Given the finite set of states with integer indices, their probabilities at time n can be collected in the $|S|$ -dimensional vector $\mathbf{p}_n$. Transition from time $n-1$ to n can be represented as

$$p_n(s') = \sum_s P(s'|s)p_{n-1}(s) = \sum_s P_{ss'}p_{n-1}(s),$$

or using matrix formalism

$$\mathbf{p}_n^T = \mathbf{p}_{n-1}^T \mathbf{P}.$$

As $\mathbf{p}_n$ and $\mathbf{P}$ represents probability, their elements are non-negative. As the i^{th} element of $\mathbf{p}_n$ represents the probability of the system of being in state i at time n , it follows

$$\sum_i \{\mathbf{p}_n\}_i = \sum_i p_n(i) = 1.$$

As the element $P_{ij} = P(i|j)$ represents the conditional probability of reaching j from state i in one step, it follows

$$\sum_j \{\mathbf{P}\}_{ij} = \sum_j P(j|i) = 1.$$

11.2.2 Properties

Eigenvalue $\lambda = 1$, and stationary state π

The sum of the elements of each row of $\mathbf{P}$ is equal to 1, as $\sum_{s'} P_{ss'} = \sum_{s'} P(s'|s) = 1$. Equivalently, state transition matrix $\mathbf{P}$ has an eigenvalue $\lambda = 1$, with the uniform right eigenvector $\mathbf{1}$, i.e. $\mathbf{P}\mathbf{1} = \mathbf{1}$.

The corresponding left eigenvector is a stationary state π ,

$$\pi^T \mathbf{P} = \pi^T.$$

Spectrum of the transition matrix and asymptotic behavior of the system

todo Are all the transition matrices diagonalizable?

Assuming the transition matrix can be diagonalized, the evolution of the system from the initial probability $\mathbf{p}_0$ can be easily represented. The initial probability is written as $\mathbf{p}_0 = c_k \mathbf{w}_k$, being $\mathbf{w}_k$ the left eigenvectors of the transition matrix, and

$$\mathbf{p}_n^T = \mathbf{p}_{n-1}^T \mathbf{P} = \mathbf{p}_0^T \underbrace{\mathbf{P} \dots \mathbf{P}}_{n \text{ times}} = a_k \underbrace{\mathbf{w}_k^T \mathbf{P}}_{=\lambda_k \mathbf{w}_k^T} \dots \mathbf{P} = a_k \lambda_k^n \mathbf{w}_k^T .$$

As $n \rightarrow +\infty$ only the contributions of the eigenvectors with $|\lambda_k| = 1$ survive. If there's only one of these eigenvalues, $\lambda_1 = 1$, the system converges to $\mathbf{1}$. If there are many eigenvalues with $\lambda_i = 1$, there system converges with probability a_i to those stationary states, being a_i the component of the initial state of the system along the i^{th} eigenvector,

$$\mathbf{p}_n \sim a_k \mathbf{w}_k .$$

Components a_i can be evaluated as the dot product between the **dual basis** $\{\mathbf{v}_i\}_i$ w.r.t. the left eigenvectors and the initial state $\mathbf{p}_0$, as

$$\mathbf{p}_0^T \mathbf{v}_i = a_k \underbrace{\mathbf{w}_k^T \mathbf{v}_i}_{=\delta_{ik}, \text{ def. of dual basis}} = a_i .$$

Existence of eigenvalues $|s_j| = 1$, $s_j = e^{i\theta_j}$ makes limit cycles possible as

$$\mathbf{p}_n \sim \mathbf{w}_j e^{in\theta_j} + \text{c.c.} = 2 \left(\text{re}\{\mathbf{w}_j\} \cos \theta_j - \text{im}\{\mathbf{w}_j\} \sin \theta_j \right) .$$

todo Discuss the possibility (?) of negative components. What's a negative probability?

Detailed balance

By definition, a **reversible Markov chain** has a positive stationary distribution π that satisfies the detailed balance equations

$$\pi_i P_{ij} = \pi_j P_{ji} ,$$

i.e. the flow between each pair of states i, j is individually balanced. Reversibility is a sufficient - even not necessary - condition for the existence of a stationary state.

Let $\pi_i \neq 0$ for $\forall i$, it's possible to define a symmetric matrix via a similarity transformation, as

$$\begin{aligned} \sqrt{\pi_i} \sqrt{\pi_i} P_{ij} &= \sqrt{\pi_j} \sqrt{\pi_j} P_{ji} \\ \sqrt{\pi_i} P_{ij} \frac{1}{\sqrt{\pi_j}} &= \sqrt{\pi_j} P_{ji} \frac{1}{\sqrt{\pi_i}} \\ S_{ij} &= S_{ji} , \end{aligned}$$

with $\mathbf{S} = \mathbf{D} \mathbf{P} \mathbf{D}^{-1}$, with $\mathbf{D} = \text{diag}\{\sqrt{\pi_i}\}$. As the matrices $\mathbf{P}$ and $\mathbf{S}$ are related via a similarity transformation, they have the same eigenvalues

todo

- Discuss the meaning of "reversible"

Ergodicity. ...**todo** start from https://en.wikipedia.org/wiki/Markov_chain#Ergodicity

11.3 Ising model - Numerics

Some references:

- <https://farside.ph.utexas.edu/teaching/329/lectures/node110.html>
- O.Rojas, Spectral mechanism and nearly reducible transfer matrices for pseudotransitions in one-dimensional systems, <https://arxiv.org/html/2511.18109v2>
- L.Onsager, 1944, <https://arxiv.org/html/2412.07328v1> *really useful?*

```
"""
Ising model on regular 2D lattice with periodic boundaries

The Hamiltonian, H, is used to define the probability of the (micro)states of the
↪system
following a Boltzmann probability distribution  $p_i \sim e^{-(E_i / T)} = e^{-(\beta * E_i)}$ 
↪i ).
The expression of the Hamiltonian reads:  $H = .5 * \sum_{\langle ij \rangle} J_{ij} s_i s_j + M \sum_i p_i s_i$ 

First with low-dimensional lattice:
Then increasing the dimensions:

Samples states from the distribution  $p_i$  using Metropolis-Hastings method

"""

import numpy as np
```

11.3.1 $m \times n$ lattice

```
#> Useful functions for ising model on 2D lattice
class IsingLattice():
    def __init__(self, m=3, n=3, J=1., M=0., T=1.):
        """ """
        self.m = m
        self.n = n
        self.J = J
        self.M = M
        self.T = T

    def initialize(self,):
        """ """
        set_random_state()
        # Other initializations? Compatible with {-1, +1} spin values

    def set_random_state(self,):
        """ """
        self.state = 2 * np.random.randint(0,2, size=(self.m, self.n)) - 1
        self.E0 = self.compute_total_energy()

    def flip_indices(self,):
        """ """
        return np.random.randint(0, self.m), np.random.randint(0, self.n)
```

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```

def flip_loc(self, i,j):
    """ """
    self.state[i,j] *= -1

def flip(self,):
    """ """
    self.state[self.flip_indices()] *= -1

def compute_local_energy(self, i,j):
    """
    Energy contribution due to [i,j] element of state reads
     $E_{local}[i,j] = -J s[i,j] \sum_{\{k,l\} \in B(i,j)} s[k,l] - M s[i,j]$ 
    Flipping the state,  $s[i,j] \rightarrow -s[i,j]$ , gives a energy difference
     $E_{local\_flip}[i,j] - E_{local}[i,j] = 2 ( J s[i,j] \sum_{\{k,l\}} s[k,l] + M s[i,j] ) =$ 
     $= - 2 E_{local}[i,j]$ 
    """
    neighbor_sum = self.state[(i+1)%self.m, j] + self.state[(i-1)%self.m, j] + \
        self.state[i, (j+1)%self.n] + self.state[i, (j-1)%self.n]
    return - self.state[i,j] * ( self.J * neighbor_sum + self.M )

def compute_total_energy(self,):
    # For efficiency, use vector operations and avoid nested for loops
    # Just shift up and left, to avoid double counting in an efficient way
    up = np.roll(self.state, shift=1, axis=0)
    left = np.roll(self.state, shift=1, axis=1)

    # Energy = -J * sum(spin * neighbors)
    return - self.J * np.sum(self.state * (up + left)) - self.M * np.sum(self.
↪state)

def compute_total_magnetization(self,):
    """ """
    return np.sum(self.state)

def compute_avg_magnetization(self,):
    """ """
    return np.mean(self.state)

def metropolis_hastings_run(self, n_steps=100, output_states=False):
    """ """
    E = self.compute_total_energy()
    Es = [ E ]
    Ms = [ self.compute_total_magnetization() ]
    states = []

    if output_states: states += [ self.state.copy() ]

    i_step = 0
    for i_step in range(n_steps):
        #> 1 step of Metropolis-Hastings algorithm
        dE = self.metropolis_hastings_step() # Updating state with flip
        E += dE # Update system energy

        #> Update output arrays
        Es += [ E ]
        Ms += [ self.compute_total_magnetization() ]

```

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```

        if output_states:
            states += [ self.state.copy() ]    # <-- ! Memory. Add some sampling_
<period,                                     # or dumping method. Don't store_
<the state                                   # at all the timesteps.

        #> Update step index
        i_step += 1

    return Es, Ms, states

    def metropolis_hastings_step(self,):
        """ """
        #> Proposal step
        i_flip, j_flip = self.flip_indices()    # random sampling of the_
<flip indices
        dE = - 2 * self.compute_local_energy(i_flip, j_flip) # evaluate delta energy

        if dE < 0:    # accept
            self.flip_loc(i_flip, j_flip)
        else:
            p_flip = np.exp( -dE / self.T )

            if np.random.random() < p_flip:    # flip
                self.flip_loc(i_flip, j_flip)
            else:    # reject flip proposal
                dE = 0

    return dE

```

```

#> Parameters
m, n = 32, 32    # 16, 16
J, M = 1., 0.    # 1.
Tc = 2.26918    # 2.26918 * J/k = T_critical
T = Tc * 1.

#> Ising Lattice Model
isi = IsingLattice(m=m, n=n, T=T, M=M, J=J)
isi.set_random_state()

print(f"Lattice dimensions: {m},{n}")
print( "Initial conditions:")
print(f"Energy          : {isi.E0}")
print(f"N. of spin up: {np.sum(isi.state[isi.state > 0])}")
print(f"Magnetization: {np.sum(isi.state)}")

#> Run Metropolis-Hastings algorithm
Es, Ms, states = isi.metropolis_hastings_run(n_steps=int(1e6), output_states=True)

```

```

Lattice dimensions: 32,32
Initial conditions:
Energy          : -16.0
N. of spin up: 493
Magnetization: -38

```

```

-----
KeyboardInterrupt                                Traceback (most recent call last)
Cell In[3], line 18
     15 print(f"Magnetization: {np.sum(isi.state)}")
     17 #> Run Metropolis-Hastings algorithm
--> 18 Es, Ms, states = isi.metropolis_hastings_run(n_steps=int(1e6), output_
    ↪states=True)

Cell In[2], line 81, in IsingLattice.metropolis_hastings_run(self, n_steps, output_
    ↪states)
     79 Ms += [ self.compute_total_magnetization() ]
     80 if output_states:
--> 81     states += [ self.state.copy() ]    # <-- ! Memory. Add some sampling_
    ↪period,
     82                                     # or dumping method. Don't store the_
    ↪state
     83                                     # at all the timesteps.
     84 #> Update step index
     85 i_step += 1

KeyboardInterrupt:

```

```

import matplotlib.pyplot as plt

fig, ax = plt.subplots(2,1, figsize=(5,5))
ax[0].plot(Es); ax[0].set_ylabel('E'); ax[0].grid()
ax[1].plot(Ms); ax[1].set_ylabel('M'); ax[1].grid()
ax[1].set_xlabel('step')

nrows_plots = 6
ncols_plots = 5
n_plots = nrows_plots * ncols_plots
i_plot = 0

#> Indices of all the flipping time steps
indices = np.where(np.diff(Es) != 0)[0] + 1
n_flips = len(indices)

d_indices = np.max([n_flips // n_plots, 1])
indices_plot = indices[d_indices * np.arange(n_plots)]
indices_plot[n_plots - 2] = len(Es) - 1

# print(d_indices)
# print(np.arange(n_plots))
# print(indices)
# print(indices_plot)

fig, ax = plt.subplots(nrows_plots, ncols_plots, figsize=(3*ncols_plots, 3*nrows_
    ↪plots))

ax[0,0].imshow(states[0])
ax[0,0].set_title(f"it: {indices_plot[i_plot]}")
i_plot += 1

while i_plot < n_plots and i_plot < len(indices):

```

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```

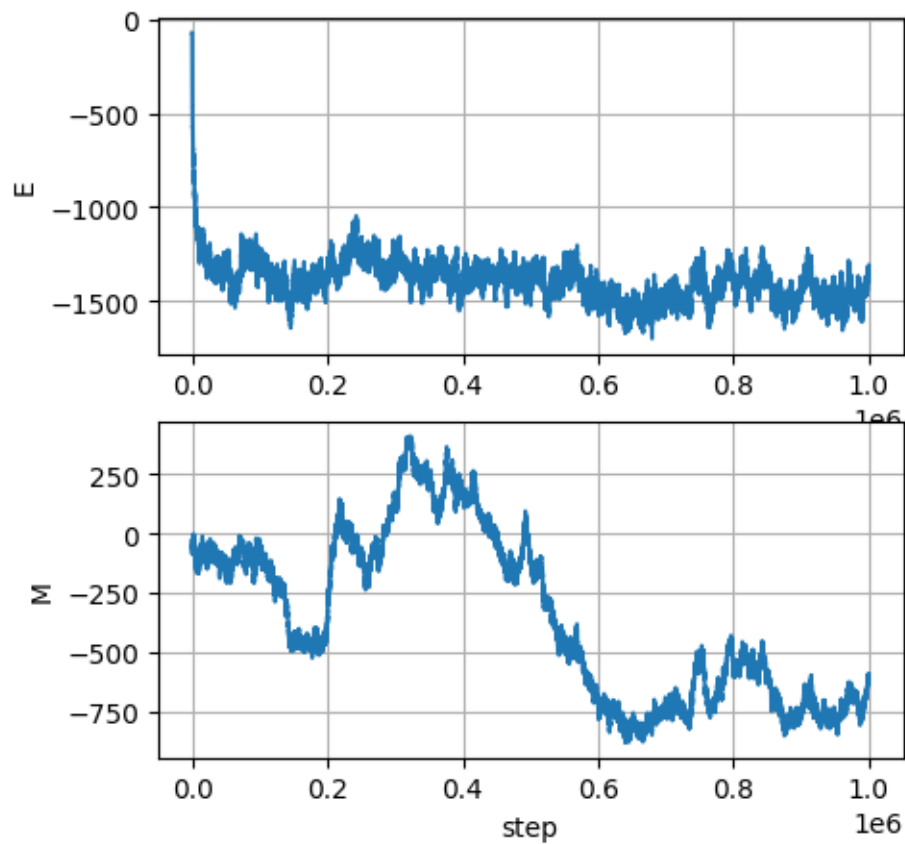
row_plot = i_plot // ncols_plots
col_plot = i_plot - row_plot * ncols_plots

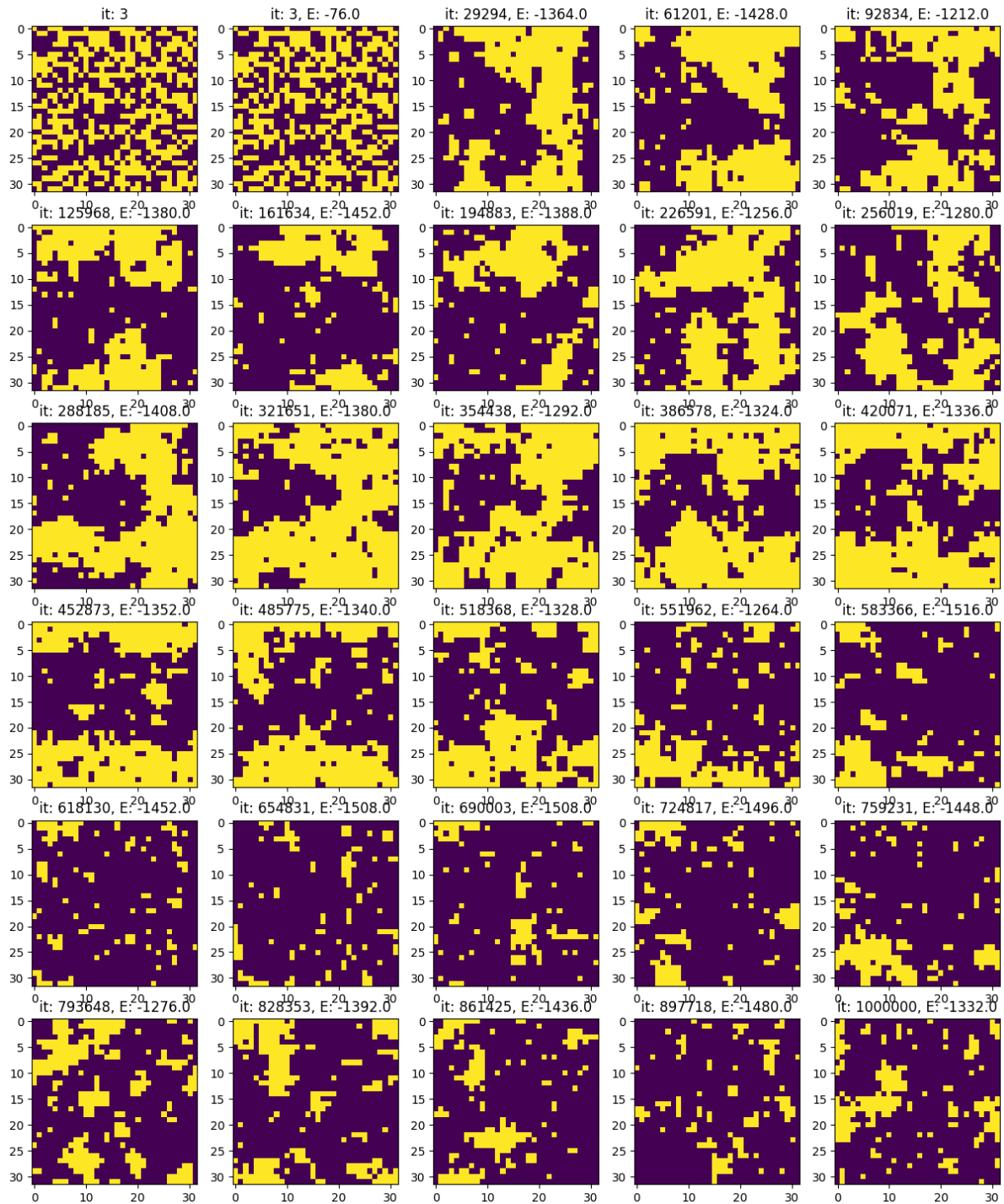
ax[row_plot, col_plot].imshow(states[indices_plot[i_plot-1]], vmin=-1, vmax=1)
ax[row_plot, col_plot].set_title(f"it: {indices_plot[i_plot-1]}, E: {Es[indices_
plot[i_plot-1]]}")

i_plot += 1

plt.show()

```





```
from scipy.special import logsumexp

#> Sample the stationary probability density function
burn_in = 5000                                # From visual inspection
Es_stationary = np.array(Es[burn_in:])         # Discard transient evolution

#> Evaluate P(E) from time history
```

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```

energy_levels, counts = np.unique(Es_stationary, return_counts=True)
pE = counts / np.sum(counts)

ln_g_tilde = np.log(pE) + energy_levels / isi.T

# N = 2 ** (isi.m, isi.n)
ln_N = isi.m * isi.m * np.log(2)

ln_Z = ln_N - logsumexp(ln_g_tilde)
ln_g = ln_Z + ln_g_tilde

#> P(M) probability distribution of magnetization, from time history
Ms_stationary = np.array(Ms[burn_in:]) # Discard transient evolution
mag_levels, mag_counts = np.unique(Ms_stationary, return_counts=True)
pM = mag_counts / np.sum(mag_counts)

```

```

fig, ax = plt.subplots(1,2, figsize=(10,4))
ax[0].semilogy(energy_levels, pE)
ax[1].plot(energy_levels, ln_g)

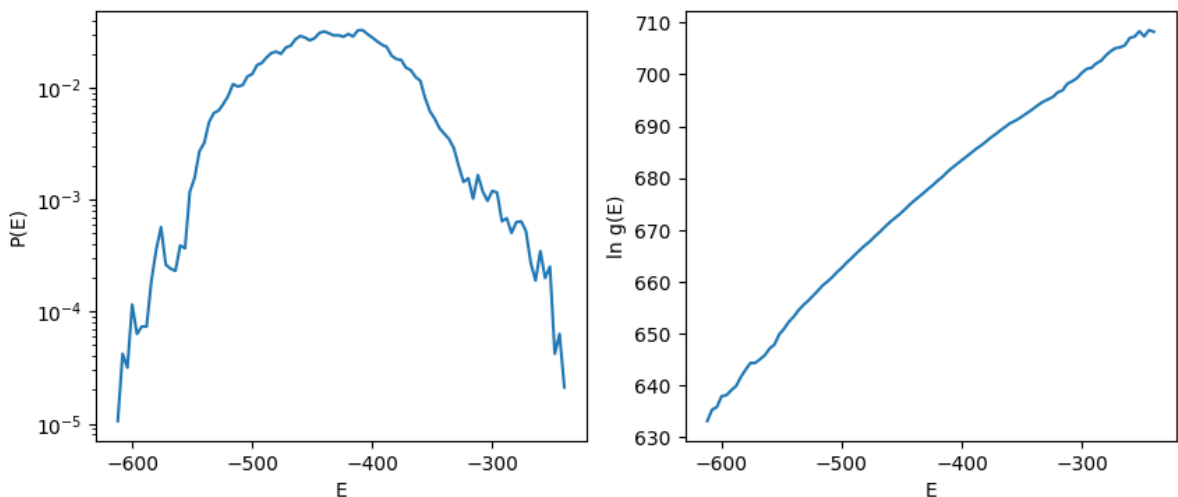
ax[0].set_xlabel('E'); ax[0].set_ylabel('P(E)')
ax[1].set_xlabel('E'); ax[1].set_ylabel('ln g(E)')

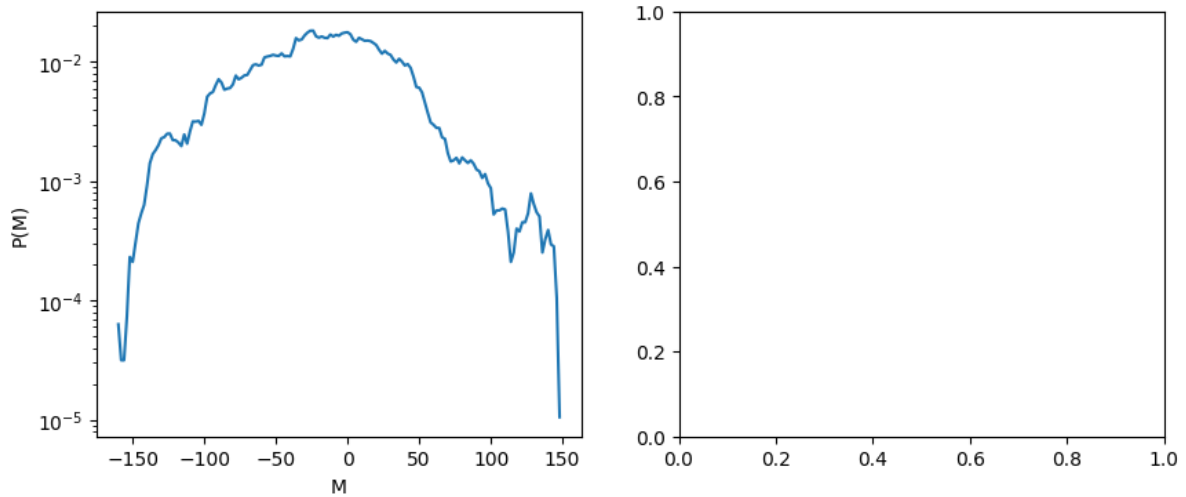
# ---
fig, ax = plt.subplots(1,2, figsize=(10,4))
ax[0].semilogy(mag_levels, pM)
# ax[1].plot(energy_levels, ln_g)

ax[0].set_xlabel('M'); ax[0].set_ylabel('P(M)')
# ax[1].set_xlabel('E'); ax[1].set_ylabel('ln g(E)')

```

```
Text(0, 0.5, 'P(M)')
```





```
"""
Old script using histograms

#> Sample the stationary probability density function
burn_in = 1000 # From visual inspection
Es_stationary = np.array(Es[burn_in:]) # Discard transient evolution

fig, ax = plt.subplots(1,1, figsize=(5,5))
ax.hist(Es_stationary, density=True)
ax.grid()
ax.set_xlabel('E')
# ax.plot(Es_stationary, np.exp(Es_stationary/isi.T), color='red')

# P(E) = g(E) * exp(-beta*E) / Z
# Z is the partition function, that works as the normalization factor
# g(E) is the number of degenerate states with energy = E
# Thus:
# - P(E) can be evaluated from samples
# - exp(-beta*E) is known for every value E
# - g(E) / Z can be evaluated as P(E) / exp(-beta*E) for every value E
#
# Z = sum_E ( g(E) * exp(-beta*E) )
# ...is it possible to find Z and thus g(E)?

"""
```

```
"\nOld script using histograms\n\n#> Sample the stationary probability density_\nfunction\nburn_in = 1000 # From visual inspection\nEs_\nstationary = np.array(Es[burn_in:]) # Discard transient evolution\nfig, ax =_\nplt.subplots(1,1, figsize=(5,5))\nax.hist(Es_stationary, density=True)\nax.\ngrid()\nax.set_xlabel('E')\n# ax.plot(Es_stationary, np.exp(Es_stationary/isi.T),\n# color='red')\n\n# P(E) = g(E) * exp(-beta*E) / Z\n# Z is the partition function,\n# g(E) is the number of degenerate_\nstates with energy = E\n# Thus:\n# - P(E) can be evaluated from samples\n# -_\nexp(-beta*E) is known for every value E\n# - g(E) / Z can be evaluated as P(E) / _\nexp(-beta*E) for every value E\n#\n# Z = sum_E ( g(E) * exp(-beta*E) )\n# ...is_\nit possible to find Z and thus g(E)?\n\n"
```

The relation between state probability $P(E)$ - i.e. probability of getting a microstate with energy E -, degeneracy number

$g(E)$ - i.e. the number of microstates with energy E - and the Boltzmann distribution $\propto e^{-\beta E}$ reads

$$P(E) = \frac{g(E)e^{-\beta E}}{Z} = \frac{g(E)e^{-\beta E}}{\sum_{E'} g(E')e^{-\beta E'}} .$$

Once the value E of energy levels is known, the exponential is known. Probability $P(E)$ is sampled from experiments. Thus

$$\frac{g(E)}{Z} = P(E)e^{\beta E} .$$

If the total number of microstate is not known, $g(E)$ can be evaluated up to a multiplicative constant.

If the total number of microstates is known, $\sum_E g(E) = N$, it's possible to compute all the values of $g(E)$ and Z . In this case, a linear system reads

$$\begin{cases} g(E) = ZP(E)e^{\beta E} \\ \sum_{E'} g(E') = N , \end{cases}$$

and thus

$$Z = \frac{N}{\sum_{E'} P(E')e^{\beta E'}} , \quad g(E) = \frac{NP(E)e^{\beta E}}{\sum_{E'} P(E')e^{\beta E'}} .$$

Log to avoid numerical exceptions.

$$\ln g(E) = \ln Z + \ln P(E) + \beta E$$

In order to evaluate Z ,

$$Z = \sum_E g(E)e^{-\beta E} = \sum_E e^{\ln g(E) - \beta E} ,$$

and avoid overflow, the largest contribution is factored using the relation

$$\ln \sum_i e^{x_i} = \ln \left(e^{x_{max}} \sum_i e^{-(x_{max} - x_i)} \right) = \ln e^{x_{max}} + \ln \sum_i e^{-(x_{max} - x_i)} = x_{max} + \ln \sum_i e^{-(x_{max} - x_i)} .$$

The largest exponential corresponds to energy value E_0 at the maximum of $P(E)$ as

$$\begin{aligned} E_0 &= \operatorname{argmax}_E (\ln g(E) - \beta E) = \\ &= \operatorname{argmax}_E (\ln Z + \ln P(E)) = \\ &= \operatorname{argmax}_E \ln P(E) = \\ &= \operatorname{argmax}_E P(E) . \end{aligned}$$

Then the value of the partition function becomes

$$Z = \sum_E e^{\ln g(E) - \beta E} = \ln g(E_0) - \beta E_0 + \sum_E e^{-\Delta(\ln g(E) - \beta E)} ,$$

begin $\Delta f = f(E) - f(E_0)$, and E_0

If N is known, $N = \sum_E g(E)$

ENTROPY AND ARROW OF TIME

Microscopic view vs. macroscopic view. Usually we don't know it accurately/we don't care to know it, and summarize it with a macroscopic view: in this description some information is lost

Usually:

- Microscopic symmetry, with uniform transition matrix
- Many microstates correspond to the same macrostate
- As the system evolves, its macrostate may evolve towards a stationary state, corresponding with the macrostate with the most microstates. This evolution has a preferred direction - defining an **arrow of time** -, the one with **non-decreasing** expected **entropy**. This evolution can also be described in terms of **increasing ignorance**
- A macroscopic equilibrium corresponds to a stationary distribution of the macrostates with quite sharp maximum. While the macroscopic view looks stationary, the system is still evolving at the microscopic level. The amplitude of oscillations around an equilibrium decreases as $\sim \frac{1}{\sqrt{N}}$

12.1 Box occupation example

Remark. In this example, the microscopic dynamics of the system is not deterministic.

A box with two rooms contains N chambers. This balls are numbered so that they are distinguishable if you can see their labels.

Macrostate definition. A macrostate of the system is defined here by the number of balls in each room, (N_L, N_R) . As the total number of balls is given and constant, the number of balls in one chamber, e.g. N_L in the left chamber, is enough to determine the macrostate being $N_R = N - N_L$.

Microstate counting. The number of microstates producing the macrostate N_L is the number of combinations, C_{N, N_L}

$$\Omega(N_L) = C_{N, N_L} = \binom{N}{N_L} = \frac{N!}{(N - N_L)! N_L!} .$$

Entropy of a macrostate. ...

$$S(n) = \ln \binom{N}{n} .$$

Probability transition between macrostates. Being the probability uniform for any transition of microstates, the probability of transition between macrostates reads

$$\begin{aligned} P_-(n) &:= P(n \rightarrow n-1) = \frac{n}{N} \\ P_+(n) &:= P(n \rightarrow n+1) = \frac{N-n}{N} \end{aligned} \tag{12.1}$$

Expected change of entropy.

$$\begin{aligned}
 \mathbb{E}[\Delta S(n)] &= \sum_i P_i(n) \Delta S_i(n) = \\
 &= P_-(n) (S(n-1) - S(n)) + P_+(n) (S(n+1) - S(n)) = \\
 &= \frac{n}{N} \ln \frac{\binom{N}{n-1}}{\binom{N}{n}} + \frac{N-n}{N} \ln \frac{\binom{N}{n+1}}{\binom{N}{n}} = \\
 &= \frac{n}{N} \ln \frac{N!}{(N-n+1)!(n-1)!} \frac{(N-n)!n!}{N!} + \frac{N-n}{N} \ln \frac{N!}{(N-n-1)!(n+1)!} \frac{(N-n)!n!}{N!} = \\
 &= \frac{n}{N} \ln \frac{(N-n)!n!}{(N-n+1)!(n-1)!} + \frac{N-n}{N} \ln \frac{(N-n)!n!}{(N-n-1)!(n+1)!} = \\
 &= \ln \frac{(N-n)!n!}{(N-n+1)!(n-1)!} + \frac{n}{N} \ln \frac{(N-n-1)!(n+1)!}{(N-n+1)!(n-1)!} = \\
 &= \ln \frac{(N-n)}{n+1} + \frac{n}{N} \ln \frac{n(n+1)}{(N-n)(N-n+1)} = \\
 &= \dots
 \end{aligned}$$

todo keep going with the algebra without the limit of large number of particles...

In the limit $N \gg 1, n \gg 1, N-n \gg 1$,

$$\begin{aligned}
 \mathbb{E}[\Delta S(n)] &\simeq \ln \frac{N-n}{n} + \frac{n}{N} \ln \frac{n^2}{(N-n)^2} = \\
 &= \ln \frac{N-n}{n} - \frac{2n}{N} \ln \frac{N-n}{n} = \\
 &= \left(1 - 2\frac{n}{N}\right) \ln \frac{1 - \frac{n}{N}}{\frac{n}{N}}.
 \end{aligned}$$

Both factors change sign when $\frac{n}{N} = \frac{1}{2}$. For $\frac{n}{N} < \frac{1}{2}$ both factors are positive, for $\frac{n}{N} > \frac{1}{2}$ both factors are negative. Thus, the expected variation of entropy of this process is non-negative for every value of $n \in [0, N]$ and thus for every transition,

$$\mathbb{E}[\Delta S(n)] \geq 0, \quad \forall n \in [0, N].$$

Markov process of macrostates. The macrostates and the transition between them can be model as a stationary [Markov process](#). The probability of the system of being in (macro)state n at time t is $p_n(t)$, and using the transition probability $P(n|m)$ from state m at $t-1$ to state n at t , the following holds

$$p_n(t) = \sum_m P(n|m) p_m(t-1),$$

or using matrix formalism, introducing transition probability matrix $\{\mathbf{P}\}_{mn} := P(n|m)$,

$$\mathbf{p}(t) = \mathbf{p}(t-1)\mathbf{P}.$$

The transition probabilities (12.1) define the elements of the transition probability matrix $\mathbf{P}$,

$$\mathbf{P} = \begin{bmatrix} \cdot & 1 & & & & & & \\ \frac{1}{N} & \cdot & 1 - \frac{1}{N} & & & & & \\ & \frac{2}{N} & \cdot & 1 - \frac{2}{N} & & & & \\ & & \dots & \cdot & \dots & & & \\ & & & 1 - \frac{2}{N} & \cdot & \dots & & \\ & & & & 1 - \frac{1}{N} & \frac{2}{N} & & \\ & & & & & \cdot & \frac{1}{N} & \\ & & & & & & 1 & \cdot \end{bmatrix}.$$

Stationary distribution. The stationary distribution $\mathbf{p}^*$ is invariant under a transition, i.e. it satisfies $\mathbf{p}^* = \mathbf{p}^* \mathbf{P}$. The stationary distribution of the process is the Binomial distribution

$$p_n^* = \frac{1}{2^N} \binom{N}{n}.$$

Proof

This can be proved by direct computation. The first element reads

$$p_0^* = \frac{1}{N} p_1^*,$$

or

$$\frac{1}{2^N} \cdot 1 = \frac{1}{N} \frac{1}{2^N} \frac{1}{N}.$$

The n^{th} element ($n \in (1, N-1)$) reads

$$p_n^* = \left(1 - \frac{n-1}{N}\right) p_{n-1}^* + \frac{n+1}{N} p_{n+1}^*.$$

or

$$\begin{aligned} \frac{1}{2^N} \frac{N!}{(N-n)!n!} &= \left(1 - \frac{n-1}{N}\right) \frac{1}{2^N} \frac{N!}{(N-n+1)!(n-1)!} + \frac{n+1}{N} \frac{1}{2^N} \frac{N!}{(N-n-1)!(n+1)!} = \\ &= \frac{1}{2^N} \left[\frac{N-n+1}{N} \frac{N!}{(N-n+1)!(n-1)!} + \frac{n+1}{N} \frac{N!}{(N-n-1)!(n+1)!} \right] = \\ &= \frac{1}{2^N} \frac{N!}{(N-n)!n!} \left[\frac{n}{N} + \frac{N-n}{N} \right] = \\ &= \frac{1}{2^N} \frac{N!}{(N-n)!n!}. \end{aligned}$$

Statistics of the stochastic process. ...

Statistics of the stationary distribution.

- Average $\mu = \frac{N}{2}$
- Variance $\sigma^2 = \frac{N}{4}$
- Standard deviation $\sigma = \frac{\sqrt{N}}{2}$

Thus the relative fluctuation around the average can be estimated with the ratio of the standard deviation and the average value,

$$\frac{\sigma}{\mu} = \frac{1}{\sqrt{N}},$$

so, the relative fluctuation decreases as N increases.

N	Relative fluctuation
10^2	10^{-1}
10^4	10^{-2}
10^6	10^{-3}
10^8	10^{-4}

Average of a binomial distribution

$$\begin{aligned}
 \mathbb{E}[n] &= \sum_{n=0}^N n p_n^* = \\
 &= \frac{1}{2^N} \sum_{n=0}^N n \binom{N}{n} = \\
 &= \frac{1}{2^N} \sum_{n=1}^N N \binom{N-1}{n-1} = \\
 &= \frac{1}{2^N} N \sum_{n=0}^{N-1} \binom{N-1}{n} = \\
 &= \frac{1}{2^N} N 2^{N-1} = \frac{N}{2}.
 \end{aligned}$$

Variance of a binomial distribution

$$\sigma_n^2 = \mathbb{E}[n^2] - (\mathbb{E}[n])^2$$

Then,

$$\mathbb{E}[n^2] = \mathbb{E}[n(n-1)] + \mathbb{E}[n],$$

where the first term follows from a similar procedure as the one used for the average, using the identity $n(n-1) \binom{N}{n} = N(N-1) \binom{N-2}{n-2}$,

$$\mathbb{E}[n(n-1)] = \frac{N(N-1)}{4}.$$

Thus the variance reads

$$\sigma_n^2 = \frac{N(N-1)}{4} + \frac{N}{2} - \frac{N^2}{4} = \frac{N}{4}.$$

Markov process of microstates. A microstate is defined by the id of the box of each of the N balls. One-step transition switch the index of one ball. If the transition probability between these microstates is uniform it has probability $\frac{1}{N}$. Not, let's consider two states i, j . If these states are adjacent, the probabilities $P^{micro}(i \rightarrow j) = P^{micro}(j \rightarrow i) = \frac{1}{N}$, and zero otherwise. Thus, the transition probability matrix $\mathbf{P}^{micro}$ between microstates is symmetric. Since the **transition matrix is symmetric**, the **stationary distribution is uniform**. This is immediately proved using the normalization condition $\mathbf{P}\mathbf{1} = \mathbf{1}$, and the definition of a stationary distribution, $\mathbf{p}^{*T} = \mathbf{p}^{*T}\mathbf{P}$, and the definition of symmetric transition matrix $\mathbf{P} = \mathbf{P}^T$. The transpose of the definition of the stationary distribution, $\mathbf{p}^* = \mathbf{P}\mathbf{p}^*$, compared with the normalization condition, provides the relation $\mathbf{p}^* = \alpha\mathbf{1}$, and normalization condition $1 = |\mathbf{p}^*|_1 = \alpha N$, and $\alpha = \frac{1}{N}$. The stationary distribution for the symmetric Markov process is $\mathbf{p}^* = \frac{1}{N}\mathbf{1}$.

Combinatorics naturally links the distribution of microstates and the distribution of the macrostates.

H-Theorem for symmetric Markov chains. A discrete version of the Boltzmann's H-Theorem exists for symmetric Markov process, assessing that the entropy $H(\mathbf{p}(t)) := -\sum_k p_k(t) \ln p_k(t)$ is non-decreasing, i.e.

$$H(\mathbf{p}(t+1)) \geq H(\mathbf{p}(t)).$$

Proof

Remark. A generic transition probability matrix $\mathbf{P}$ satisfies the normalization condition $\sum_j P_{ij} = 1$, for every i (row sum). For a symmetric probability matrix $\mathbf{P} = \mathbf{P}^T$, also the column sum holds $\sum_i P_{ij} = 1$.

Remark. The function $f(x) = x \ln x$ is convex, i.e. $f''(x) > 0$, for $x > 0$.

$$f'(x) = \ln x + 1$$

$$f''(x) = \frac{1}{x}.$$

$$\begin{aligned} H(t+1) &= - \sum_k p_k(t+1) \ln p_k(t+1) = \\ &= - \sum_k f(p_k(t+1)) = \\ &= - \sum_k f\left(\sum_i p_i(t) P_{ik}\right). \end{aligned}$$

Jensen's Inequality. As f is convex, and $\sum_i P_{ik} = 1$, with $P_{ik} \in (0, 1)$,

$$f\left(\sum_i p_i P_{ik}\right) \leq \sum_i P_{ik} f(p_i).$$

Using Jensen's inequality in the expression of $H(t+1)$,

$$\begin{aligned} H(t+1) &\geq - \sum_k \sum_i P_{ik} f(p_i) = \\ &= - \sum_i f(p_i) \underbrace{\sum_k P_{ik}}_{=1} = H(t), \end{aligned}$$

i.e. $H(t+1) \geq H(t)$.

Part IV

Quantum Mechanics

QUANTUM MECHANICS

- Principles and postulates
 - statistics and measurements outcomes (Heisenberg built its matrix mechanics only on observables...)
 - CCR
- angular momentum, spin, and atom

13.1 Mathematical tools for quantum mechanics

Definition 13.1.1 (Operator)

Definition 13.1.2 (Adjoint operator)

Given an operator $\hat{A} : U \rightarrow V$, its adjoint operator $\hat{A}^* : V \rightarrow U$ is the operator s.t.

$$(\mathbf{v}, \hat{A}\mathbf{u})_V = (\mathbf{u}, \hat{A}^*\mathbf{v})_U$$

holds for $\forall \mathbf{u} \in U, \mathbf{v} \in V$.

Definition 13.1.3 (Hermitian (self-adjoint) operator)

The operator $\hat{A} : U \rightarrow U$ is a self-adjoint operator if

$$\hat{A}^* = \hat{A}.$$

Self-adjoint operators have real eigenvalues, and orthogonal eigenvectors (at least those associated to different eigenvalues; those associated with the same eigenvalues can be used to build an orthogonal set of vectors with orthogonalization process).

13.2 Postulates of Quantum Mechanics

- ...
- Canonical Commutation Relation (CCR) and Canonical Anti-Commutation Relation...
- ...

13.3 Non-relativistic Mechanics

13.3.1 Wave function

The state of a system is described by a wave function $|\Psi\rangle$

todo

- properties: domain, image,...
- unitary $1 = \langle\Psi|\Psi\rangle = |\Psi|^2$, for statistical interpretation of $|\Psi|^2$ as a density probability function

13.3.2 Operators and Observables

Physical **observable** quantities are represented by *Hermitian operators*. Possible outcomes of measurement are the eigenvalues of the operator

Given $\hat{A}$ and the set of its eigenvectors $\{|A_i\rangle\}_i$ (**todo** continuous or discrete spectrum..., need to treat this difference quite in details), with associated eigenvalues $\{a_i\}_i$

$$\hat{A}|A_i\rangle = a_i|A_i\rangle$$

$$|\Psi\rangle = |A_i\rangle\langle A_i|\Psi\rangle = |A_i\rangle\Psi_i^A$$

$$\langle A_j|\Psi\rangle = \langle A_j|A_i\rangle\langle A_i|\Psi\rangle = \Psi_j^A$$

and thus

$$\Psi_j^A = \langle A_j|\Psi\rangle$$

$$\Psi_j^{A*} = \langle\Psi|A_j\rangle$$

- identity operator $\sum_i |A_i\rangle\langle A_i| = \mathbb{I}$, since

$$\sum_i |A_i\rangle\langle A_i|\Psi\rangle = \sum_i |A_i\rangle\langle A_i|\Psi_j^A A_j\rangle = \sum_i |A_i\rangle\delta_{ij}\Psi_j^A = \sum_i |A_i\rangle\Psi_i^A = |\Psi\rangle$$

- Normalization:

$$1 = \langle\Psi|\Psi\rangle = \Psi_j^{A*} \underbrace{\langle A_j|A_i\rangle}_{\delta_{ij}} \Psi_i^A = \sum_i |\Psi_i^A|^2$$

with $|\Psi_i^A|^2$ that can be interpreted as the probability of finding the system in state $|A_i\rangle$

- Expected value of the physical quantity in the a state $|\Psi\rangle$, with possible values a_i with probability $|\Psi_i^A|^2$

$$\begin{aligned}
 \bar{A}_\Psi &= \sum_i a_i |\Psi_i^A|^2 = \\
 &= \sum_i a_i \Psi_i^{A*} \Psi_i^A = \\
 &= \sum_i a_i \langle \Psi | A_i \rangle \langle A_i | \Psi \rangle = \\
 &= \langle \Psi | \left(\sum_i a_i | A_i \rangle \langle A_i | \right) | \Psi \rangle = \\
 &= \langle \Psi | \hat{A} | \Psi \rangle =
 \end{aligned}$$

since an operator $\hat{A}$ can be written as a function of its eigenvalues and eigenvectors

$$\begin{aligned}
 \left(\sum_i a_i | A_i \rangle \langle A_i | \right) \Psi &= \left(\sum_i a_i | A_i \rangle \langle A_i | \right) c_k | A_k \rangle = \\
 &= \sum_i a_i | A_i \rangle c_i = \\
 &= \sum_i \hat{A} | A_i \rangle c_i = \\
 &= \hat{A} \sum_i | A_i \rangle c_i = \hat{A} | \Psi \rangle .
 \end{aligned}$$

13.3.3 Space and Momentum Representation

Space Representation

Position operator $\hat{\mathbf{r}}$ has eigenvalues $\mathbf{r}$ identifying the possible measurements of the position

$$\hat{\mathbf{r}} |\mathbf{r}\rangle = \mathbf{r} |\mathbf{r}\rangle ,$$

being $\mathbf{r}$ the result of the measurement (position in space, mathematically it could be a vector), and $|\mathbf{r}\rangle$ the state function corresponding to the measurement $\mathbf{r}$ of the position.

- Result of measurement, $\mathbf{r}$, is a position in space. As an example, it could be a point in an Euclidean space $P \in E^n$. It could be written using properties of Dirac's delta "function"

$$\mathbf{r} = \int_{\mathbf{r}'} \delta(\mathbf{r}' - \mathbf{r}) \mathbf{r}' d\mathbf{r}'$$

- Projection of wave function over eigenstates of position operator

$$\begin{aligned}
 \langle \mathbf{r} | \Psi \rangle(t) &= \Psi(\mathbf{r}, t) = \int_{\mathbf{r}'} \delta(\mathbf{r} - \mathbf{r}') \Psi(\mathbf{r}', t) d\mathbf{r}' = \\
 &= \int_{\mathbf{r}'} \langle \mathbf{r} | \mathbf{r}' \rangle \Psi(\mathbf{r}', t) d\mathbf{r}' = \\
 &= \int_{\mathbf{r}'} \langle \mathbf{r} | \mathbf{r}' \rangle \langle \mathbf{r}' | \Psi \rangle(t) d\mathbf{r}' = \\
 &= \langle \mathbf{r} | \underbrace{\left(\int_{\mathbf{r}'} |\mathbf{r}'\rangle \langle \mathbf{r}'| d\mathbf{r}' \right)}_{=\hat{\mathbf{I}}} | \Psi \rangle(t) .
 \end{aligned}$$

- having used orthogonality (**todo** why? provide definition and examples of operators with continuous spectrum)

$$\langle \mathbf{r}' | \mathbf{r} \rangle = \delta(\mathbf{r}' - \mathbf{r})$$

- Expansion of a state function $|\Psi\rangle(t)$ over the basis of the position operator

$$|\Psi\rangle(t) = \hat{\mathbf{1}}|\Psi\rangle(t) = \left(\int_{\mathbf{r}'} |\mathbf{r}'\rangle \langle \mathbf{r}'| d\mathbf{r}' \right) |\Psi\rangle(t) = \int_{\mathbf{r}'} |\mathbf{r}'\rangle \langle \mathbf{r}' | \Psi \rangle(t) d\mathbf{r}' .$$

- Unitarity and probability density

$$\begin{aligned} 1 &= \langle \Psi | \Psi \rangle(t) = \langle \Psi | \left(\int_{\mathbf{r}'} |\mathbf{r}'\rangle \langle \mathbf{r}'| d\mathbf{r}' \right) | \Psi \rangle \\ &= \int_{\mathbf{r}'} \langle \Psi | \mathbf{r}' \rangle \langle \mathbf{r}' | \Psi \rangle d\mathbf{r}' \\ &= \int_{\mathbf{r}'} \Psi^*(\mathbf{r}', t) \Psi(\mathbf{r}', t) d\mathbf{r}' \\ &= \int_{\mathbf{r}'} |\Psi(\mathbf{r}', t)|^2 d\mathbf{r}' \end{aligned}$$

and thus $|\Psi(\mathbf{r}, t)|^2$ can be interpreted as the **probability density function** of measuring position of the system equal to $\mathbf{r}'$.

- Average value of the operator

$$\begin{aligned} \bar{\mathbf{r}} &= \langle \Psi | \hat{\mathbf{r}} | \Psi \rangle = \\ &= \int_{\mathbf{r}'} \langle \Psi | \mathbf{r}' \rangle \langle \mathbf{r}' | d\mathbf{r}' | \hat{\mathbf{r}} | \int_{\mathbf{r}''} |\mathbf{r}''\rangle \langle \mathbf{r}'' | \Psi \rangle d\mathbf{r}'' \\ &= \int_{\mathbf{r}'} \int_{\mathbf{r}''} \langle \Psi | \mathbf{r}' \rangle \langle \mathbf{r}' | \hat{\mathbf{r}} | \mathbf{r}'' \rangle \langle \mathbf{r}'' | \Psi \rangle d\mathbf{r}' d\mathbf{r}'' = \\ &= \int_{\mathbf{r}'} \int_{\mathbf{r}''} \langle \Psi | \mathbf{r}' \rangle \underbrace{\langle \mathbf{r}' | \mathbf{r}'' \rangle}_{=\delta(\mathbf{r}' - \mathbf{r}'')} \mathbf{r}'' \langle \mathbf{r}'' | \Psi \rangle d\mathbf{r}' d\mathbf{r}'' = \\ &= \int_{\mathbf{r}'} \langle \Psi | \mathbf{r}' \rangle \mathbf{r}' \langle \mathbf{r}' | \Psi \rangle d\mathbf{r}' = \\ &= \int_{\mathbf{r}'} \Psi^*(\mathbf{r}', t) \mathbf{r}' \Psi(\mathbf{r}', t) d\mathbf{r}' = \\ &= \int_{\mathbf{r}'} |\Psi(\mathbf{r}', t)|^2 \mathbf{r}' d\mathbf{r}' . \end{aligned}$$

Momentum Representation

Momentum operator as the limit of ... **todo** prove the expression of the momentum operator as the limit of the generator of translation

$$\langle \mathbf{r} | \hat{\mathbf{p}} = -i\hbar \nabla \langle \mathbf{r} | \quad (13.1)$$

- Spectrum

$$\hat{\mathbf{p}}|\mathbf{p}\rangle = \mathbf{p}|\mathbf{p}\rangle$$

$$\langle \mathbf{r} | \hat{\mathbf{p}} | \mathbf{p} \rangle = -i\hbar \nabla \langle \mathbf{r} | \mathbf{p} \rangle = \mathbf{p} \langle \mathbf{r} | \mathbf{p} \rangle$$

and thus the eigenvectors in space base $\mathbf{p}(\mathbf{r}) = \langle \mathbf{r} | \mathbf{p} \rangle$ are the solution of the differential equation

$$-i\hbar \nabla \mathbf{p}(\mathbf{r}) = \mathbf{p} \mathbf{p}(\mathbf{r}) ,$$

that in Cartesian coordinates reads

$$-i\hbar \partial_j p_k(\mathbf{r}) = p_j p_k(\mathbf{r})$$

$$p_k(\mathbf{r}) = p_{k,0} \exp \left[i \frac{p_j}{\hbar} r_j \right]$$

or

$$\langle \mathbf{r} | \mathbf{p} \rangle = \mathbf{p}(\mathbf{r}) = \mathbf{p}_0 \exp \left[i \frac{\mathbf{p} \cdot \mathbf{r}}{\hbar} \right]$$

todo

- normalization factor $\frac{1}{(2\pi)^{\frac{3}{2}}}$

$$\mathcal{F}\{\delta(x)\}(k) = \int_{-\infty}^{\infty} \delta(x) e^{-ikx} dx = 1$$

- Fourier transform and inverse Fourier transform: definitions and proofs (link to a math section)
- representation in basis of wave vector operator $\hat{\mathbf{k}}, \hat{\mathbf{p}} = \hbar \hat{\mathbf{k}}$

From position to momentum representation

Momentum and wave vector, $\mathbf{p} = \hbar \mathbf{k}$

$$\begin{aligned} \langle \mathbf{p} | \Psi \rangle &= \langle \mathbf{p} | \int_{\mathbf{r}'} |\mathbf{r}'\rangle \langle \mathbf{r}' | \Psi \rangle d\mathbf{r}' = \\ &= \int_{\mathbf{r}'} \langle \mathbf{p} | \mathbf{r}' \rangle \langle \mathbf{r}' | \Psi \rangle d\mathbf{r}' = \\ &= \frac{1}{(2\pi)^{3/2}} \int_{\mathbf{r}'} \exp \left[i \frac{\mathbf{p} \cdot \mathbf{r}'}{\hbar} \right] \langle \mathbf{r}' | \Psi \rangle d\mathbf{r}' = \end{aligned}$$

Relation between position and wave-number representation can be represented with a Fourier transform

$$\begin{aligned} \langle \mathbf{k} | \Psi \rangle &= \langle \mathbf{k} | \int_{\mathbf{r}'} |\mathbf{r}'\rangle \langle \mathbf{r}' | \Psi \rangle d\mathbf{r}' = \\ &= \int_{\mathbf{r}'} \langle \mathbf{k} | \mathbf{r}' \rangle \langle \mathbf{r}' | \Psi \rangle d\mathbf{r}' = \\ &= \frac{1}{(2\pi)^{3/2}} \int_{\mathbf{r}'} \exp [i \mathbf{k} \cdot \mathbf{r}'] \langle \mathbf{r}' | \Psi \rangle d\mathbf{r}' = \\ &= \frac{1}{(2\pi)^{3/2}} \int_{\mathbf{r}'} \exp [i \mathbf{k} \cdot \mathbf{r}'] \Psi(\mathbf{r}') d\mathbf{r}' = \\ &= \mathcal{F}\{\Psi(\mathbf{r})\}(\mathbf{k}) \end{aligned}$$

13.3.4 Schrodinger Equation

$$i\hbar \frac{d}{dt} |\Psi\rangle = \hat{H} |\Psi\rangle$$

being $\hat{H}$ the Hamiltonian operator and $|\Psi\rangle$ the wave function, as a function of time t as an independent variable.

Stationary States

Eigenspace of the Hamiltonian operator

$$\hat{H} |\Psi_k\rangle = E_k |\Psi_k\rangle ,$$

with E_k possible values of energy measurements. *If no eigenstates with the same eigenvalue exists, then...otherwise... Without external influence **todo** be more detailed!*, energy values and eigenstates of the systems are constant in time.

Thus, expanding the state of the system $|\Psi\rangle$ over the stationary states gives $|\Psi_k\rangle$, $|\Psi\rangle = |\Psi_k\rangle c_k(t)$, and inserting in Schrodinger equation

$$i\hbar \dot{c}_k |\Psi_k\rangle = c_k E_k |\Psi_k\rangle$$

and exploiting orthogonality of eigenstates, a diagonal system for the amplitudes of stationary states arises,

$$i\hbar \dot{c}_k = c_k E_k .$$

whose solution reads

$$c_k(t) = c_{k,0} \exp \left[-i \frac{E_k}{\hbar} t \right]$$

Thus the state of the system evolves like a superposition of monochromatic waves with frequencies $\omega_k = \frac{E_k}{\hbar}$,

$$|\Psi\rangle = |\Psi_k\rangle c_k(t) = |\Psi_k\rangle c_{k,0} \exp \left[-i \frac{E_k}{\hbar} t \right] .$$

$$\begin{aligned} \frac{d}{dt} \bar{A} &= \frac{d}{dt} (\langle \Psi | \hat{A} | \Psi \rangle) = \\ &= \frac{d}{dt} \langle \Psi | \hat{A} | \Psi \rangle + \langle \Psi | \frac{d\hat{A}}{dt} | \Psi \rangle + \langle \Psi | \hat{A} \frac{d}{dt} | \Psi \rangle = \\ &= \langle \Psi | \frac{d\hat{A}}{dt} | \Psi \rangle + \frac{i}{\hbar} \langle \Psi | \hat{H} \hat{A} | \Psi \rangle - \frac{i}{\hbar} \langle \Psi | \hat{A} \hat{H} | \Psi \rangle = \\ &= \langle \Psi | \left(\frac{i}{\hbar} [\hat{H}, \hat{A}] + \frac{d\hat{A}}{dt} \right) | \Psi \rangle . \end{aligned}$$

Pictures

- Schrodinger
- Heisenberg
- Interaction

Schrodinger

If $\hat{H}$ not function of time

$$|\Psi\rangle(t) = \exp \left[-i \frac{\hat{H}}{\hbar} (t - t_0) \right] |\Psi\rangle(t_0) = \hat{U}(t, t_0) |\Psi\rangle(t_0)$$

$$\bar{A} = \langle \Psi | \hat{A} | \Psi \rangle = \langle \Psi_0 | \hat{U}^*(t, t_0) \hat{A} \hat{U}(t, t_0) | \Psi_0 \rangle$$

Heisenberg

While Schrodinger picture works with (usually) time-independent operators and time-dependent wave functions, Heisenberg picture works with time-dependent operators and time-independent wave functions. The state of a quantum system at two different times is connected by a unitary operator U_{t,t_0} ,

$$|\Psi_t\rangle = U_{t,t_0} |\Psi_{t_0}\rangle$$

Expected value. The expected value of the physical quantity represented by the Hermitian operator $\hat{A}$ reads

$$\begin{aligned} \langle \hat{A} \rangle &= \langle \Psi_t | \hat{A} | \Psi_t \rangle = \\ &= \langle \Psi_{t_0} | U_{t,t_0}^\dagger \hat{A} U_{t,t_0} | \Psi_{t_0} \rangle = \\ &= \langle \Psi_{t_0} | \hat{A}_H(t) | \Psi_{t_0} \rangle, \end{aligned}$$

having defined the counterpart in Heisenberg picture, $\hat{A}_H(t) = U_{t,t_0}^\dagger \hat{A} U_{t,t_0}$ of the operator $\hat{A}$ in Schrodinger picture.

Hamiltonian operator in Heisenberg picture.

$$\hat{H}_H = U^\dagger \hat{H} U.$$

If the Hamiltonian operator is **time-independent in Schrodinger picture**, the evolution operator can be written as $U_t = \exp \left(-i \frac{\hat{H}}{\hbar} t \right)$. In this case, U and $\hat{H}$ commute, and the Hamiltonian operator in Heisenberg picture has the same expression as the Hamiltonian operator in Schrodinger equation,

$$\hat{H}_H = U^\dagger \hat{H} U = \left(\text{if } \partial_t \hat{H} = 0 \right) = U^\dagger U \hat{H} = \hat{H}.$$

Time derivative of the operator $\hat{A}_H(t)$.

$$\begin{aligned} \frac{d}{dt} \hat{A}_H(t) &= \frac{d}{dt} (U_{t,t_0}^\dagger \hat{A} U_{t,t_0}) = \\ &= \partial_t U_{t,t_0}^\dagger \hat{A} U_{t,t_0} + U_{t,t_0}^\dagger \partial_t \hat{A} U_{t,t_0} + U_{t,t_0}^\dagger \hat{A} \partial_t U_{t,t_0} = \\ &= -\frac{1}{i\hbar} U_{t,t_0}^\dagger \hat{H} \hat{A} U_{t,t_0} + U_{t,t_0}^\dagger \partial_t \hat{A} U_{t,t_0} + \frac{1}{i\hbar} U_{t,t_0}^\dagger \hat{A} \hat{H} U_{t,t_0} = \\ &= -\frac{1}{i\hbar} U_{t,t_0}^\dagger \hat{H} U_{t,t_0} U_{t,t_0}^\dagger \hat{A} U_{t,t_0} + U_{t,t_0}^\dagger \partial_t \hat{A} U_{t,t_0} + \frac{1}{i\hbar} U_{t,t_0}^\dagger \hat{A} U_{t,t_0} U_{t,t_0}^\dagger \hat{H} U_{t,t_0} = \\ &= -\frac{1}{i\hbar} [\hat{H}_H, \hat{A}_H] + (\partial_t \hat{A})_H, \end{aligned}$$

as the relation (15.2) gives $\partial_t U_{t,t_0} U_{t,t_0}^\dagger = \frac{1}{i\hbar} \hat{H}$,

old ... for $\hat{H}$ independent from time t ,

$$\begin{aligned} \frac{d}{dt} \hat{\mathbf{r}} &= \frac{i}{\hbar} [\hat{H}, \hat{\mathbf{r}}] = -\frac{1}{i\hbar} [\hat{H}, \hat{\mathbf{r}}] \\ \frac{d}{dt} \hat{\mathbf{p}} &= \frac{i}{\hbar} [\hat{H}, \hat{\mathbf{p}}] = -\frac{1}{i\hbar} [\hat{H}, \hat{\mathbf{p}}] \end{aligned}$$

Hamiltonian Mechanics

From Lagrange equations

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = Q_q$$

q generalized coordinates, $p := \frac{\partial L}{\partial \dot{q}}$ generalized momenta.

Hamiltonian

$$H(p, q, t) = p\dot{q} - L(\dot{q}, q, t)$$

Increment of the Hamiltonian

$$\begin{aligned} dH &= \partial_p H dp + \partial_q H dq + \partial_t H dt \\ dH &= \dot{q} dp + p d\dot{q} - \partial_{\dot{q}} L d\dot{q} - \partial_q L dq - \partial_t L dt = \\ &= \dot{q} dp - \partial_q L dq - \partial_t L dt = \\ &= \dot{q} dp - (\dot{p} + Q_q) dq - \partial_t L dt = \\ &\quad \begin{cases} \frac{\partial H}{\partial p} = \dot{q} \\ \frac{\partial H}{\partial q} = -\frac{\partial L}{\partial q} = -\dot{p} + Q_q \\ \frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t} \end{cases} \end{aligned}$$

Physical quantity $f(q(t), p(t), t)$. Its time derivative reads

$$\begin{aligned} \frac{df}{dt} &= \frac{\partial f}{\partial q} \dot{q} + \frac{\partial f}{\partial p} \dot{p} + \frac{\partial f}{\partial t} = \\ &= \frac{\partial f}{\partial q} \frac{\partial H}{\partial p} + \frac{\partial f}{\partial p} \left[-\frac{\partial H}{\partial q} + Q_q \right] + \frac{\partial f}{\partial t} = \\ &= -\partial_q H \partial_p f + \partial_p H \partial_q f + \partial_t f + Q_q \partial_p f = \\ &= -\{H, f\} + \partial_t f + Q_q \partial_p f \end{aligned}$$

If $Q_q = 0$, the correspondence between quantum mechanics and classical mechanics

$$\begin{aligned} \frac{df}{dt} &= -\{H, f\} + \partial_t f \quad \leftrightarrow \quad \frac{d}{dt} \hat{f} = -\frac{1}{i\hbar} [\hat{H}, \hat{f}] + \frac{\partial \hat{f}}{\partial t} \\ \{H, f\} &\quad \leftrightarrow \quad \frac{i}{\hbar} [\hat{H}, \hat{f}] \end{aligned}$$

Interaction

13.3.5 Operators

Commuting operators, $0 = [\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$. Their eigenvalue problems read

$$\hat{A}|a\rangle = a|a\rangle$$

$$\hat{B}|b\rangle = b|b\rangle$$

$$a\hat{B}|a\rangle = \hat{B}\hat{A}|a\rangle = \hat{A}(\hat{B}|a\rangle)$$

Thus the state $\hat{B}|a\rangle$ is an eigenstate of $\hat{A}$, with the same eigenvalue a . If the eigenvalue is non-degenerate, this eigenstate must be proportional to the original eigenstate, and thus $\hat{B}|a\rangle = b|a\rangle$. If the eigenvalue is degenerate, the state $\hat{B}|a\rangle$ can be a linear combination of all the eigenstates $|a_i\rangle$ with eigenvalue $a_i = a$.

13.3.6 Matrix Mechanics

Attualization of 1925 papers

...to find the canonical commutation relation,

$$[\hat{\mathbf{r}}, \hat{\mathbf{p}}] = i\hbar \mathbf{1}.$$

$$\begin{aligned} [\hat{\mathbf{r}}, \hat{\mathbf{p}}] &= \hat{\mathbf{r}}\hat{\mathbf{p}} - \hat{\mathbf{p}}\hat{\mathbf{r}} = \\ &= \hat{\mathbf{r}} \int_{\mathbf{r}} |\mathbf{r}\rangle \langle \mathbf{r}| d\mathbf{r} \hat{\mathbf{p}} - \hat{\mathbf{p}} \int_{\mathbf{r}} |\mathbf{r}\rangle \langle \mathbf{r}| d\mathbf{r} \hat{\mathbf{r}} \int_{\mathbf{r}'} |\mathbf{r}'\rangle \langle \mathbf{r}'| d\mathbf{r}' = \\ &= - \int_{\mathbf{r}} |\mathbf{r}\rangle \langle \mathbf{r}| i\hbar \nabla \langle \mathbf{r}| d\mathbf{r} - \hat{\mathbf{p}} \int_{\mathbf{r}} \int_{\mathbf{r}'} |\mathbf{r}\rangle \mathbf{r}' \underbrace{\langle \mathbf{r}|\mathbf{r}'\rangle}_{\delta(\mathbf{r}-\mathbf{r}')} \langle \mathbf{r}'| d\mathbf{r}' = \\ &= - \int_{\mathbf{r}} |\mathbf{r}\rangle \langle \mathbf{r}| i\hbar \nabla \langle \mathbf{r}| d\mathbf{r} - \hat{\mathbf{p}} \int_{\mathbf{r}} |\mathbf{r}\rangle \langle \mathbf{r}| d\mathbf{r} = \\ &= - \int_{\mathbf{r}} |\mathbf{r}\rangle \langle \mathbf{r}| i\hbar \nabla \langle \mathbf{r}| d\mathbf{r} - \int_{\mathbf{r}'} |\mathbf{r}\rangle \langle \mathbf{r}| d\mathbf{r} \hat{\mathbf{p}} \int_{\mathbf{r}'} |\mathbf{r}'\rangle \langle \mathbf{r}'| d\mathbf{r}' = \\ &= - \int_{\mathbf{r}} |\mathbf{r}\rangle \langle \mathbf{r}| i\hbar \nabla \langle \mathbf{r}| d\mathbf{r} + \int_{\mathbf{r}} |\mathbf{r}\rangle \langle \mathbf{r}| i\hbar \nabla \langle \mathbf{r}| d\mathbf{r} \int_{\mathbf{r}'} |\mathbf{r}'\rangle \langle \mathbf{r}'| d\mathbf{r}' = \dots \\ [\hat{\mathbf{r}}, \hat{\mathbf{p}}] |\Psi\rangle &= - \int_{\mathbf{r}} |\mathbf{r}\rangle \langle \mathbf{r}| i\hbar \nabla \Psi(\mathbf{r}, t) + \int_{\mathbf{r}} |\mathbf{r}\rangle i\hbar \nabla (\mathbf{r} \Psi(\mathbf{r}, t)) = \\ &= - \int_{\mathbf{r}} |\mathbf{r}\rangle i\hbar [\mathbf{r} \nabla \Psi(\mathbf{r}, t) + \Psi(\mathbf{r}, t) + \mathbf{r} \nabla \Psi(\mathbf{r}, t)] = \\ &= i\hbar \underbrace{\int_{\mathbf{r}} |\mathbf{r}\rangle \langle \mathbf{r}| d\mathbf{r}}_{\mathbf{1}} |\Psi\rangle, \end{aligned}$$

and since $|\Psi\rangle$ is arbitrary

$$[\hat{\mathbf{r}}, \hat{\mathbf{p}}] = i\hbar \mathbf{1}.$$

$$[\hat{r}_a, \hat{p}_b] = i\hbar \delta_{ab}.$$

13.3.7 Heisenberg Uncertainty relation

Uncertainty principle is a relation that holds for “wave descriptions” as it can be proved in the generic framework of [Fourier transform](#), see [Fourier transform:Uncertainty relation](#).

- Heisenberg uncertainty relation is a relation between product of the variance of two physical quantities and their commutator,
- **todo** relation with measurement process and outcomes. Commutation as a measurement process: first measure B and then A , or first measure A and then B

$$\sigma_A \sigma_B \geq \frac{1}{2} |\overline{[\hat{A}, \hat{B}]}|.$$

Proof of Heisenberg uncertainty “principle”

$$\begin{aligned}
 \sigma_A^2 \sigma_B^2 &= \langle \Psi | (\hat{A} - \bar{A})^2 | \Psi \rangle \langle \Psi | (\hat{B} - \bar{B})^2 | \Psi \rangle = \\
 &= \langle (\hat{A} - \bar{A})\Psi | (\hat{A} - \bar{A})\Psi \rangle \langle (\hat{B} - \bar{B})\Psi | (\hat{B} - \bar{B})\Psi \rangle = \\
 &= \|(\hat{A} - \bar{A})\Psi\|^2 \|(\hat{B} - \bar{B})\Psi\|^2 = \\
 &\geq \left| \langle (\hat{A} - \bar{A})\Psi | (\hat{B} - \bar{B})\Psi \rangle \right|^2 = \\
 &= \left| \langle \Psi | (\hat{A} - \bar{A})(\hat{B} - \bar{B}) | \Psi \rangle \right|^2 = \\
 &= \left| \langle \Psi | \hat{A}\hat{B} - \hat{A}\bar{B} - \bar{A}\hat{B} + \bar{A}\bar{B} | \Psi \rangle \right|^2 = \\
 &= \left| \langle \Psi | \hat{A}\hat{B} - \bar{A}\bar{B} | \Psi \rangle \right|^2 \geq \tag{1} \\
 &= \left| \frac{\langle \Psi | \hat{A}\hat{B} - \hat{B}\hat{A} | \Psi \rangle}{2i} \right|^2 = \\
 &= \frac{|\langle \Psi | [\hat{A}, \hat{B}] | \Psi \rangle|^2}{4} = \frac{1}{4} \left| [\hat{A}, \hat{B}] \right|^2
 \end{aligned}$$

having used Cauchy-Schwartz triangle inequality in (1),

$$|z| \geq |\text{im}(z)| = \frac{z - z^*}{2i} .$$

Heisenberg uncertainty principles applied to position and momentum reads

$$\sigma_{r_a} \sigma_{p_b} \geq \frac{1}{2} \left| [\hat{r}_a, \hat{p}_b] \right| = \frac{\hbar}{2} \delta_{ab} .$$

13.4 Many-body problem

Wave function with symmetries: Fermions and Bosons

QUANTUM MECHANICS - NOTES

WAVE QUANTUM MECHANICS

If the system behaves like a wave, it must satisfy a wave function

As the geometrical optics is the short wave-length approximation of the ondulatory optics, classical mechanics may be the short wave-length approximation of ondulatory mechanics

15.1 De Broglie

15.2 Schrodinger

15.2.1 Derivation

From the probability interpretation of the wave function $|\Psi_t\rangle$, and the unitary condition $\langle\Psi_t|\Psi_t\rangle = 1$, for every t , the evolution of the system must be governed by a unitary operator U_{t,t_0} ,

$$|\Psi_t\rangle = U_{t,t_0} |\Psi_{t_0}\rangle ,$$

for the unitary condition to hold at every t ,

$$1 = \langle\Psi_t|\Psi_t\rangle = \langle U_{t,t_0} \Psi_{t_0} | U_{t,t_0} \Psi_{t_0} \rangle = \langle\Psi_{t_0} | \underbrace{U_{t,t_0}^\dagger U_{t,t_0}}_{=\mathbf{I}} |\Psi_{t_0}\rangle = \langle\Psi_{t_0}|\Psi_{t_0}\rangle ,$$

i.e.

$$U_{t,t_0}^{-1} = U_{t,t_0}^\dagger .$$

Taking the time derivative w.r.t. t of the evolution relation, it follows

$$|\dot{\Psi}_t\rangle = \partial_t U_{t,t_0} |\Psi_{t_0}\rangle = \underbrace{\partial_t U_{t,t_0} U_{t,t_0}^\dagger}_{\hat{A}(t,t_0)} |\Psi_t\rangle .$$

Composition. As $|\Psi_t\rangle = U_{t,t_1} |\Psi_{t_1}\rangle = U_{t,t_1} U_{t_1,t_0} |\Psi_{t_0}\rangle$,

$$U_{t,t_0} = U_{t,t_1} U_{t_1,t_0} .$$

Independence of $\hat{A}(t, t_0) = \partial_t U_{t,t_0} U_{t,t_0}^\dagger$ from t_0 . The operator $\hat{A}(t, t_0) = \partial_t U_{t,t_0} U_{t,t_0}^\dagger$ can thus be written as

$$\hat{A}(t, t_0) = \partial_t U_{t,t_0} U_{t,t_0}^\dagger = \partial_t U_{t,t_1} \underbrace{U_{t_1,t_0} U_{t_1,t_0}^\dagger}_{=\mathbf{I}} U_{t,t_1}^\dagger = \partial_t U_{t,t_1} U_{t,t_1}^\dagger = \hat{A}(t, t_1) ,$$

showing that it's independent from the initial time, $\hat{A}(t, t_0) = \hat{A}(t)$.

$\hat{A}(t)$ is anti-Hermitian. As

$$0 = \partial_t \underbrace{(U_{t,t_0} U_{t,t_0}^\dagger)}_{=\hat{1}} = \underbrace{\partial_t U_{t,t_0} U_{t,t_0}^\dagger}_{=\hat{A}(t)} + \underbrace{U_{t,t_0} \partial_t U_{t,t_0}^\dagger}_{=\hat{A}^\dagger(t)} = \hat{A}(t) + \hat{A}^\dagger(t) ,$$

and thus $\hat{A} = -\hat{A}^\dagger$. Thus the operator $\hat{A}$ can be written as $\hat{A} = i\hat{H}$, being $\hat{H}$ Hermitian operator. For the correspondence principle (classical limit for $\hbar \rightarrow 0$) the Hermitian operator is found to be $\hat{H} = -\frac{1}{\hbar}\hat{H}$, being $\hat{H}$ the Hamiltonian operator. **todo** Add a link to “correspondence principle”, either Ehrenfest theorem and/or equations in Heisenberg picture

Thus, the wave equation becomes

$$|\dot{\Psi}\rangle = \frac{1}{i\hbar} \hat{H}|\Psi\rangle \quad \text{or} \quad i\hbar|\dot{\Psi}\rangle = \hat{H}|\Psi\rangle . \quad (15.1)$$

The Hamiltonian can be written as a function of the unitary operator U_{t,t_0} and its time derivative as

$$\hat{H} = i\hbar \partial_t U_{t,t_0} U_{t,t_0}^\dagger . \quad (15.2)$$

15.2.2 Energy eigen-states

$$\hat{H}|\Psi_k\rangle = E_k|\Psi_k\rangle .$$

Let the Hamiltonian be independent from time, thus also the eigenvalues and eigenfunctions are independent from time. Let a state be a superposition (linear combination) of eigenfunctions of the Hamiltonian operator

$$|\Psi_t\rangle = c_k(t)|\Psi_k\rangle ,$$

the time evolution immediately follows as

$$i\hbar\dot{c}_k|\Psi_k\rangle = c_k\hat{H}|\Psi_k\rangle = c_k E_k|\Psi_k\rangle ,$$

from orthogonality condition $\langle\Psi_j|\Psi_k\rangle = \delta_{jk}$, that gives the dynamical equation for the coefficients c_i ,

$$\dot{c}_k = -i\frac{E_k}{\hbar}c_k ,$$

whose solution is $c_k(t) = c_{k,0} \exp\left(-i\frac{E_k}{\hbar}t\right)$. The evolution of the state is

$$|\Psi_t\rangle = |\Psi_k\rangle c_{k,0} \exp\left(-i\frac{E_k}{\hbar}t\right) = |\Psi_k\rangle \langle\Psi_k|\Psi_0\rangle \exp\left(-i\frac{E_k}{\hbar}t\right)$$

Properties. If an **isolated system** starts in an eigen-state of the Hamiltonian operator, i.e. $|\Psi_0\rangle = |\Psi_a\rangle$, $|c_{k,0}|^2 = \delta_{ka}$, then the evolution of the system reads

$$|\Psi_t\rangle = |\Psi_a\rangle \exp\left(-i\frac{E_a}{\hbar}t\right) ,$$

i.e. the probability of measuring the k^{th} energy is

$$p(E = E_k) = |\langle\Psi_k|\Psi_t\rangle|^2 = \left| \underbrace{\langle\Psi_k|\Psi_a\rangle}_{\delta_{ka}} \exp\left(-i\frac{E_a}{\hbar}t\right) \right|^2 = \delta_{ka} ,$$

i.e. there's probability of finding it in the same state as the initial state a is $p(E = E_a) = 1$, while the probability of finding in any other state is zero, $p(E = E_k) = 0$, $k \neq a$.

15.2.3 Space and momentum operators

Position operator

Definition

$$\hat{\mathbf{r}}|\mathbf{r}\rangle = \mathbf{r}|\mathbf{r}\rangle$$

Wave function in space basis

$$\Psi(\mathbf{r}, t) := \langle \mathbf{r} | \Psi_t \rangle .$$

Orthonality, and identity operator

$$\begin{aligned} 1 &= \int_{\mathbf{r} \in \Omega} \Psi^*(\mathbf{r}, t) \Psi(\mathbf{r}, t) d\mathbf{r} = \\ &= \langle \Psi_t | \underbrace{\int_{\mathbf{r} \in \Omega} |\mathbf{r}\rangle \langle \mathbf{r}| d\mathbf{r}}_{=\hat{\mathbf{1}}} | \Psi_t \rangle = \\ &= \langle \Psi_t | \Psi_t \rangle . \end{aligned}$$

So that the identity operator in space basis reads $\hat{\mathbf{1}} = \int_{\mathbf{r} \in \Omega} |\mathbf{r}\rangle \langle \mathbf{r}| d\mathbf{r}$. A wave function can be thus written as

$$|\Psi\rangle = \int_{\mathbf{r}' \in \Omega} |\mathbf{r}'\rangle \langle \mathbf{r}' | d\mathbf{r}' | \Psi \rangle . \quad (15.3)$$

Expansion in space basis

$$\langle \mathbf{r} | \Psi \rangle = \Psi(\mathbf{r}, t) = \int_{\mathbf{r}' \in \Omega} \delta(\mathbf{r} - \mathbf{r}') \Psi(\mathbf{r}', t) d\mathbf{r}'$$

Comparing the last expression, with the projection of the expression (15.3) over $\langle \mathbf{r} |$,

$$\langle \mathbf{r} | \Psi \rangle = \langle \mathbf{r} | , \int_{\mathbf{r}' \in \Omega} |\mathbf{r}'\rangle \langle \mathbf{r}' | d\mathbf{r}' | \Psi \rangle = \int_{\mathbf{r}' \in \Omega} \langle \mathbf{r} | \mathbf{r}' \rangle \langle \mathbf{r}' | \Psi \rangle d\mathbf{r}' = \int_{\mathbf{r}' \in \Omega} \langle \mathbf{r} | \mathbf{r}' \rangle \Psi(\mathbf{r}', t) d\mathbf{r}' ,$$

if follows the orthogonality condition $\langle \mathbf{r} | \mathbf{r}' \rangle = \delta(\mathbf{r} - \mathbf{r}')$.

The classical Hamiltonian of the system reads

$$H(\mathbf{r}, \mathbf{p}) = \frac{|\mathbf{p}|^2}{2m} - \frac{q^2}{4\pi\epsilon|\mathbf{r}|} ,$$

the promotion to the Hamiltonian operator is thus

$$\hat{H} = \frac{|\hat{\mathbf{p}}|^2}{2m} - \frac{q^2}{4\pi\epsilon} |\hat{\mathbf{r}}|^{-1} .$$

Power of an operator

Positive integer power

$$\hat{A}^n = \underbrace{\hat{A} \dots \hat{A}}_{n \text{ times}}$$

Let the eigenproblem of the operator be $\hat{A}|a\rangle = a|a\rangle$, with $a \in \mathbb{R}$. Then, the eigenproblem for the operator $\hat{A}^n$ reads

$$\hat{A}^n|a\rangle = a^n|a\rangle .$$

The inverse of an operator has the eigenproblem

$$\hat{A}^{-1}|a\rangle = \frac{1}{a}|a\rangle ,$$

as $|a\rangle = \hat{A}(\hat{A}^{-1}|a\rangle) = \frac{1}{a}\hat{A}|a\rangle = \frac{1}{a}a|a\rangle = |a\rangle$.

Vector operators

Momentum operator, $\hat{\mathbf{p}}$, in space base $\langle \mathbf{r} | \hat{\mathbf{p}} = -i\hbar \nabla_{\mathbf{r}} \langle \mathbf{r} |$. Using Cartesian coordinates

$$\hat{\mathbf{e}}_a \langle \mathbf{r} | \hat{p}_a = -\hat{\mathbf{e}}_a i\hbar \partial_a \langle \mathbf{r} |$$

The operator of the square magnitude reads

$$(|\hat{\mathbf{p}}|^2) = (p_x^2 + p_y^2 + p_z^2) = \hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2 ,$$

and in space coordinates

$$\langle \mathbf{r} | (|\hat{\mathbf{p}}|^2) = -\hbar^2 \partial_{aa} \langle \mathbf{r} | = -\hbar^2 \nabla^2 \langle \mathbf{r} |$$

ANGULAR MOMENTUM

Before going into details, an introduction is likely to be required.

Spatial (orbital) angular momentum, $\hat{\mathbf{L}}$. Spatial angular motion is the quantum mechanical counterpart of the angular momentum in classical mechanics, $\hat{\mathbf{L}} = \mathbf{r} \times \mathbf{p}$, due to the spatial motion of the system w.r.t. a point. The angular momentum operator in quantum mechanics can be defined promoting the classical space and momentum variables to the corresponding operators,

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} \quad \rightarrow \quad \hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}} ,$$

whose representation using spatial basis reads

$$\langle \mathbf{r} | \hat{\mathbf{L}} = -i\hbar \mathbf{r} \times \nabla_{\mathbf{r}} \langle \mathbf{r} | .$$

This operator has several properties that will be investigated in details in the dedicated section. This operator applies to a wave function representing the position, momentum, and physical quantities with classical analogous, here called $|\psi\rangle$.

Spin (intrinsic) angular momentum, $\hat{\mathbf{S}}$. Quantum systems may have intrinsic properties, with no classical counterpart. Spin momentum is one of these variables. Spin angular momentum $\hat{\mathbf{S}}$ is defined in analogy with the spatial orbital momentum, with the same properties and acts on a spin state vector $|s\rangle$.

Thus, the full state of the system belongs to a space state that is a tensor product of a space and a spin state space,

$$\mathcal{H} = \mathcal{H}_{space} \otimes \mathcal{H}_{spin} ,$$

defined as the tensor product of a spatial state vector and a spin state vector,

$$|\Psi\rangle = |\psi\rangle \otimes |s\rangle .$$

Total angular momentum $\hat{\mathbf{J}}$. Total angular momentum can be defined as the sum of the spatial and the spin angular momentum,

$$\hat{\mathbf{J}} = (\hat{\mathbf{L}} \otimes \hat{\mathbf{1}}) + (\hat{\mathbf{1}} \otimes \hat{\mathbf{S}}) ,$$

or, remembering that $\hat{\mathbf{L}}$ only acts on $|\psi\rangle$ and $\hat{\mathbf{S}}$ only acts on $|s\rangle$, it's usually written as $\hat{\mathbf{J}} = \hat{\mathbf{L}} + \hat{\mathbf{S}}$.

16.1 Spatial Angular Momentum

Promoting the classical definition of angular momentum $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, the angular momentum operator in quantum mechanics reads

$$\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}} = \hat{\mathbf{e}}_a \varepsilon_{abc} \hat{r}_b \hat{p}_c ,$$

and using space basis

$$\langle \mathbf{r} | \hat{\mathbf{L}} = \hat{\mathbf{e}}_a \underbrace{(-i\hbar \varepsilon_{abc} r_b \partial_c \langle \mathbf{r} |)}_{= \langle \mathbf{r} | \hat{L}_a} .$$

Properties

Composition of 2 components of the angular momentum operator. Using Cartesian coordinates

$$\hat{L}_a \hat{L}_b = -\hbar^2 r_b \partial_a + \hbar^2 \delta_{ab} r_c \partial_c - \hbar^2 \varepsilon_{acd} \varepsilon_{bef} r_c r_e \partial_{df} =$$

Details

$$\begin{aligned} \hat{L}_a \hat{L}_b &= -\hbar^2 \varepsilon_{acd} r_c \partial_d (\varepsilon_{bef} r_e \partial_f) = \\ &= -\hbar^2 \varepsilon_{acd} \varepsilon_{bef} r_c (\delta_{de} \partial_f + r_e \partial_{df}) = \\ &= -\hbar^2 \varepsilon_{acd} \varepsilon_{bdf} r_c \partial_f - \hbar^2 \varepsilon_{acd} \varepsilon_{bdf} r_c r_e \partial_{df} = \\ &= -\hbar^2 (\delta_{af} \delta_{cb} - \delta_{ab} \delta_{cf}) r_c \partial_f - \hbar^2 \varepsilon_{acd} \varepsilon_{bef} r_c r_e \partial_{df} = \\ &= -\hbar^2 r_b \partial_a + \hbar^2 \delta_{ab} r_c \partial_c - \hbar^2 \varepsilon_{acd} \varepsilon_{bef} r_c r_e \partial_{df} = \end{aligned}$$

The last term is symmetric w.r.t. a, b .

Commutators.

$$[\hat{L}_a, \hat{L}_b] = \hat{L}_a \hat{L}_b - \hat{L}_b \hat{L}_a = -\hbar^2 (r_b \partial_a - r_a \partial_b)$$

If $a = b$, the obvious result follows, as an operator commutes with itself. If $a \neq b$,

$$[\hat{L}_a, \hat{L}_b] = i\varepsilon_{abc} \hbar \hat{L}_c,$$

or with a compact “vector notation”, $\hat{\mathbf{L}} \times \hat{\mathbf{L}} = i\hbar \hat{\mathbf{L}}$.

Magnitude of the angular momentum, $|\hat{L}|^2$. Using Cartesian coordinates,

$$|\hat{L}|^2 = 2\hbar^2 r_a \partial_a - \hbar^2 r_b r_b \partial_{aa} + \hbar^2 r_a r_b \partial_{ab}.$$

Details

$$\begin{aligned} |\hat{L}|^2 &= \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2 = \\ &= \hat{L}_a \hat{L}_a = \\ &= -\hbar^2 r_a \partial_a + 3\hbar^2 r_a \partial_a - \hbar^2 (\delta_{ce} \delta_{df} - \delta_{cf} \delta_{de}) r_c r_e \partial_{df} = \\ &= 2\hbar^2 r_a \partial_a - \hbar^2 r_b r_b \partial_{aa} + \hbar^2 r_a r_b \partial_{ab}. \end{aligned}$$

Commutations with magnitude.

$$[\hat{L}^2, \hat{L}_a] = 0$$

Details

The same hold for any integer power of $\hat{L}_a$, like $\hat{L}_a^2$, as it can be easily proved

$$\begin{aligned} [\hat{L}^2, \hat{L}_a^2] &= \hat{L}^2 \hat{L}_a^2 - \hat{L}_a^2 \hat{L}^2 = \\ &= \hat{L}^2 \hat{L}_a \hat{L}_a - \hat{L}_a^2 \hat{L}^2 = \\ &= \hat{L}_a \hat{L}^2 \hat{L}_a - \hat{L}_a \hat{L}_a \hat{L}^2 = \\ &= \hat{L}_a [\hat{L}^2, \hat{L}_a] = 0. \end{aligned}$$

Analogous relations in **classical mechanics**

$$[L_a, L_b] = i\hbar \varepsilon_{abc} \hat{L}_c, \quad \{L_a, L_b\} = \varepsilon_{abc} L_c$$

provides another example of the relation between Poisson brackets in classical mechanics and commutator in quantum mechanics (**Dirac**)

$$\{A, B\} \leftrightarrow \frac{1}{i\hbar} [\hat{A}, \hat{B}]$$

16.1.1 Eigenproblem of angular momentum operators

In this section, the eigenproblems of $\hat{L}^2$ and $\hat{L}_z$ are discussed. Here, all the operators and state functions are written in components using space basis, without explicitly writing the projection $\langle \mathbf{r} |$.

- Using spherical coordinates, the 2π -periodicity of the problem in ϕ implies quantization in the spectral decomposition of $\hat{L}_z$,
-

$\langle \mathbf{r} | \hat{\mathbf{L}}$ using Cartesian coordinates.

$$\ell\psi = \hat{L}_a\psi = -i\hbar \varepsilon_{abc} r_b \partial_c \psi,$$

$$\text{or } \ell\psi = -i\hbar \mathbf{r} \times \nabla \psi.$$

...

$\langle \mathbf{r} | \hat{\mathbf{L}}$ using spherical coordinates.

With the set of spherical coordinates $\psi(r, \theta, \phi)$

$$\begin{aligned} \mathbf{r} \times \nabla \psi &= \mathbf{r} \times \left[\partial_r \psi \hat{\mathbf{r}} + \frac{1}{r} \partial_\theta \psi \hat{\boldsymbol{\theta}} + \frac{1}{r \sin \theta} \partial_\phi \psi \hat{\boldsymbol{\phi}} \right] = \\ &= \partial_\theta \psi \hat{\boldsymbol{\phi}} - \frac{1}{\sin \theta} \partial_\phi \psi \hat{\boldsymbol{\theta}}. \end{aligned}$$

Thus the z -component, with $\hat{\mathbf{z}} \cdot \hat{\boldsymbol{\theta}} = -\sin \theta$, $\hat{\mathbf{z}} \cdot \hat{\boldsymbol{\phi}} = 0$, can be written as

$$\hat{\mathbf{z}} \cdot (\mathbf{r} \times \nabla \psi) = \partial_\phi \psi$$

Spectral decomposition of the z -component operator, $\hat{L}_z$

$$\ell_z \psi_z(r, \theta, \phi) = -i\hbar \partial_\phi \psi_z(r, \theta, \phi)$$

The eigenfunctions read

$$\psi_{z,\ell}(r, \theta, \phi) = A(r, \theta) \exp\left(i \frac{\ell_z}{\hbar} \phi\right).$$

Periodic condition $\psi_z(0) = \psi_z(2\pi)$ implies $\frac{\ell_z}{\hbar} = m_l$, $m_l \in \mathbb{Z}$.

Remark. The constraint $m_l \in \mathbb{Z}$ is just **one of the constraints** on m_l .

In order to keep the notation as intuitive as possible, the eigenvalue ℓ_z and the corresponding eigenstate are denoted as $\{L_z, |L_z\rangle\} = \{\hbar m_l, |m_l\rangle\}$, s.t. $\hat{L}_z |m_l\rangle = \hbar m_l |m_l\rangle$.

Spectral decomposition of commuting operators

Let $\hat{A}|a\rangle = a|a\rangle$, and $\hat{B}|b\rangle = b|b\rangle$. If $0 = [\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$. Pre-multiplying the first one by $\hat{B}$,

$$a\hat{B}|a\rangle = \hat{B}\hat{A}|a\rangle = \hat{A}\hat{B}|a\rangle .$$

So that $\hat{B}|a\rangle$ is an eigenstate of $\hat{A}$ with eigenvalue a .

- If a is an eigenvalue with algebraic multiplicity equal to 1, then $\hat{B}|a\rangle$ must be proportional to the only eigenstate $|a\rangle$ of the operator $\hat{A}$ with eigenvalue a , i.e.

$$\hat{B}|a\rangle = \mu|a\rangle ,$$

and thus $|a\rangle$ is also an eigenstate of $\hat{B}$.

- If a has algebraic multiplicity larger than 1, then $\hat{B}|a_i\rangle$, can be a linear combination of all the eigenvectors $|a_k\rangle$ with eigenvalues $a_k = a$,

$$\hat{B}|a_i\rangle = c_{ik}|a_k\rangle ,$$

for $\forall i$.

...

Relation between $\hat{L}_z$ and $\hat{L}^2$

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$$

As $\hat{L}^2$ commutes with the Cartesian components $\hat{L}_a$, they share the same eigenvalues. The same holds for $\hat{L}_a^2$

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2 ,$$

implies

$$\hat{L}^2 - \hat{L}_z^2 = \hat{L}_x^2 + \hat{L}_y^2 .$$

Let ψ an eigenstate of $\hat{L}^2$, $\hat{L}_z$ with eigenvalues ℓ^2 , ℓ_z , it follows

$$(\hat{L}_x^2 + \hat{L}_y^2) \psi = (\hat{L}^2 - \hat{L}_z^2) \psi = (\ell^2 - \ell_z^2) \psi$$

As $(\ell^2 - \ell_z^2)|\psi|^2 = \psi^* \hat{L}_x^2 \psi = |\hat{L}_x \psi|^2 \geq 0$, it follows that $\ell^2 - \ell_z^2 \geq 0$, and thus $-\ell \leq \ell_z \leq \ell$.

Combining

$$\begin{aligned} [\hat{L}_y, \hat{L}_z] &= \hat{L}_y \hat{L}_z - \hat{L}_z \hat{L}_y = i\hbar \hat{L}_x \\ [\hat{L}_z, \hat{L}_x] &= \hat{L}_z \hat{L}_x - \hat{L}_x \hat{L}_z = i\hbar \hat{L}_y \end{aligned}$$

summing and subtracting (1) and (2) them

$$\begin{aligned} 0 &= -\hbar \hat{L}_x \mp i\hbar \hat{L}_y - i\hat{L}_y \hat{L}_z + i\hat{L}_z \hat{L}_y \pm \hat{L}_z \hat{L}_x \mp \hat{L}_x \hat{L}_z = \\ &= -(\hat{L}_x \pm i\hat{L}_y)(\hbar \pm \hat{L}_z) + \hat{L}_z(i\hat{L}_y \pm \hat{L}_x) , \end{aligned}$$

and thus

$$\hat{L}_z(i\hat{L}_y \pm \hat{L}_x) = (\hat{L}_x \pm i\hat{L}_y)(\hbar \pm \hat{L}_z)$$

or

$$\hat{L}_z(\hat{L}_x \pm i\hat{L}_y) = (\hat{L}_x \pm i\hat{L}_y)(\hat{L}_z \pm \hbar)$$

Applying this operator to the eigenstate ψ

$$\hat{L}_z(\hat{L}_x \pm i\hat{L}_y)\psi = (\hat{L}_x \pm i\hat{L}_y)(\hat{L}_z \pm \hbar)\psi = (\ell_z \pm \hbar)(\hat{L}_x \pm i\hat{L}_y)\psi \quad (16.1)$$

Thus, the state functions $\psi^\pm = (\hat{L}_x \pm i\hat{L}_y)\psi$ are eigenstates of the operator $\hat{L}_z$ with eigenvalues $\ell_z \pm \hbar$. As $\ell_z \in [-\ell, \ell]$, let's call $\ell_{z,min}$ and $\ell_{z,max}$ the minimum and maximum values of ℓ_z . It follows that the operator $\hat{L}^-$ has the smallest eigenvalue equal to $\ell_{z,min} - \hbar$ and $\hat{L}^+$ has the largest eigenvalue equal to $\ell_{z,max} + \hbar$.

Thus, for $\ell_{z,min}$, $\ell_z - \hbar$ can't be an eigenvalue of $\hat{L}_z$, and thus the relation

$$\hat{L}_z(\hat{L}_x - i\hat{L}_y)\psi = (\ell_{z,min} - \hbar)(\hat{L}_x - i\hat{L}_y)\psi$$

implies that $(\hat{L}_x - i\hat{L}_y)\psi$ can't be an eigenfunction of $\hat{L}_z$, and thus it must be in the kernel or $\hat{L}_z$, i.e.

$$0 = \hat{L}_z(\hat{L}_x - i\hat{L}_y)\psi_{\ell_{z,min}} = (\hat{L}_x - i\hat{L}_y)(\hat{L}_z + \hbar)\psi_{\ell_{z,min}} = (\ell_{z,min} + \hbar)(\hat{L}_x - i\hat{L}_y)\psi_{\ell_{z,min}}$$

A similar relation holds for the largest eigenvalue

$$0 = \hat{L}_z(\hat{L}_x + i\hat{L}_y)\psi_{\ell_{z,max}} = (\hat{L}_x + i\hat{L}_y)(\hat{L}_z - \hbar)\psi_{\ell_{z,max}} = (\ell_{z,max} - \hbar)(\hat{L}_x + i\hat{L}_y)\psi_{\ell_{z,max}}$$

Applying $\hat{L}_x + i\hat{L}_y$ to the first equation, the operator becomes

$$(\hat{L}_x + i\hat{L}_y)(\hat{L}_x - i\hat{L}_y) = \hat{L}_x^2 + \hat{L}_y^2 - i[\hat{L}_x, \hat{L}_y] = \hat{L}^2 - \hat{L}_z^2 + \hbar\hat{L}_z.$$

Applying $\hat{L}_x - i\hat{L}_y$ to the second equation, the operator becomes

$$(\hat{L}_x - i\hat{L}_y)(\hat{L}_x + i\hat{L}_y) = \hat{L}_x^2 + \hat{L}_y^2 + i[\hat{L}_x, \hat{L}_y] = \hat{L}^2 - \hat{L}_z^2 - \hbar\hat{L}_z.$$

Thus, the 2 equations are, for a given value of ℓ

$$0 = (\ell_{z,min} + \hbar)(\ell^2 - \ell_{z,min}^2 + \hbar\ell_{z,min})$$

$$0 = (\ell_{z,max} - \hbar)(\ell^2 - \ell_{z,max}^2 - \hbar\ell_{z,max})$$

Either $\ell_{z,min,max} = \mp\hbar$, or

$$0 = \ell^2 - \ell_{z,min}^2 + \hbar\ell_{z,min}$$

$$0 = \ell^2 - \ell_{z,max}^2 - \hbar\ell_{z,max},$$

and subtracting $0 = (\ell_{z,max} + \ell_{z,min})(\ell_{z,max} - \ell_{z,min} + \hbar)$. As the content of the second bracket is always positive, then $\ell_{z,min} = -\ell_{z,max}$.

Following (16.1), if $|\psi\rangle$ is the eigenvector of $\hat{L}_z$ with eigenvalue ℓ_z , the wave functions $\hat{L}^\mp|\psi\rangle$ are eigenvectors of $\hat{L}_z$ with eigenvalues $\ell_z \mp \hbar$. Thus, the set

$$\{\ell_{z,min}, \ell_{z,min} + \hbar, \ell_{z,min} + 2\hbar, \dots, \ell_{z,max} - \hbar, \ell_{z,max}\} = \{-m_{l,max}\hbar, \dots, m_{l,max}\hbar\}$$

contains possible values of the eigenvalues of $\hat{L}_z$.

$$\lambda(L_z) = \hbar m_l \quad m_l \in$$

Now, for $\ell_{z,min} = -m_{l,max}\hbar$, and $\ell_{z,max} = m_{l,max}\hbar$, the corresponding eigenvalues ℓ^2 are

$$\ell^2 = \hbar^2 m_{l,max} (m_{l,max} + 1).$$

16.2 Spin

16.2.1 Properties

In analogy with the spatial angular momentum, and using Cartesian coordinates

$$\begin{aligned}[S_a, S_b] &= i\hbar \varepsilon_{abc} S_c \\ [S_a, S^2] &= 0.\end{aligned}$$

As the operators S_z and S^2 commute, they share the same eigenvectors $|s, m_s\rangle$ (all the eigvecs?) and their eigenproblems read

$$\begin{aligned}S_z |s, m_s\rangle &= \hbar m_s |s, m_s\rangle \\ S^2 |s, m_s\rangle &= \hbar^2 s(s+1) |s, m_s\rangle,\end{aligned}$$

The value of s depends on the system of interest: for an electron $s = \frac{1}{2}$. The value of the spin projection quantum number m_s belongs to $m_s \in \{-s, -s+1, \dots, s-1, s\}$. For an electron, $m_s \in \{-\frac{1}{2}, \frac{1}{2}\}$.

16.3 Total Angular Momentum

State function

$$|\Psi\rangle = |\psi\rangle \otimes |\sigma\rangle \quad \in \quad \mathcal{H} = \mathcal{H}_{space} \otimes \mathcal{H}_{spin}.$$

Total angular momentum operator

$$\hat{\mathbf{J}} = \hat{\mathbf{L}} + \hat{\mathbf{S}} = \hat{\mathbf{L}} \otimes \hat{\mathbf{1}}_{spin} + \hat{\mathbf{1}}_{space} \otimes \hat{\mathbf{S}}$$

Dot product of $\hat{\mathbf{L}}$ and $\hat{\mathbf{S}}$,

$$\hat{\mathbf{L}} \cdot \hat{\mathbf{S}} = L_x \otimes S_x + L_y \otimes S_y + L_z \otimes S_z.$$

Magnitude square of $\hat{\mathbf{J}}$, $\hat{J}^2$,

$$\begin{aligned}\hat{J}^2 &= (\hat{\mathbf{L}} + \hat{\mathbf{S}}) \cdot (\hat{\mathbf{L}} + \hat{\mathbf{S}}) = \\ &= \hat{L}^2 \otimes \hat{\mathbf{1}} + 2\hat{\mathbf{L}} \cdot \hat{\mathbf{S}} + \hat{\mathbf{1}} \otimes \hat{S}^2.\end{aligned}$$

Orbital angular momentum.

$$\begin{aligned}\hat{L}^2 |\ell, m_\ell\rangle &= \hbar^2 \ell(\ell+1) |\ell, m_\ell\rangle \quad , \quad \ell \in \{0, 1, 2, \dots\} \\ \hat{L}_z |\ell, m_\ell\rangle &= \hbar m_\ell |\ell, m_\ell\rangle \quad , \quad m_\ell \in \{-\ell, -\ell+1, \dots, \ell-1, \ell\}\end{aligned}$$

Spin angular momentum.

$$\begin{aligned}\hat{S}^2 |s, m_s\rangle &= \hbar^2 s(s+1) |s, m_s\rangle \quad , \quad s \in \left\{0, \frac{1}{2}, 1, \dots\right\} \\ \hat{S}_z |s, m_s\rangle &= \hbar m_s |s, m_s\rangle \quad , \quad m_s \in \{-s, -s+1, \dots, s-1, s\}\end{aligned}$$

For an electron $s = \frac{1}{2}$, so that $m_s \in \{-\frac{1}{2}, \frac{1}{2}\}$, and thus, without explicitly writing s in the eigenvectors - being that a given parameter -,

$$\begin{aligned}\hat{S}^2 |m_s\rangle &= \frac{3}{4} \hbar^2 |m_s\rangle \\ \hat{S}_z |m_s\rangle &= \hbar m_s |m_s\rangle \quad , \quad m_s \in \left\{-\frac{1}{2}, \frac{1}{2}\right\}\end{aligned}$$

Total angular momentum. The z -component of the total angular momentum, and its magnitude squared are defined as

$$\begin{aligned}\hat{J}^2|j, m_j\rangle &= \hbar^2 j(j+1)|j, m_j\rangle \quad , \quad j \in \left\{0, \frac{1}{2}, 1, \dots\right\} \\ \hat{J}_z|j, m_j\rangle &= \hbar m_j|j, m_j\rangle \quad , \quad m_j \in \{-j, -j+1, \dots, j-1, j\}\end{aligned}$$

16.3.1 Eigenvalue problem of the total angular momentum for an electron

Commutation of operators. The operators $\hat{J}^2$, $\hat{J}_z$, $\hat{L}^2$, $\hat{S}^2$ commute, and thus they share common eigenvectors

$$|j, m_j, \ell, s\rangle .$$

The operator $\hat{J}_z$ commutes with $\hat{L}^2$ and $\hat{S}^2$, $[\hat{J}_z, \hat{L}^2] = 0$, $[\hat{J}_z, \hat{S}^2] = 0$.

Proof

$$\begin{aligned}[\hat{J}_z, \hat{L}^2] &= \hat{J}_z \hat{L}^2 - \hat{L}^2 \hat{J}_z = \\ &= \hat{L}_z \hat{L}^2 + \hat{S}_z \hat{L}^2 - \hat{L}^2 \hat{L}_z - \hat{L}^2 \hat{S}_z = \\ &= \underbrace{[\hat{L}_z, \hat{L}^2]}_{=0} + \underbrace{[\hat{S}_z, \hat{L}^2]}_{=0} = 0 .\end{aligned}$$

The last term holds because $\hat{S}_z$ and $\hat{L}^2$ acts on two different sub-spaces, or explicitly for any $|\Psi\rangle = |\psi\rangle \otimes |\sigma\rangle$,

$$\begin{aligned}\hat{S}_z \hat{L}^2 |\Psi\rangle &= (\hat{\mathbf{1}} \otimes \hat{S}_z)(\hat{L}^2 \otimes \hat{\mathbf{1}})|\psi\rangle \otimes |\sigma\rangle = \\ &= (\hat{\mathbf{1}} \otimes \hat{S}_z)(\hat{L}^2 |\psi\rangle \otimes |\sigma\rangle) = \\ &= \hat{L}^2 |\psi\rangle \otimes \hat{S}_z |\sigma\rangle ,\end{aligned}$$

and $\hat{L}^2 \hat{S}_z |\Psi\rangle$ provides the same result for any $|\Psi\rangle = |\psi\rangle \otimes |\sigma\rangle$.

The operator $\hat{L}_z$ doesn't commute with $\hat{J}^2$, $[\hat{L}_z, \hat{J}^2] = -2i\hbar \hat{\mathbf{z}} \cdot \hat{\mathbf{L}} \times \hat{\mathbf{S}}$.

Proof

$$\begin{aligned}[\hat{L}_z, \hat{J}^2] &= \hat{L}_z \hat{J}^2 - \hat{J}^2 \hat{L}_z = \\ &= \hat{L}_z (\hat{L}^2 + \hat{S}^2 + 2\hat{\mathbf{L}} \cdot \hat{\mathbf{S}}) - (\hat{L}^2 + \hat{S}^2 + 2\hat{\mathbf{L}} \cdot \hat{\mathbf{S}}) \hat{L}_z = \\ &= \underbrace{[\hat{L}_z, \hat{L}^2]}_{=0} + \underbrace{[\hat{L}_z, \hat{S}^2]}_{=0} + 2[\hat{L}_z, \hat{L}_a] \hat{S}_a = \\ &= 2i\hbar \hat{L}_y \hat{S}_x - 2i\hbar \hat{L}_x \hat{S}_y = \\ &= -2i\hbar \hat{\mathbf{z}} \cdot \hat{\mathbf{L}} \times \hat{\mathbf{S}} ,\end{aligned}$$

as $[\hat{L}_z, \hat{L}_x] = i\hbar \hat{L}_y$, $[\hat{L}_y, \hat{L}_z] = i\hbar \hat{L}_x$, $[\hat{L}_z, \hat{L}_z] = 0$,

z -component operators.

$$\begin{aligned}\hat{J}_z |\Psi_{m_\ell, m_s}\rangle &= (\hat{L}_z + \hat{S}_z) |m_\ell\rangle \otimes |m_s\rangle = \\ &= \hat{L}_z |m_\ell\rangle \otimes |m_s\rangle + \hat{S}_z |m_\ell\rangle \otimes |m_s\rangle = \\ &= \hbar m_\ell |m_\ell\rangle \otimes |m_s\rangle + \hbar m_s |m_\ell\rangle \otimes |m_s\rangle = \\ &= \hbar (m_\ell + m_s) |m_\ell\rangle \otimes |m_s\rangle = \\ &= \hbar m_j |m_\ell\rangle \otimes |m_s\rangle ,\end{aligned}$$

and thus the relation $m_j = m_\ell + m_s$ for every pair of eigenvectors $|m_\ell\rangle$ and $|m_s\rangle$ of the operators $\hat{L}_z$, and $\hat{S}_z$. It also follows, that the extreme values of m_j are $-j$ and $j = \ell + s$.

DENSITY OPERATOR

Let's start from an example of the preparation of a system of subsystems in pure states $|\psi_i\rangle$, with probability p_i , $\sum_i p_i = 1$. Let M an observable, with the corresponding Hermitian operator $\hat{M}$ with discrete values m_μ and corresponding states $|m_\mu\rangle$.

Now, the probability of measuring the value m_μ for a system in state $|\psi_i\rangle$, i.e. the conditional probability $p(m = m_\mu || \psi) = |\psi_i\rangle$ reads

$$p(m = m_\mu || \psi) = |\psi_i\rangle = |\langle m_\mu | \psi_i \rangle|^2 = \langle \psi_i | m_\mu \rangle \langle m_\mu | \psi_i \rangle .$$

The probability of measuring m from the ensemble is the marginal probability,

$$\begin{aligned} p(m = m_\mu) &= \sum_i p(m = m_\mu || \psi) p(|\psi\rangle = |\psi_i\rangle) = \\ &= \sum_i \langle \psi_i | m_\mu \rangle \langle m_\mu | \psi_i \rangle p_i = \\ &= \sum_i \langle \psi_i | \hat{\Pi}_{m_\mu} | \psi_i \rangle p_i , \end{aligned}$$

having introduced the orthogonal projector over the μ^{th} eigenfunction of the operator $\hat{M}$, i.e. $\hat{\Pi}_{m_\mu} = |m_\mu\rangle \langle m_\mu|$.

Let's define the **density operator** as

$$\hat{\rho} = \sum_i p_i |\psi_i\rangle \langle \psi_i| .$$

Using an orthonormal basis $\{|e_k\rangle\}_k$, it's easy to show that

$$\begin{aligned} \text{Tr}(\hat{\rho} \hat{\Pi}_{m_\mu}) &:= \sum_k \langle e_k | \hat{\rho} \hat{\Pi}_{m_\mu} | e_k \rangle = \\ &= \sum_{k,i} p_i \langle e_k | \psi_i \rangle \langle \psi_i | \hat{\Pi}_{m_\mu} | e_k \rangle = \\ &= \sum_i p_i \langle \psi_i | \hat{\Pi}_{m_\mu} \underbrace{\sum_k |e_k\rangle \langle e_k|}_{=\hat{\mathbf{I}}} | \psi_i \rangle = \\ &= \sum_i p_i \langle \psi_i | \hat{\Pi}_{m_\mu} | \psi_i \rangle = \\ &= p(m = m_\mu) . \end{aligned}$$

Expected value.

$$\begin{aligned}
 \mathbb{E}[M] &= \sum_{\mu} m_{\mu} p(m = m_{\mu}) = \\
 &= \sum_i \sum_{\mu} m_{\mu} p_i \langle \psi_i | m_{\mu} \rangle \langle m_{\mu} | \psi_i \rangle = \\
 &= \sum_i \sum_{\mu} p_i \langle \psi_i | \hat{M} | m_{\mu} \rangle \langle m_{\mu} | \psi_i \rangle = \\
 &= \sum_{\mu} \langle m_{\mu} | \sum_i p_i | \psi_i \rangle \langle \psi_i | \hat{M} | m_{\mu} \rangle = \\
 &= \sum_{\mu} \langle m_{\mu} | \hat{\rho} \hat{M} | m_{\mu} \rangle = \\
 &= \text{Tr}(\hat{\rho} \hat{M}) .
 \end{aligned}$$

Trace of an operator

Choosing a set of **orthogonal unit vectors** $|\psi_i\rangle$, the trace of an operator $\hat{A}$ can be defined as

$$\text{Tr}(\hat{A}) = \sum_i \langle \psi_i | \hat{A} | \psi_i \rangle .$$

Properties

•

$$\text{Tr}(\hat{A} |\psi_j\rangle \langle \psi_j|) = \sum_i \langle \psi_i | \hat{A} | \psi_j \rangle \underbrace{\langle \psi_j | \psi_i \rangle}_{\delta_{ij}} = \langle \psi_j | \hat{A} | \psi_j \rangle .$$

•

$$\text{Tr}(\hat{A}) = \text{Tr}\left(\hat{A} \sum_j |\psi_j\rangle \langle \psi_j|\right) = \sum_{i,j} \langle \psi_i | \hat{A} | \psi_j \rangle \underbrace{\langle \psi_j | \psi_i \rangle}_{\delta_{ij}} = \sum_i \langle \psi_i | \hat{A} | \psi_i \rangle .$$

Choosing a **generic basis** $\{|\phi_k\rangle\}_k$, the k^{th} vector of this basis can be written as a linear combination of the vectors of a unit orthogonal basis,

As done in Differential Geometry, a reciprocal basis $\{|\phi^j\rangle\}_j$ exists s.t. $\langle \phi^j | \phi_k \rangle = \delta_k^j$. Defining the components of the metric tensor $g_{ij} = \langle \phi_i | \phi_j \rangle$, $g^{ij} = \langle \phi^i | \phi^j \rangle$ the relations between the original basis and its reciprocal follows

$$|\phi^i\rangle = g^{ij} |\phi_j\rangle \quad , \quad |\phi_i\rangle = g_{ij} |\phi^j\rangle .$$

The identity operator can be written as $\hat{\mathbf{1}} = \sum_j |\phi_j\rangle \langle \phi_j| = \sum_j |\phi^j\rangle \langle \phi_j|$, as

$$|v\rangle = \sum_i v^i |\phi_i\rangle = \sum_{i,j} v^i |\phi_j\rangle \underbrace{\langle \phi^j | \phi_i \rangle}_{=\delta_i^j} = \sum_j |\phi_j\rangle \langle \phi^j | \sum_i v^i |\phi_i\rangle = \hat{\mathbf{1}} |v\rangle .$$

The vectors of the basis can be written as a linear combination of the vectors of an orthonormal basis $\{|\psi_k\rangle\}_k$,

$$|\phi_i\rangle = \sum_k T_i^{\ k} |\psi_k\rangle .$$

The dual vectors are written as $|\phi^j\rangle = \sum_l R^{jl} |\psi_l\rangle$. As the dual vectors should be orthogonal to the vectors of the original basis, it follows

$$\delta_i^j = \langle \phi^j | \phi_i \rangle = \sum_{k,l} (R^{jl})^* T_i^{\ k} \underbrace{\langle \psi_l | \psi_k \rangle}_{=\delta_{lk}} = \sum_k (R^{jk})^* T_i^{\ k} ,$$

i.e. the transformation matrix $\mathbf{R}$ of the vectors of the reciprocal basis is the inverse of the adjoint of the matrix $\mathbf{T}$, i.e.

$$\begin{aligned}(\mathbf{T})_{ik} &= T_i^k \\ (\mathbf{R})_{jk} &= R^{ij} \\ (\mathbf{T}^{-1})_{kj} &= (R^{jk})^* = (\mathbf{R}^H)_{kj}\end{aligned}$$

Using matrix formalism, the definition of the inverse matrix gives $\mathbf{I} = \mathbf{T}\mathbf{R}^H = \mathbf{R}^H\mathbf{T}$. (**todo** what happens for infinite dimensional spaces?)

Thus, the relation

$$\sum_i \langle \phi^i | \hat{A} | \phi_i \rangle = \sum_{l,k} \underbrace{\sum_i (R^{il})^* T_i^k}_{=(\mathbf{R}^H \mathbf{T})_{lk} = \delta^{lk}} \langle \psi_l | \hat{A} | \psi_k \rangle = \sum_k \langle \psi_k | \hat{A} | \psi_k \rangle = \text{Tr}(\hat{A}) .$$

todo Uncomment or delete (more likely)

Example 17.1 (Difference between mixed states and pure state in superposition)

Pure state in superposition of two orthogonal states $|\psi_1\rangle, |\psi_2\rangle$,

$$|\psi\rangle = a_1|\psi_1\rangle + a_2|\psi_2\rangle ,$$

with $|a_1|^2 + |a_2|^2 = 1$. A system prepared in this pure state with probability $p = 1$ has density operator

$$\hat{\rho} = |\psi\rangle\langle\psi| = |a_1|^2|\psi_1\rangle\langle\psi_1| + a_1a_2^*|\psi_1\rangle\langle\psi_2| + a_2a_1^*|\psi_2\rangle\langle\psi_1| + |a_2|^2|\psi_2\rangle\langle\psi_2| .$$

If $a = b = \frac{1}{\sqrt{2}}$, the components of the density operator in the basis $\{|\psi_1\rangle, |\psi_2\rangle\}$ are

$$\begin{bmatrix} |a_1|^2 & a_1a_2^* \\ a_1^*a_2 & |a_2|^2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{bmatrix} .$$

Mixed state. An ensemble prepared in state $|\psi_1\rangle$ with probability p_1 , $p_2 = 1 - p_1$ has density operator

$$\hat{\rho} = p_1|\psi_1\rangle\langle\psi_1| + p_2|\psi_2\rangle\langle\psi_2| ,$$

whose components in the $\{|\psi_1\rangle, |\psi_2\rangle\}$ basis are

$$\begin{bmatrix} p_1 & 0 \\ 0 & p_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix} .$$

DECOHERENCE

18.1 Example: two-dimensional system interacting with a large-dimensional environment

18.1.1 State of the system

A 2-dimensional system is interacting with a large-dimensional environment. The state of the 2-dimensional system is identified by the wave function

$$|\Psi_S\rangle = c_0|0\rangle + c_1|1\rangle ,$$

the state of the environment by the wave function $|\Psi_E\rangle$, and the state of the whole system (S+E) by the tensor product of the two,

$$|\Psi_{S+E}\rangle = |\Psi_S\rangle \otimes |\Psi_E\rangle .$$

18.1.2 Evolution of the system

The evolution of the system is governed by the Schrodinger equation, with the Hamiltonian of the whole system

$$i\hbar \frac{d}{dt} |\Psi_{S+E}\rangle = \hat{H}_{S+E} |\Psi_{S+E}\rangle ,$$

whose evolution can be represented by an unitary operator $\hat{U}_{S+E;t,0}$,

$$|\Psi_{S+E}\rangle_t = U_{S+E;t,0} |\Psi_{S+E}\rangle_0 = \dots = \alpha|0\rangle \otimes |\Psi_{E;0}\rangle + \beta|1\rangle \otimes |\Psi_{E;1}\rangle$$

Density operator reads

$$\hat{\rho} := |\Psi_{S+E}\rangle \langle \Psi_{S+E}| .$$

Tracing out the environment states,

$$\begin{aligned} \hat{\rho}_E &= \text{Tr}_E (\hat{\rho}) = \\ &= \sum_k (\alpha|0\rangle \langle \phi_k| \Psi_{E;0}\rangle + \beta|1\rangle \langle \phi_k| \Psi_{E;1}\rangle) (\alpha^* \langle 0| \langle \Psi_{E;0}| \phi_k\rangle + \beta^* \langle 1| \langle \phi_k| \Psi_{E;1}\rangle) = \\ &= |\alpha|^2 |0\rangle \langle 0| \underbrace{\langle 0| \langle \Psi_{E;0}| \sum_k |\phi_k\rangle \langle \phi_k| \Psi_{E;0}\rangle}_{=\hat{\mathbf{1}}} + \alpha\beta^* |0\rangle \langle 1| \underbrace{\langle 0| \langle \Psi_{E;0}| \sum_k |\phi_k\rangle \langle \phi_k| \Psi_{E;1}\rangle}_{=\hat{\mathbf{1}}} + \\ &\quad + \alpha^*\beta |1\rangle \langle 0| \underbrace{\langle 1| \langle \Psi_{E;1}| \sum_k |\phi_k\rangle \langle \phi_k| \Psi_{E;0}\rangle}_{=\hat{\mathbf{1}}} + |\beta|^2 |1\rangle \langle 1| \underbrace{\langle 1| \langle \Psi_{E;1}| \sum_k |\phi_k\rangle \langle \phi_k| \Psi_{E;1}\rangle}_{=\hat{\mathbf{1}}} = \\ &= |\alpha|^2 |0\rangle \langle 0| + \alpha\beta^* \langle \Psi_{E;0}| \Psi_{E;1}\rangle |0\rangle \langle 1| + \alpha^*\beta \langle \Psi_{E;1}| \Psi_{E;0}\rangle |1\rangle \langle 0| + |\beta|^2 |1\rangle \langle 1| . \end{aligned}$$

The components of the reduced density operator in the system basis are

$$\begin{bmatrix} |\alpha|^2 & \alpha\beta^* \langle \Psi_{E;0} | \Psi_{E;1} \rangle \\ \alpha^* \beta \langle \Psi_{E;1} | \Psi_{E;0} \rangle & |\beta|^2 \end{bmatrix}.$$

18.1.3 Decoherence time

todo *check and uncomment*

HISTORICAL APPROACH TO QUANTUM MECHANICS

References

- JK⁰ YouTube channel

19.1 Topics

Atomic models.

- 1891, J.J.Thomson find electrons in cathodic rays. He found out that cathodic rays are made of negatively charged particles¹, deflected by a transverse electromagnetic field. He:
 - evaluated the ratio $\frac{\text{mass}}{\text{el.charge}}$ of these particles. See [Scoperta dell'elettrone - J.J.Thomson](#)
 - proposed the pudding atomic model, with negatively charged particles in a positive cloud.
- 1909, Rutherford scattering experiment and Rutherford atomic model: electrons around a compact nucleus
- 1913, Bohr...
- 1916, Sommerfeld
 - relativistic model
 - elliptial orbits
 - introduced another quantum number and degeneracy
 - describe energy level split by magnetic field (Zeeman effect)
- 1925, Heisenberg and Schrodinger first theories about QM; Pauli solution of the H atom system

Spectroscopy.

- 1707, Newton "Optics" described his experiments about diffraction of light with glass prisms
- In the second half of the XIX century, experimental studies on the emission/absorption spectra in chemistry. Each element showed characteristics black lines in otherwise continuous absorption spectra (and corresponding bright lines in emission spectra, "hot spectra")
- Empirical laws for the refractive index followed:

¹ While J.J.Thomson won the Nobel prize in Physics (1906) for having found the first sub-atomic particle, the electron, his son G.P.Thomson won the Noble prize (1937) for its discovery of the diffraction of electrons by crystals in 1927, thus showing wave behavior. Particle-wave duality never ends...

- 1836, Cauchy proposed a relation between the refractive index n of a material and the wave-length of the radiation

$$n^2 = a + \frac{b}{\lambda^2} + \frac{c}{\lambda^4} + \dots,$$

with material-dependent parameters $a, b, c, \dots$. This model proposed $n(\lambda)$ as a continuous function

- 1870, Christiansen observed anomalous dispersion, with apparently discontinuous refractive index at wave lengths that are characteristics of the element under investigation
- 1872, Sellmeier proposed that anomalous dispersion should be described in the context of light-matter interaction, in terms of resonance modes of the matter and proposed his law for the refractive index

$$n^2 - 1 = \sum_k \frac{a_k \lambda_k^2}{\lambda^2 - \lambda_k^2}$$

- 1875, Helmholtz modified Sellmeier formula, introducing a (wrong, this was not a dampening factor, as the one shown by Drude-Lorentz model) dampening factor.
- *Drude-Lorentz model*. Following the idea of Sellmeier, P. Drude (1900) and then H. Lorentz (1905) proposed a theoretical model for the refractive index of a medium, using a second-order damped oscillator, forced by the electric field of an incoming electromagnetic wave, and evaluated the transmitted wave as the sum of the incident wave and the one generated by the oscillating charges of the matter.
- Generalization of the Drude-Lorentz model for systems with multiple modes became the general semi-empirical classical model for light dispersion

$$n(\omega) - 1 = \frac{q^2}{2\varepsilon_0 m} \sum_k \frac{N_k}{\omega_k^2 - \omega^2 + i\gamma_k \omega_k}$$

or

$$n^2(\omega) - 1 \simeq 2\delta.$$

- Dispersion becomes quantum:
 - 1905, Einstein explanation of photoelectric effect
 - 1906-1908, **Ladenburg** and Loria began experiments about spectral lines, anomalous dispersion, and the microstructure of the light-matter interaction. They found a way to measure the number density of the electrons responsible for spectral lines in absorption spectra in H atom (H_α line only, while they had not enough accuracy for other weaker lines). Only few electrons resonate while the majority appear to be unaffected. They then focused on Na for better resolution. They came back to H few years later (1911), to find $\frac{N_{H_\alpha}}{N_H} \sim \frac{1}{50.000}$, $\frac{N_{H_\beta}}{N_H} \sim \frac{1}{200.000}$.
 - 1913, Bohr model (incompatible with classical mechanics, no e^- fast collapse on the nucleus, due to classical radiation):
 - * circular stable orbits with quantized angular momentum $n\hbar$
 - * lines are generated by quantum jumps, with $\nu = \frac{\Delta E}{h}$
 - * results compatible with the empirical Rydberg formula (1888)
 - Ladenburg: how do both Bohr model and Drude-Lorentz model work so well in atomic spectra?
 - 1914, H. Kohn: experimental confirmed radiation emitted by incandescent bodies can be treated as thermal radiation. Ladenburg then used thermodynamics to explain thermal radiation.
 - Still Ladenburg

- * 1914, procedure to compute the density of dispersion electrons
- * 1921, quantum counterpart of the classical dispersion, and link with Einstein coefficients (1916)
 - the **physical atom is replaced by a representation of oscillators**: only frequencies and intensities of spectral lines can be observed. They abandoned the idea of building a “physical model of the atom” following the electron along an orbit, and built a model that’s just compatible with the physical quantities they can measure
- 1924, **Kramers** (with theoretical derivation from Born, 1924, and Kramers and Heisenberg, 1925):
 - * rule for **polarization** with transition frequencies (both to higher and lower energy levels, modeling both emission and absorption)
 - * Kramers formula and the idea of virtual oscillators are used by Heisenberg in his formulation of matrix mechanics: this formula set the origin for the non-commutativity of physical quantities and, thus, Heisenberg uncertainty principle
 - * tested by Ladenburg, 1928

Radiation and Light-matter interaction.

- 1896, Wien’s law
- 1900, Planck described the **continuous spectrum** of black-body radiation, introducing of h , and the idea that light radiation with frequency ν have discrete amount of energy, $h\nu$. He first introduced h as a trick for matching experimental results of the spectrum $\rho(\nu, T)$; a model of statistical mechanics of radiation with discrete levels of energy followed, see *Black-body radiation using Bose-Einstein statistics*, even though that was the original way Planck first found a theoretical model
- 1905, Einstein explained the photoelectric effects using Planck constant h .
- 1916, Einstein
 - derived Planck’s law from Bohr’s atomic model, using a (quantum) statistical approach to the interaction of atoms (and their electrons) and electromagnetic radiation in a box. He postulated the existence of three processes, introducing proportionality **Einstein coefficients**, for transition probability (of a e^-) between states: *absorption* (rate: $B_{kj}N_k\rho(\nu)$) *spontaneous emission* (rate: $A_{jk}N_j$), and **stimulated emission** (rate: $B_{jk}N_j\rho(\nu)$). Stimulated emissions was found to be necessary in order for this model to be compatible with Planck’s law $\rho \propto \frac{1}{e^{h\nu/kT}-1}$ for the system at thermal equilibrium (and not with Wien’s law $\rho(\nu) \propto e^{-h\nu/kT}$, the law one would get without stimulated emission)
 - Einstein couldn’t calculate his coefficients but he was positive that they could be computed once a theory about the atom had been established
 - recognized that light quanta have momentum, $p = \hbar k = h\frac{\nu}{c}$, then found by Compton (1923) in experiments about X-rays scattering
 - in 1927, Dirac calculated Einstein coefficients (The Quantum Theory of Emission and Absorption of Radiation)

Origin of quantum mechanics.

- **Matrix mechanics.** (Heisenberg, Born, Jordan), three papers in 1925
- **Ondulatory mechanics.** (Schrodinger)
 - 1923, L.de Broglie: idea of guiding wave for matter particles
 - 1925, E.Schrodinger: wave equation. If matter behaves as a wave, a wave equation should be the governing equation of the system
 - 1923-1927, C.Davisson and L.Germer found a diffraction pattern in an experiment about electron scattering
- **Equivalence.** (Dirac)

19.2 Timeline

19.3 Light dispersion

19.3.1 First experiences

19.3.2 Classical dispersion

Drude-Lorentz model

...

$$n(\omega) - 1 = \frac{q^2 N}{2\varepsilon_0 m} \frac{1}{\omega_0^2 - \omega^2 + i\gamma\omega}$$

Remark. The refractive index n is a complex number, with

$$\begin{aligned} \operatorname{re}\{n\}(\omega) &= 1 + \Delta n_0 \omega_0^2 \frac{\omega_0^2 - \omega^2}{(\omega_0^2 - \omega^2)^2 + \gamma^2 \omega^2} \\ \operatorname{im}\{n\}(\omega) &= -\Delta n_0 \omega_0^2 \frac{\gamma \omega}{(\omega_0^2 - \omega^2)^2 + \gamma^2 \omega^2} \leq 0, \quad \forall \omega \geq 0 \end{aligned}$$

Here the condition $\omega \geq 0$ is required by the condition that the incoming wave travels from left to right.

Thus, a travelling wave

$$\varphi(x, t) = e^{i(\omega t - kx)} = e^{i\omega(t - \frac{x}{c})} = e^{i\omega(t - n \frac{x}{c_0})},$$

thus contains both an oscillating and a dampening contribution by $n = \operatorname{re}\{n\} + i \operatorname{im}\{n\}$,

$$\varphi(x, t) = e^{\frac{i\omega}{c_0} \operatorname{im}\{n(\omega)\} x} e^{i\omega(t - \operatorname{re}\{n(\omega)\} \frac{x}{c_0})}$$

As the imaginary part of the refractive index is negative for all the positive frequencies, then the real exponential represents a damping term. The minimum value of $\operatorname{im}\{n\}(\omega)$ for a slightly damped second-order oscillator occurs approximately at the natural frequency, $\omega_{\max \text{ damp}} \simeq \omega_0$. This represents an **absorption of radiation** by the matter.

Remark. Drude-Lorentz model in the form $n - 1 = \delta$ is compatible with empirical formulas in the form $n^2 - 1 = \tilde{\delta}$, as the correction is “small enough” for a linear approximation

$$n^2 = (1 + \delta)^2 \simeq 1 + 2\delta.$$

Einstein

Ladenburg

Ladenburg wrote the energy balance at thermodynamical equilibrium, from transitions between each pair of states i , $k > i$. The total amount of energy emitted by N_k molecules in the state k transitioning to a lower state i is

$$J = \underbrace{h\nu_{ik}}_{\Delta E_{ik}} N_k (A_{ki} + B_{ki} u(\nu_{ik}))$$

At thermal equilibrium, this energy is equal to the energy absorbed by N_i molecules in the state i ,

$$A = h\nu_{ik}N_iB_{ik}u(\nu_{ik}) .$$

Exploiting the relations between Einstein coefficients (), Ladenburg wrote the emitted and absorbed energy as a function of the spontaneous emission coefficient A_{ki} (allowing to relate dispersion with other phenomena involving emission),

$$J = A = h\nu_{ik}u(\nu_{ik})N_iB_{ik} =$$

19.4 Light matter interaction

19.4.1 Einstein: quantum theory of radiation interacting with matter

Three mechanisms, whose probabilities are

- Spontaneous emission of radiation, for a photon going from level j to level i ($E_j > E_i$),

$$\left(\frac{dn_{ij}}{dt} \right)_{spont} = A_{ji}n_j ,$$

- Stimulated emission of radiation, for a photon going from level j to level i ($E_j > E_i$),

$$\left(\frac{dn_{ij}}{dt} \right) = B_{ji}n_j\rho(\nu_{ji}) ,$$

with $\nu_{ji} = \frac{E_j - E_i}{h}$, and $\rho(\nu)$ the density of radiation in the system at frequency ν

- Absorption, for a photon going from level j to level i ($E_j > E_i$),

$$\left(\frac{dn_{ij}}{dt} \right) = -B_{ij}n_i\rho(\nu_{ij}) .$$

The overall rate reads

$$\frac{dn_i}{dt} = \sum_{j, E_j > E_i} \frac{dn_{ij}}{dt} = \sum_{j, E_j > E_i} \{A_{ji}n_j + B_{ji}n_j\rho(\nu_{ij}) - B_{ij}n_i\rho(\nu_{ij})\}$$

At thermodynamic equilibrium between all the states

$$0 = \frac{dn_{ij}}{dt} = A_{ji}n_j + B_{ji}n_j\rho(\nu_{ij}) - B_{ij}n_i\rho(\nu_{ij})$$

Using Boltzmann distribution

$$\frac{n_i}{n} = \frac{g_i e^{-\frac{E_i}{kT}}}{Z} ,$$

the expression of the radiation density follows

$$\begin{aligned} \rho(\nu_{ij}) &= \frac{A_{ji}n_j}{n_iB_{ij} - n_jB_{ji}} = \\ &= \frac{A_{ji}g_j}{B_{ij}g_i \exp(-E_i/kT) - \frac{g_jB_{ji}}{g_iB_{ij}} \exp(-E_j/kT)} = \\ &= \frac{A_{ji}g_j}{B_{ij}g_i \exp((E_j - E_i)/kT) - \frac{g_jB_{ji}}{g_iB_{ij}}} = \\ &= \frac{A_{ji}g_j}{B_{ij}g_i \exp(h\nu_{ji}/kT) - \frac{g_jB_{ji}}{g_iB_{ij}}} . \end{aligned}$$

Comparing with Planck's law (with the **radiance**?) $\rho(\nu, T) = \frac{2h\nu^3}{c^3} \frac{1}{\exp(h\nu/kT) - 1}$, the relations between Einstein coefficients ($E_j > E_i$) follows

$$\frac{g_j A_{ji}}{g_i B_{ij}} = \frac{2h\nu_{ji}^3}{c^3} \quad , \quad \frac{g_j B_{ji}}{g_i B_{ij}} = 1 \quad .$$

Planck's law

- Spectral energy density (energy per unit volume, per unit frequency):

$$u_\nu(\nu, T) = \frac{8\pi h\nu^3}{c^3} \frac{1}{\exp(h\nu/kT) - 1}$$

Dimensional analysis:

$$[u] = \frac{[h][\nu]^3}{[c]^3} = \frac{Js \cdot s^{-3}}{m^3 c^{-3}} = \frac{J}{m^3 \cdot \text{Hz}} = \frac{\text{energy}}{\text{length}^3 \cdot \text{freq.}} \quad .$$

- Radiance, $B_\nu(\nu, T) = \frac{c}{4\pi} u_\nu(\nu, T)$ **todo** Justify this relation. Is this right? Or should the factor be $\frac{c}{8\pi}$ ¹. Thus

$$B_\nu(\nu, T) = \frac{2h\nu^3}{c^2} \frac{1}{\exp(h\nu/kT) - 1}$$

Dimensional analysis

$$[B] = \frac{\text{power}}{\text{solid angle} \cdot \text{surface} \cdot \text{Hz}}$$

19.5 Ratiation of moving charges

19.5.1 Green's function solution of wave equations: Liénard-Wiechert potential

Wave equations in electromagnetism. See [Classical Electromagnetism: Wave Equations in Electromagnetism](#). Using **Lorentz gauge**, $\nabla \cdot \mathbf{a} + \frac{1}{c^2} \partial_t \varphi = 0$, the wave equations for the vector potential $\mathbf{a}$ and the scalar potential φ read

$$\begin{aligned} \frac{1}{c^2} \partial_{tt} \mathbf{a} - \Delta \mathbf{a} &= \mu \mathbf{j} \\ \frac{1}{c^2} \partial_{tt} \varphi - \Delta \varphi &= \frac{\rho}{\varepsilon} \end{aligned}$$

Solution of wave equation with Green's function

The solution of a wave equation

$$\frac{1}{c^2} \partial_{tt} u - \Delta u = f \quad ,$$

¹ Taltavull, Rudolf Ladenburg and the first quantum interpretation of optical dispersion, Eur. Phys. J. H.

can be written using **Math: Green's function method** (with assumptions of causality, and no contribution from i.c. and b.c.),

$$\begin{aligned} u(\mathbf{r}, t) &= \frac{1}{4\pi} \int_{t' \in T} \int_{\mathbf{r}' \in V} \frac{\delta\left(t' - t + \frac{|\mathbf{r}' - \mathbf{r}|}{c}\right)}{|\mathbf{r}' - \mathbf{r}|} f(\mathbf{r}', t') d\mathbf{r}' dt' = \\ &= \frac{1}{4\pi} \int_{\mathbf{r}' \in V} \frac{1}{|\mathbf{r}' - \mathbf{r}|} f(\mathbf{r}', t_{ret}(\mathbf{r}, t; \mathbf{r}')) d\mathbf{r}' \end{aligned}$$

with the retarded time $t_{ret}(\mathbf{r}, t; \mathbf{r}') = t - \frac{|\mathbf{r}' - \mathbf{r}|}{c}$.

The solution of the wave equations reads

$$\begin{aligned} \mathbf{a}(\mathbf{r}, t) &= \frac{\mu}{4\pi} \int_{\mathbf{r}' \in V} \frac{1}{|\mathbf{r}' - \mathbf{r}|} \mathbf{j}(\mathbf{r}', t_{ret}(\mathbf{r}, t; \mathbf{r}')) d\mathbf{r}' \\ \varphi(\mathbf{r}, t) &= \frac{1}{4\pi\epsilon} \int_{\mathbf{r}' \in V} \frac{1}{|\mathbf{r}' - \mathbf{r}|} \rho(\mathbf{r}', t_{ret}(\mathbf{r}, t; \mathbf{r}')) d\mathbf{r}' \end{aligned}$$

with the retarded time $t_{ret}(\mathbf{r}, t; \mathbf{r}') = t - \frac{|\mathbf{r}' - \mathbf{r}|}{c}$.

Point charge. The charge density and the current density of a point charge with electrical charge q , position $\mathbf{r}_s(t)$ and velocity $\mathbf{v}_s = \dot{\mathbf{r}}_s$ are respectively

$$\begin{aligned} \rho(\mathbf{r}, t) &= q \delta(\mathbf{r} - \mathbf{r}_s(t)) \\ \mathbf{j}(\mathbf{r}, t) &= q \mathbf{v}_s(t) \delta(\mathbf{r} - \mathbf{r}_s(t)) \end{aligned}$$

Thus, the electromagnetic potentials generated by a point charge are

$$\begin{aligned} \mathbf{a}(\mathbf{r}, t) &= \frac{\mu}{4\pi} \frac{1}{|\mathbf{r}_s(t_{ret}) - \mathbf{r}|} q \mathbf{v}_s(t_{ret}) \\ \varphi(\mathbf{r}, t) &= \frac{1}{4\pi\epsilon} \frac{1}{|\mathbf{r}_s(t_{ret}) - \mathbf{r}|} q \end{aligned}$$

Derivatives of the retarded time

$$\begin{aligned} \partial_t t_{ret} &= \frac{1}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} \\ \nabla t_{ret} &= -\frac{1}{c} \frac{\hat{\mathbf{r}}}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}}. \end{aligned}$$

Details

Time derivative

$$\begin{aligned} \partial_t t_{ret}(\mathbf{r}, t, \mathbf{r}_s(t)) &= \partial_t \left(t - \frac{|\mathbf{r}_s(t_{ret}(\mathbf{r}, t, \mathbf{r}_s(t_{ret}))) - \mathbf{r}|}{c} \right) = \\ &= 1 - \frac{1}{c} \frac{d|\mathbf{r}_s - \mathbf{r}|}{dt_{ret}} \partial_t t_{ret} = \\ &= 1 + \frac{x_k - x_{s,k}}{|\mathbf{r}_s - \mathbf{r}|} \frac{v_{s,k}}{c} \partial_t t_{ret} = \\ &= 1 + \hat{\mathbf{r}} \cdot \boldsymbol{\beta} \partial_t t_{ret}, \end{aligned}$$

and thus

$$\partial_t t_{ret} = \frac{1}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}}.$$

Space derivatives

$$\begin{aligned}
 \partial_k t_{ret}(\mathbf{r}, t, \mathbf{r}_s(t)) &= \partial_k \left(t - \frac{|\mathbf{r}_s(t_{ret}) - \mathbf{r}|}{c} \right) = \\
 &= -\frac{1}{c} \partial_k |\mathbf{r}_s(t_{ret}) - \mathbf{r}| = \\
 &= \frac{1}{c} \frac{r_i - r_{s,i}(t_{ret})}{|\mathbf{r}_s(t_{ret}) - \mathbf{r}|} \partial_k (r_{s,i}(t_{ret}) - r_i) = \\
 &= \frac{1}{c} \frac{r_i - r_{s,i}(t_{ret})}{|\mathbf{r}_s(t_{ret}) - \mathbf{r}|} (v_{s,i}(t_{ret}) \partial_k t_{ret} - \delta_{ik}) = \\
 &= \hat{\mathbf{r}} \cdot \boldsymbol{\beta}, \partial_k t_{ret} - \frac{1}{c} \frac{r_k - r_{s,k}(t_{ret})}{|\mathbf{r}_s(t_{ret}) - \mathbf{r}|},
 \end{aligned}$$

so that

$$\partial_k t_{ret} = -\frac{1}{c} \frac{1}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} \frac{r_k - r_{s,k}(t_{ret})}{|\mathbf{r}_s(t_{ret}) - \mathbf{r}|}$$

or, using vector notation,

$$\nabla t_{ret} = -\frac{1}{c} \frac{\hat{\mathbf{r}}}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}}.$$

Derivatives of $|\mathbf{r} - \mathbf{r}_s(t_{ret})|^{-1}$

$$\begin{aligned}
 \partial_t |\mathbf{r} - \mathbf{r}_s(t_{ret})|^n &= -n |\mathbf{r} - \mathbf{r}_s(t_{ret})|^{n-1} \frac{\hat{\mathbf{r}} \cdot \boldsymbol{\beta}}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} c \\
 \partial_k |\mathbf{r} - \mathbf{r}_s(t_{ret})|^n &= n |\mathbf{r}_s(t_{ret}) - \mathbf{r}|^{n-1} \left[\frac{\hat{\mathbf{r}} \cdot \boldsymbol{\beta}}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} + 1 \right] \hat{r}_k
 \end{aligned}$$

Details

$$\begin{aligned}
 \partial_k |\mathbf{r} - \mathbf{r}_s(t_{ret})|^n &= \partial_k [(\mathbf{r} - \mathbf{r}_s(t_{ret})) \cdot (\mathbf{r} - \mathbf{r}_s(t_{ret}))]^{\frac{n}{2}} = \\
 &= n |\mathbf{r} - \mathbf{r}_s(t_{ret})|^{n-2} (\mathbf{r} - \mathbf{r}_s(t_{ret})) \cdot \partial_k (\mathbf{r} - \mathbf{r}_s(t_{ret}))
 \end{aligned}$$

Now,

$$\begin{aligned}
 \partial_t (r_i - r_{s,i}(t_{ret})) &= -\partial_t t_{ret} v_{s,i}(t_{ret}) = \\
 &= -\frac{v_{s,i}(t_{ret})}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} \\
 \partial_k (r_i - r_{s,i}(t_{ret})) &= \delta_{ik} - \partial_k t_{ret} v_{s,i}(t_{ret}) = \\
 &= \delta_{ik} + \frac{1}{c} \frac{\hat{r}_k}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} v_{s,i}(t_{ret}),
 \end{aligned}$$

and thus

$$\begin{aligned}
 \partial_t |\mathbf{r} - \mathbf{r}_s(t_{ret})|^n &= \partial_t [(\mathbf{r} - \mathbf{r}_s(t_{ret})) \cdot (\mathbf{r} - \mathbf{r}_s(t_{ret}))]^{\frac{n}{2}} = \\
 &= n |\mathbf{r} - \mathbf{r}_s(t_{ret})|^{n-2} (\mathbf{r} - \mathbf{r}_s(t_{ret})) \cdot \partial_t (\mathbf{r} - \mathbf{r}_s(t_{ret})) = \\
 &= -n |\mathbf{r} - \mathbf{r}_s(t_{ret})|^{n-1} \frac{\hat{\mathbf{r}} \cdot \mathbf{v}_s(t_{ret})}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}}
 \end{aligned}$$

and

$$\begin{aligned}
 \partial_k |\mathbf{r} - \mathbf{r}_s(t_{ret})|^n &= \partial_k [(\mathbf{r} - \mathbf{r}_s(t_{ret})) \cdot (\mathbf{r} - \mathbf{r}_s(t_{ret}))]^{\frac{n}{2}} = \\
 &= n |\mathbf{r} - \mathbf{r}_s(t_{ret})|^{n-2} (\mathbf{r} - \mathbf{r}_s(t_{ret})) \cdot \partial_k (\mathbf{r} - \mathbf{r}_s(t_{ret})) = \\
 &= n |\mathbf{r} - \mathbf{r}_s(t_{ret})|^{n-1} \left[\frac{1}{c} \frac{\hat{\mathbf{r}} \cdot \mathbf{v}_s(t_{ret})}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} + 1 \right] \hat{r}_k
 \end{aligned}$$

From potentials to the electromagnetic field

$$\begin{aligned}
 \mathbf{e} &= -\frac{q}{4\pi\epsilon} \frac{\mathbf{a}_s(t_{ret})}{|\mathbf{r}|c^2} \frac{1}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} + \frac{q}{4\pi\epsilon} \frac{1}{|\mathbf{r}|^2} \left(\frac{\hat{\mathbf{r}} - \boldsymbol{\beta}(\hat{\mathbf{r}} \cdot \boldsymbol{\beta})}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} \right) . \\
 \mathbf{b} &= \frac{q}{4\pi\epsilon} \frac{1}{c} \left[-\frac{\hat{\mathbf{r}} \times \mathbf{a}_s(t_{ret})}{|\mathbf{r}|c^2} \frac{1}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} - \frac{\hat{\mathbf{r}}}{|\mathbf{r}|^2} \times \boldsymbol{\beta} \left(\frac{1}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} \right) \right] .
 \end{aligned}$$

It's easy to realize that $\mathbf{b} = \frac{1}{c} \hat{\mathbf{r}} \times \mathbf{e}$.

Details

$$\begin{aligned}
 \mathbf{e} &= -\partial_t \mathbf{a} - \nabla \varphi = \\
 &= -\partial_t \left(\frac{\mu}{4\pi} \frac{q \mathbf{v}_s(t_{ret})}{|\mathbf{r}_s(t_{ret}) - \mathbf{r}|} \right) - \nabla \left(\frac{1}{4\pi\epsilon} \frac{q}{|\mathbf{r}_s(t_{ret}) - \mathbf{r}|} \right) = \\
 &= -\frac{q\mu}{4\pi} \left[\frac{1}{|\mathbf{r}|} \mathbf{a}_s(t_{ret}) \partial_t t_{ret} + \mathbf{v}_s |\mathbf{r}|^{-2} \frac{\hat{\mathbf{r}} \cdot \boldsymbol{\beta}}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} c \right] - \frac{q}{4\pi\epsilon} |\mathbf{r}|^{-2} \left(-1 - \frac{\hat{\mathbf{r}} \cdot \boldsymbol{\beta}}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} \right) \hat{\mathbf{r}} = \\
 &= -\frac{q\mu c^2}{4\pi} \left[\frac{\mathbf{a}_s(t_{ret})}{|\mathbf{r}|c^2} \frac{1}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} + \frac{\boldsymbol{\beta}}{|\mathbf{r}|^2} \frac{\hat{\mathbf{r}} \cdot \boldsymbol{\beta}}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} \right] + \frac{q}{4\pi\epsilon} \frac{1}{|\mathbf{r}|^2} \left(1 + \frac{\hat{\mathbf{r}} \cdot \boldsymbol{\beta}}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} \right) \hat{\mathbf{r}} = \\
 &= -\frac{q}{4\pi\epsilon} \frac{\mathbf{a}_s(t_{ret})}{|\mathbf{r}|c^2} \frac{1}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} + \frac{q}{4\pi\epsilon} \frac{1}{|\mathbf{r}|^2} \left(\frac{-\boldsymbol{\beta}(\hat{\mathbf{r}} \cdot \boldsymbol{\beta}) + \hat{\mathbf{r}}(\hat{\mathbf{r}} \cdot \boldsymbol{\beta})}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} + \hat{\mathbf{r}} \right) = \\
 &= -\frac{q}{4\pi\epsilon} \frac{\mathbf{a}_s(t_{ret})}{|\mathbf{r}|c^2} \frac{1}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} + \frac{q}{4\pi\epsilon} \frac{1}{|\mathbf{r}|^2} \left(\frac{\hat{\mathbf{r}} - \boldsymbol{\beta}(\hat{\mathbf{r}} \cdot \boldsymbol{\beta})}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} \right) . \\
 \mathbf{b} &= \nabla \times \mathbf{a} = \\
 &= \nabla \times \left(\frac{\mu}{4\pi} \frac{q \mathbf{v}_s(t_{ret})}{|\mathbf{r} - \mathbf{r}_s(t_{ret})|} \right) = \\
 &= \hat{\mathbf{e}}_k \epsilon_{klm} \partial_l \left(\frac{\mu}{4\pi} \frac{q v_{s,m}(t_{ret})}{|\mathbf{r} - \mathbf{r}_s(t_{ret})|} \right) = \\
 &= \hat{\mathbf{e}}_k \epsilon_{klm} \frac{\mu q}{4\pi} \partial_l \left(\frac{v_{s,m}(t_{ret})}{|\mathbf{r} - \mathbf{r}_s(t_{ret})|} \right) = \\
 &= \hat{\mathbf{e}}_k \epsilon_{klm} \frac{\mu q}{4\pi} \left[\frac{a_{s,m}(t_{ret}) \partial_l t_{ret}}{|\mathbf{r} - \mathbf{r}_s(t_{ret})|} - v_{s,m}(t_{ret}) \frac{\hat{r}_l}{|\mathbf{r}|^2} \left(1 + \frac{\hat{\mathbf{r}} \cdot \boldsymbol{\beta}}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} \right) \right] = \\
 &= \hat{\mathbf{e}}_k \epsilon_{klm} \frac{\mu q}{4\pi} \left[-\frac{a_{s,m}(t_{ret})}{|\mathbf{r} - \mathbf{r}_s(t_{ret})|} \frac{1}{c} \frac{\hat{r}_l}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} - v_{s,m}(t_{ret}) \frac{\hat{r}_l}{|\mathbf{r}|^2} \left(1 + \frac{\hat{\mathbf{r}} \cdot \boldsymbol{\beta}}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} \right) \right] = \\
 &= \frac{q\mu c^2}{4\pi} \frac{1}{c^2} \left[-\frac{\hat{\mathbf{r}} \times \mathbf{a}_s(t_{ret})}{|\mathbf{r}|c} \frac{1}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} - \frac{\hat{\mathbf{r}}}{|\mathbf{r}|^2} \times \mathbf{v}_s(t_{ret}) \left(1 + \frac{\hat{\mathbf{r}} \cdot \boldsymbol{\beta}}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} \right) \right] = \\
 &= \frac{q}{4\pi\epsilon} \frac{1}{c} \left[-\frac{\hat{\mathbf{r}} \times \mathbf{a}_s(t_{ret})}{|\mathbf{r}|c^2} \frac{1}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} - \frac{\hat{\mathbf{r}}}{|\mathbf{r}|^2} \times \boldsymbol{\beta} \left(1 + \frac{\hat{\mathbf{r}} \cdot \boldsymbol{\beta}}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} \right) \right] = \\
 &= \frac{q}{4\pi\epsilon} \frac{1}{c} \left[-\frac{\hat{\mathbf{r}} \times \mathbf{a}_s(t_{ret})}{|\mathbf{r}|c^2} \frac{1}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} - \frac{\hat{\mathbf{r}}}{|\mathbf{r}|^2} \times \boldsymbol{\beta} \left(\frac{1}{1 - \hat{\mathbf{r}} \cdot \boldsymbol{\beta}} \right) \right] .
 \end{aligned}$$

19.5.2 Larmor formula

Poynting vector reads

$$\mathbf{s}(\mathbf{r}, t) := \frac{\mathbf{e}(\mathbf{r}, t) \times \mathbf{b}(\mathbf{r}, t)}{\mu},$$

and thus, using the electromagnetic field of a moving charge,

$$\mathbf{s} = \frac{1}{\mu} \mathbf{e} \times \mathbf{b} = \frac{1}{c\mu} \mathbf{e} \times (\hat{\mathbf{r}} \times \mathbf{e}) = \frac{1}{\mu c} [|\mathbf{e}|^2 \hat{\mathbf{r}} - (\mathbf{e} \cdot \hat{\mathbf{r}}) \mathbf{e}].$$

Far-field approximation (also assuming that $\beta \sim 0$. *slow charge? Non-relativistic limit?*),

$$\mathbf{e} \sim -\frac{q}{4\pi\epsilon} \frac{\mathbf{a}_{s,ret}}{|\mathbf{r}|c^2}$$

and thus

$$\begin{aligned} \mathbf{s} &= \frac{1}{\mu c} \left(\frac{q}{4\pi\epsilon|\mathbf{r}|c^2} \right)^2 \mathbf{a} \times (\hat{\mathbf{r}} \times \mathbf{a}) = \\ &= \frac{1}{\mu c} \left(\frac{q}{4\pi\epsilon|\mathbf{r}|c^2} \right)^2 (|\mathbf{a}|^2 \hat{\mathbf{r}} - (\mathbf{a} \cdot \hat{\mathbf{r}}) \mathbf{a}) = \end{aligned}$$

The elementary power flux at $\mathbf{r} = R\hat{\mathbf{r}}$, assuming that the motion is in a region much smaller than R , so that $t_{ret} = t - \frac{R}{c}$, is

$$\hat{\mathbf{r}} \cdot \mathbf{s} = \frac{1}{\mu c} \left(\frac{q}{4\pi\epsilon Rc^2} \right)^2 (|\mathbf{a}|^2 - (\mathbf{a} \cdot \hat{\mathbf{r}})^2) =$$

Let's define the direction of $\mathbf{a}_{ret} = a\hat{\mathbf{t}}$, and spherical coordinates so that $\hat{\mathbf{r}} \cdot \hat{\mathbf{t}} = \cos \theta$. Integration over the sphere of radius R gives

$$\begin{aligned} \Phi &:= \int_S \hat{\mathbf{r}} \cdot \mathbf{s} = \\ &= \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \frac{1}{\mu c} \left(\frac{q}{4\pi\epsilon Rc^2} \right)^2 (1 - \cos^2 \theta)^2 |\mathbf{a}|^2 R^2 \sin \theta d\theta d\phi = \\ &= \frac{1}{\mu c} \left(\frac{q|\mathbf{a}|}{4\pi\epsilon c^2} \right)^2 \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} (1 - \cos^2 \theta) \sin \theta d\phi d\theta d\phi = \\ &= \frac{1}{\mu c} \left(\frac{q|\mathbf{a}|}{4\pi\epsilon c^2} \right)^2 \cdot \frac{4}{3} \cdot 2\pi = \\ &= \frac{q^2 |\mathbf{a}_{s,ret}|^2}{6\pi\epsilon c^3}. \end{aligned}$$

Average power.

$$\langle \Phi \rangle = \frac{q^2}{6\pi\epsilon c^3} \langle |\mathbf{a}|^2 \rangle$$

Examples:

- **harmonic oscillator** $\mathbf{a}(t) = -A\omega^2 \cos(\omega t)\hat{\mathbf{x}}$, $\langle |\mathbf{a}(t)|^2 \rangle = \frac{1}{2}A^2\omega^4$
- **circular orbits**, with constant speed. Acceleration is $\mathbf{a}(t) = -R\omega^2 \hat{\mathbf{r}}(t)$, and has constant magnitude $|\mathbf{a}(t)| = \omega^2 R$

$$\int_0^{\pi} (1 - \cos^2 x) \sin x dx = \left[-\cos x + \frac{\cos^3 x}{3} \right]_0^{\pi} = 2 - \frac{2}{3} = \frac{4}{3}.$$

$$\epsilon_{abc} A_b \epsilon_{cde} B_d C_e = (\delta_{ad} \delta_{be} - \delta_{ae} \delta_{bd}) A_b B_d C_e = A_b B_a C_b - A_b B_b C_a = \mathbf{A} \cdot \mathbf{C} \mathbf{B} - \mathbf{A} \cdot \mathbf{B} \mathbf{C}$$

ATOMIC MODELS

- *Thomson*: electrons in cathode rays
- *Rutherford*: small nucleus, from scattering
- *Bohr*: atomic model, consistent with the relation $\Delta E = h\nu$ and emission/absorption of light
- *Sommerfeld*: ...
- *Schrodinger*: atomic model from Schrodinger equation $i\hbar|\dot{\Psi}\rangle = \hat{H}|\Psi\rangle$

20.1 Thomson model

20.2 Rutherford model

20.3 Bohr model

Assumption:

- Electron moves on circular orbits
- All the possible orbits are identified by discrete values of the angular momentum, $L = n\hbar$
- Light emission/absorption occurs when an electron transitions between two orbits

Some mathematical details, using the equations of classical mechanics. The problem of determining the orbit of an electron around the positive nucleus is a two-body problem, subject to a force $\vec{F} \propto \frac{\vec{r}}{|\vec{r}|^3}$ (here the assumption of the center of mass in the nucleus; otherwise change of coordinates, and equivalent masses...). For an electrically neutral atom with one electron only (H atom?) the dynamical equation governing the motion of the electron

$$m\ddot{\vec{r}} = -\frac{q^2}{4\pi\epsilon} \frac{\vec{r}}{|\vec{r}|^3},$$

being q the charge of the electron.

- Using polar coordinates, $\vec{r} = r\hat{r}$
- Taking time derivative of the position vector $\vec{r}$ identifying the position of the electron w.r.t. the nucleus

$$\begin{aligned}\frac{d}{dt}(r\hat{r}) &= \dot{r}\hat{r} + r\dot{\theta}\hat{\theta} \\ \frac{d^2}{dt^2}(r\hat{r}) &= (\ddot{r} - r\dot{\theta}^2)\hat{r} + (2\dot{r}\dot{\theta} + r\ddot{\theta})\hat{\theta}\end{aligned}$$

- Assuming circular orbits, $\dot{r} = 0$

the radial and angular components of the dynamical equation read

$$\begin{aligned} r : \quad -mr\dot{\theta}^2 &= -\frac{q^2}{4\pi\epsilon} \frac{1}{r^2} \\ \theta : \quad \ddot{\theta} &= 0 \end{aligned}$$

The angular momentum and the energy of the system are

- Angular momentum,

$$\begin{aligned} n\hbar &= L_n = mr_n^2 \dot{\theta}_n = && \text{(radial component)} \\ &= mr_n^2 \left(\frac{q^2}{4\pi\epsilon r_n^3 m} \right)^{\frac{1}{2}} = \\ &= \frac{\sqrt{mq}}{\sqrt{4\pi\epsilon}} r_n^{\frac{1}{2}} \\ \rightarrow r_n &= 4\pi\epsilon \frac{n^2 \hbar^2}{mq^2} \end{aligned}$$

- Energy,

$$\begin{aligned} E_n &= \frac{1}{2} m |\dot{r}_n|^2 - \frac{q^2}{4\pi\epsilon} \frac{1}{r_n} = && \text{(radial component of the eom)} \\ &= \frac{q^2}{8\pi\epsilon} \frac{1}{r_n} - \frac{q^2}{4\pi\epsilon} \frac{1}{r_n} = \\ &= -\frac{q^2}{8\pi\epsilon} \frac{1}{r_n} = && (r_n \text{ from angular momentum}) \quad (20.1) \\ &= -\frac{mq^4}{32\pi^2\epsilon^2\hbar^2} \frac{1}{n^2} = && (h = 2\pi\hbar) \\ &= -\frac{mq^4}{8\epsilon^2\hbar^2} \frac{1}{n^2}. \end{aligned}$$

Remark. The value of energy E_n of Bohr model coincides with the value (20.2) of the energy produced by *Schrodinger model*.

From the relation $\Delta E = \nu h$, the relation between the frequency of the emitted (if $E_n > E_m$, n initial state, m final state) or absorbed (if $E_n < E_m$, n initial state, m final state) radiation, follows

$$\nu_{nm} = \frac{E_n - E_m}{h} = -\frac{mq^4}{8\epsilon^2\hbar^3} \left(\frac{1}{n^2} - \frac{1}{m^2} \right)$$

20.4 Sommerfeld model

20.5 Schrodinger model

Contents

- *Mathematical details of the analytical solution*
- *Gallery of orbitals*

- *Auxiliary notes*

Schrodinger equation. Schrodinger equation in space basis for the electron in a H atom reads

$$i\hbar\partial_t\Psi(\mathbf{r},t) = \left[-\frac{\hbar^2}{2m}\nabla^2 - \frac{q^2}{4\pi\epsilon} \frac{1}{|\mathbf{r}|} \right] \Psi(\mathbf{r},t)$$

where the Hamiltonian is the sum of the kinetic energy and a Coulomb potential contribution, $\hat{V}$, so that $\langle \mathbf{r} | \hat{V} | \Psi \rangle = -\frac{q^2}{4\pi\epsilon|\mathbf{r}|} \Psi(\mathbf{r},t)$

Spectral decomposition of the Hamiltonian operator. The stationary states, and the corresponding energy levels - i.e. the eigenfunctions, and the eigenvalues of the Hamiltonian operator $\hat{H}$ - are the result of the eigenvalue problem, $\hat{H}|\Psi\rangle = E|\Psi\rangle$, or using space basis,

$$\left[-\frac{\hbar^2}{2m}\nabla^2 - \frac{q^2}{4\pi\epsilon} \frac{1}{|\mathbf{r}|} \right] \Psi(\mathbf{r}) = E\Psi(\mathbf{r}) .$$

Quantum numbers. The stationary states depend on **three** - in this model, no spin exists - **quantum numbers**:

- **principal** quantum number $n \in \{1, 2, \dots\}$
- **azimuthal** quantum number $\ell \in \{0, 1, \dots, n-1\}$
- **magnetic** quantum number $m_\ell \in \{-\ell, -\ell+1, \dots, \ell\}$

Energy levels only depend on the principal quantum number, $E(n)$. Thus, Schrodinger model of the atom has **degenerate** stationary states, i.e. stationary states with the same energy value.

Commutation of $\hat{H}$, $\hat{L}^2$, $\hat{L}_z$. As angular momentum operators $\hat{L}^2$, $\hat{L}_z$ commute with the Hamiltonian operator $\hat{H}$ of the H atom, these three operators share common eigenvectors. These relations are proved in the [note section](#).

20.5.1 Schrodinger model - Mathematical details of the analytical solution

Schrodinger equation for the hydrogen atom in space basis reads

$$i\hbar\Psi(\mathbf{r},t) = \left[-\frac{\hbar^2}{2m}\nabla^2 - \frac{q^2}{4\pi\epsilon} \frac{1}{r} \right] \Psi(\mathbf{r},t) .$$

The **eigenproblem of the Hermitian operator** (projected onto space basis, as well) reads

$$E_n \Psi_n(\mathbf{r}) = -\frac{\hbar^2}{2m}\nabla^2\Psi_n(\mathbf{r}) - \frac{q^2}{4\pi\epsilon r}\Psi_n(\mathbf{r}) .$$

Given the symmetries of the system, the solution of the eigenproblem is evaluated using spherical coordinates $\{r, \theta, \varphi\}$, and the method of separation of variables. Using **spherical coordinates**,

$$-\frac{\hbar^2}{2m} \left\{ \frac{1}{r^2} \partial_r (r^2 \partial_r) + \frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta) + \frac{1}{r^2 \sin^2 \theta} \partial_\varphi \partial_\varphi \right\} \Psi_n - \frac{q^2}{4\pi\epsilon_0 r} \Psi_n = E_n \Psi_n$$

Separation of variables, $\Psi_{nlm}(r, \theta, \varphi) = R_n(r)Y_{lm}(\theta, \varphi)$

With the **method of separation of variables**, we look for solutions in product form:

$$\Psi_{nlm}(r, \theta, \varphi) = R_n(r)Y_{lm}(\theta, \varphi) ,$$

and substituting into the Schrödinger equation:

$$\left(E_n + \frac{q^2}{4\pi\epsilon_0 r} \right) R_n Y_{lm} = -\frac{\hbar^2}{2m} \left[\frac{1}{r^2} (r^2 R_n')' Y_{lm} + R_n \left(\frac{1}{r^2 \sin \theta} \partial_\theta (\sin \theta \partial_\theta Y_{lm}) + \frac{1}{r^2 \sin^2 \theta} \partial_\varphi \partial_\varphi Y_{lm} \right) \right]$$

Dividing by $R_n Y_{lm}$, and multiplying by $r^2 \frac{2m}{\hbar^2}$,

$$0 = \underbrace{\frac{1}{R_n} (r^2 R'_n)' + \frac{2mr^2}{\hbar^2} \left(E_n + \frac{q^2}{4\pi\epsilon_0 r} \right)}_{f(r)} + \underbrace{\frac{1}{Y_{lm}} \left\{ \frac{1}{\sin\theta} \partial_\theta (\sin\theta \partial_\theta Y_{lm}) + \frac{1}{\sin^2\theta} \partial_\varphi^2 Y_{lm} \right\}}_{g(\theta, \varphi)}$$

Since $f(r) + g(\theta, \varphi) = 0$, both terms must equal equal-and-opposite constant C for all valid domains $r \in (0, +\infty)$, $\theta \in (0, \pi)$, $\varphi \in (0, 2\pi)$

$$C = f(r) = -g(\theta, \varphi) .$$

Radial equation, and energy levels

The radial component determines the energy level,

$$E_n = -\frac{mq^4}{32\pi^2\epsilon_0^2\hbar^2} \frac{1}{n^2} . \quad (20.2)$$

being $n \in \{1, 2, \dots\}$.

The radial distribution of the energy state functions is

$$R(\rho) = e^{-\rho/2} \rho^\ell L_{n-\ell-1}^{2\ell+1}(\rho) ,$$

with $\rho = r \frac{\sqrt{-8mE_n}}{\hbar}$ the non dimensional radius, $\ell \in \{0, 1, \dots, n-1\}$, and $L_p^k(x)$ the solution of the associated Laguerre differential equation $xy'' + (k+1-x)y' + py = 0$.

Change of variables, and asymptotic behavior

Defining $C = \ell(\ell+1)$, and dividing the equation $f(r) = C$ by r^2 , for the states with $E_n < 0$,

$$\frac{1}{r^2} \frac{d}{dr} (r^2 R') + \left[\frac{2m}{\hbar^2} \left(E + \frac{q^2}{4\pi\epsilon_0 r} \right) - \frac{\ell(\ell+1)}{r^2} \right] R = 0$$

Defining the **dimensionless radial variable** $\rho = \kappa r$, so $\frac{d}{dr} = \kappa \frac{d}{d\rho}$, and factoring out κ^2

$$0 = \kappa^2 \left\{ \frac{1}{\rho^2} \frac{d}{d\rho} \left(\rho^2 \frac{dR}{d\rho} \right) + \left(\frac{2mE}{\hbar^2 \kappa^2} + \frac{2mq^2}{\hbar^2 \kappa \cdot 4\pi\epsilon_0 \rho} - \frac{\ell(\ell+1)}{\rho^2} \right) \right\} R$$

By choosing $\kappa^2 = -\frac{8mE}{\hbar^2}$ (where $\kappa = \frac{\sqrt{-8mE}}{\hbar}$ for $E < 0$), the first term inside the parentheses becomes -1 . Defining the constant parameter ρ_0 :

$$\rho_0 = \frac{2mq^2}{\hbar^2 \kappa \cdot 4\pi\epsilon_0} = \frac{2mq^2}{\hbar^2 \sqrt{-8mE} \cdot 4\pi\epsilon_0} = \frac{q^2}{8\pi\epsilon_0 \hbar} \sqrt{\frac{2m}{-E}} ,$$

the dimensionless radial equation simplifies to:

$$0 = \frac{1}{\rho^2} \frac{d}{d\rho} \left(\rho^2 \frac{dR}{d\rho} \right) + R \left[-\frac{\ell(\ell+1)}{\rho^2} - \frac{1}{4} + \frac{\rho_0}{\rho} \right]$$

Boundary conditions required for regular wavefunctions:

$$R(\rho = 0) < \infty, \quad R(\rho \rightarrow +\infty) = 0$$

Asymptotic behavior for $r \rightarrow 0$, and $r \rightarrow +\infty$

Large Distance Limit ($\rho \rightarrow +\infty$). For large ρ , terms proportional to $\frac{1}{\rho}$ and $\frac{1}{\rho^2}$ become negligible:

$$\frac{1}{\rho^2} \frac{d}{d\rho} (\rho^2 R') = R'' + \frac{2}{\rho} R' \sim R''$$

The equation reduces to:

$$R'' - \frac{1}{4} R \sim 0$$

General solution:

$$R(\rho) = A e^{-\rho/2} + B e^{\rho/2}$$

To satisfy the non-divergent boundary condition at infinity, set $B = 0$:

$$R(\rho) \sim e^{-\rho/2} \quad \text{as } \rho \rightarrow +\infty$$

Small Distance Limit ($\rho \rightarrow 0$). For $\rho \rightarrow 0$, the centrifugal term $\frac{\ell(\ell+1)}{\rho^2}$ dominates over the constant and $\frac{1}{\rho}$ terms:

$$R'' + \frac{2}{\rho} R' - \frac{\ell(\ell+1)}{\rho^2} R \sim 0$$

Proposing a power-law ansatz $R(\rho) \sim \rho^s$:

$$s(s-1)\rho^{s-2} + 2s\rho^{s-2} - \ell(\ell+1)\rho^{s-2} = 0$$

$$s^2 + s - \ell(\ell+1) = 0$$

Solving the quadratic characteristic equation for s :

$$s_{1,2} = \frac{-1 \mp \sqrt{1 + 4\ell(\ell+1)}}{2} = \frac{-1 \mp (1 + 2\ell)}{2}$$

$$s_1 = \ell, \quad s_2 = -(\ell+1)$$

Since $\ell \geq 0$, $s_2 = -(\ell+1)$ leads to divergence at the origin $\rho \rightarrow 0$. Retaining only the physically acceptable solution $s = \ell$:

$$R(\rho) \sim \rho^\ell \quad \text{as } \rho \rightarrow 0$$

Transformation & Differential Equation for $v(\rho)$. Combining both asymptotic behaviors, we represent the full radial solution as:

$$R(\rho) = \rho^\ell e^{-\rho/2} v(\rho)$$

Writing out derivatives of $R(\rho)$:

$$R' = \left(\ell \rho^{\ell-1} - \frac{1}{2} \rho^\ell \right) e^{-\rho/2} v + \rho^\ell e^{-\rho/2} v'$$

After expanding $\frac{1}{\rho^2} \frac{d}{d\rho} (\rho^2 R') + R \left[-\frac{\ell(\ell+1)}{\rho^2} - \frac{1}{4} + \frac{\rho_0}{\rho} \right] = 0$:

$$= e^{-\rho/2} \left\{ \rho^\ell v'' + v' \left[2\ell \rho^{\ell-1} - \frac{1}{2} \rho^\ell + 2\rho^{\ell-1} \right] + v \left[\rho^\ell \ell(\ell+1) - \rho^{\ell+1} (\ell+1) + \frac{1}{4} \rho^{\ell+2} - \ell(\ell+1) \rho^\ell - \frac{1}{4} \rho^{\ell+2} + \rho_0 \rho^{\ell+1} \right] \right\} = 0$$

Dividing out non-zero factor $\rho^{\ell-1} e^{-\rho/2}$:

$$\rho v''(\rho) + [2(\ell+1) - \rho] v'(\rho) + (\rho_0 - \ell - 1) v(\rho) = 0$$

Polynomial expansion, energy quantization E_n and radial distribution $R_n(\rho)$

Starting from the transformed radial equation:

$$\rho v''(\rho) + [2(\ell + 1) - \rho]v'(\rho) + (\rho_0 - \ell - 1)v(\rho) = 0$$

Looking for a power series solution:

$$\begin{aligned} v(\rho) &= \sum_{k=0}^{\infty} c_k \rho^k \\ v'(\rho) &= \sum_{k=0}^{\infty} k c_k \rho^{k-1} \\ v''(\rho) &= \sum_{k=0}^{\infty} k(k-1) c_k \rho^{k-2} \end{aligned}$$

Substituting into the differential equation:

$$\sum_{k=0}^{\infty} k(k-1) c_k \rho^{k-1} + \sum_{k=0}^{\infty} 2(\ell + 1) k c_k \rho^{k-1} - \sum_{k=0}^{\infty} k c_k \rho^k + (\rho_0 - \ell - 1) \sum_{k=0}^{\infty} c_k \rho^k = 0$$

Shift indices to group terms by power ρ^k :

$$\sum_{k=0}^{\infty} \left\{ (k+1)(k) c_{k+1} + 2(\ell + 1)(k+1) c_{k+1} - k c_k + (\rho_0 - \ell - 1) c_k \right\} \rho^k = 0$$

Recurrence Relation. Equating coefficients to zero yields the recurrence relation for c_k :

$$c_{k+1} = c_k \frac{k + \ell + 1 - \rho_0}{(k+1)(k + 2\ell + 2)}$$

Asymptotic Behavior & Series Termination. If the power series does not terminate, as $k \rightarrow +\infty$:

$$\frac{c_{k+1}}{c_k} \rightarrow \frac{1}{k}$$

This asymptotic behavior matches that of the exponential expansion $e^\rho \sim \sum \frac{\rho^k}{k!}$ and thus $\frac{c_{k+1}}{c_k} = \frac{1}{k+1} \sim \frac{1}{k}$. Thus, an infinite series $v(\rho) \sim e^\rho$, would lead to $R(\rho) \sim e^{-\rho/2} e^\rho = e^{\rho/2}$, which diverges as $\rho \rightarrow +\infty$. Thus, in order to maintain physical normalizability, the series must terminate at a maximum power $k_{\max} = N$:

$$c_{N+1} = 0 \implies N + \ell + 1 - \rho_0 = 0 \implies \rho_0 = N + \ell + 1 \equiv n$$

where:

- $N \geq 0$ is the radial quantum number.
- $n = N + \ell + 1$ is the **Principal Quantum Number** ($n \geq 1$).
- $\forall n$, the orbital angular momentum quantum number satisfies $0 \leq \ell \leq n - 1$.

Energy Quantization. Recalling the definition of ρ_0 , $n = \rho_0 = \sqrt{\frac{2m}{-E}} \frac{q^2}{4\pi\epsilon_0\hbar}$, and solving for the bound state energies E_n ($E < 0$), the relation between the energy E_n of the n^{th} energy level and the principal quantum number n follows

$$E_n = -\frac{mq^4}{32\pi^2\epsilon_0^2\hbar^2} \frac{1}{n^2}. \quad (20.3)$$

Remark. The value of energy E_n of Schrodinger model coincides with the value (20.1) of the energy produced by *Bohr model*.

Radial solution. Putting everything together, the radial part of the solution reads

$$R(\rho) = e^{-\rho/2} \rho^\ell L_{n-\ell-1}^{2\ell+1}(\rho),$$

being $L_p^k(x)$ the solution of the associated Laguerre differential equation $xy'' + (k+1-x)y' + py = 0$.

Azimuthal and polar equations

Further separation of variables, $Y_{\ell m}(\theta, \varphi) = \Theta(\theta)\Phi(\varphi)$

With $Y(\theta, \varphi) = \Theta(\theta)\Psi(\varphi)$, the equation

$$\frac{1}{Y_{\ell m}} \left\{ \frac{1}{\sin \theta} \partial_{\theta} (\sin \theta \partial_{\theta} Y_{\ell m}) + \frac{1}{\sin^2 \theta} \partial_{\varphi \varphi} Y_{\ell m} \right\} = -\ell(\ell + 1) .$$

becomes - after multiplying by $\sin^2 \theta$,

$$\underbrace{\frac{\sin \theta}{\Theta} (\sin \theta \Theta')' + \ell(\ell + 1) \sin^2 \theta}_{F(\theta)} = - \underbrace{\frac{\Phi''}{\Phi}}_{G(\varphi)} ,$$

and thus

$$F(\theta) = -G(\varphi) = m_{\ell}^2 ,$$

with a non-negative constant m_{ℓ}^2 in order to match periodic conditions for $\Phi(\varphi = 0) = \Phi(\varphi = 2\pi)$.

Azimuthal equation

$$\Phi'' + m_{\ell}^2 \Phi = 0 ,$$

whose solution is $\Phi(\varphi) \propto \exp(\mp i m_{\ell} \varphi)$. In order to satisfy periodic conditions, $m_{\ell} \in \mathbb{Z}$. This parameter will be further constrained later, in the solution of the polar equation, to be $m_{\ell} \in \{-\ell, -\ell + 1, \dots, \ell\}$, with $\ell \in \mathbb{N}$. Thus, selecting only the independent solutions, the azimuthal distribution of the eigenfunctions of the Hamiltonian is

$$\Phi(\varphi) \propto \exp(i m_{\ell} \varphi) , \quad m_{\ell} \in \{-\ell, -\ell + 1, \dots, \ell\} .$$

Polar equation

Multiplying the equation $F(\theta) = m_{\ell}^2$ by Θ , the polar equation becomes,

$$\sin \theta (\sin \theta \Theta')' + [\ell(\ell + 1) \sin^2 \theta - m_{\ell}^2] \Theta = 0 .$$

Polar equation as an associated Legendre equation

Dividing by $\sin^2 \theta = 1 - \cos^2 \theta$, the second term can be immediately recast as the second term in the associated Legendre equation, if $x = \cos \theta$.

As $\frac{dx}{d\theta} = -\sin \theta$, then $\frac{d}{d\theta} = \frac{dx}{d\theta} \frac{d}{dx} = -\sin \theta \frac{d}{dx}$, the first term - divided by $\sin^2 \theta$ - becomes

$$-\frac{1}{\sin^2 \theta} \sin^2 \theta \frac{d}{dx} \left(-\sin^2 \theta \frac{d}{dx} \Theta \right) = \frac{d}{dx} \left[(1 - x^2) \frac{d}{dx} \Theta \right]$$

Thus the polar equation can be recast as

$$0 = \frac{d}{dx} \left[(1 - x^2) \frac{d\Theta}{dx} \right] + \left[\ell(\ell + 1) - \frac{m_{\ell}^2}{1 - x^2} \right] \Theta ,$$

i.e. as the associated Legendre equation whose solution (see below) is $\Theta(x) = P_{\ell}^{m_{\ell}}(x)$, for $m_{\ell} \in \{-\ell, \dots, \ell\}$, $\ell \in \mathbb{N}$.

Legendre differential equation and associated Legendre equation

Legendre differential equations read

$$0 = (1 - x^2)P_n''(x) - 2xP_n'(x) + n(n+1)P_n(x) = , \quad x \in [-1, 1]$$

$$= \frac{d}{dx} [(1 - x^2)P_n'(x)] + n(n+1)P_n(x) ,$$

with the solution $P_n(x)$ be the Legendre polynomial. The **associated Legendre differential equations** are defined as

$$0 = \frac{d}{dx} [(1 - x^2)P_n'(x)] + \left[n(n+1) - \frac{m^2}{1 - x^2} \right] P_n(x) , \quad x \in [-1, 1]$$

and the solution reads $P_n^m(x) = (-1)^m (1 - x^2)^{\frac{m}{2}} \frac{d^m}{dx^m} P_n(x)$, for $m \geq 0$. The solutions for $m < 0$ are defined via symmetry conditions, or Rodrigues' formula, as $P_\ell^{-m} = (-1)^m \frac{(\ell-m)!}{(\ell+m)!} P_\ell^m(x)$, see [Wikipedia: associated Legendre polynomials](#).

20.5.2 Schrodinger model - Gallery

Stationary states

$$\psi_{n,\ell,m_\ell}(r, \theta, \phi) = R_{n\ell}(r) Y_\ell^{m_\ell}(\theta, \phi)$$

Radial Wavefunctions, $R_{n\ell}(r)$. Using the dimensionless variable $\rho = \frac{2r}{na_0}$,

$$R_{n\ell}(r) = -\sqrt{\left(\frac{2}{na_0}\right)^3 \frac{(n-\ell-1)!}{2n[(n+\ell)!]^3}} e^{-\rho/2} \rho^\ell L_{n-\ell-1}^{2\ell+1}(\rho) ,$$

where $a_0 = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2}$ is the **Bohr radius**, and $L_p^k(\rho)$ the solution of the associated Laguerre equation.

Spherical Harmonics, $Y_\ell^{m_\ell}(\theta, \phi) = \Theta_{\ell,m_\ell}(\theta) \Phi_{m_\ell}(\phi)$.

$$Y_\ell^{m_\ell}(\theta, \phi) = (-1)^{m_\ell} \sqrt{\frac{(2\ell+1)(\ell-m_\ell)!}{4\pi(\ell+m_\ell)!}} P_\ell^{m_\ell}(\cos \theta) e^{im_\ell\phi}$$

with $P_\ell^{m_\ell}$ the associated Legendre polynomials. The polar functions are real-valued and by definition $\Theta_{\ell,-m_\ell}(\theta) = (-1)^{m_\ell} \Theta_{\ell,m_\ell}(\theta)$.

```
#> Spatial resolution and default parameters
```

```
#> Functions
```

```
#> Evaluate stationary states
```

```
#> Plot functions
```

```
#> Isosurface of probability density of stationary states, colored with phase
```

```
#> Plot a set of stationary states
```

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Real orbitals

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#> Plot a set of real orbitals
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Some details

As $\Theta_{\ell,-m_\ell} = (-1)^{m_\ell} \Theta_{\ell,m_\ell}$, it follows that

$$\begin{cases} Y_\ell^{m_\ell} + (-1)^{m_\ell} Y_\ell^{-m_\ell} = \Theta_{\ell,m_\ell}(\theta) [e^{im_\ell\phi} + e^{-im_\ell\phi}] \propto \cos(m_\ell\theta) \\ Y_\ell^{m_\ell} - (-1)^{m_\ell} Y_\ell^{-m_\ell} = \Theta_{\ell,m_\ell}(\theta) [e^{im_\ell\phi} - e^{-im_\ell\phi}] \propto i \sin(m_\ell\theta) \end{cases}$$

and thus the real orbitals as the linear combinations

$$\psi_{n,\ell,m_\ell,\text{real}} = \begin{cases} \psi_{n,\ell,0} & \text{for } m_\ell = 0 \\ \frac{1}{\sqrt{2}} (\psi_{n,\ell,-|m_\ell|} + (-1)^{m_\ell} \psi_{n,\ell,|m_\ell|}) \propto \cos(|m_\ell|\phi) & \text{for real orbital } (x, xy, x^2 - y^2, \dots) \\ \frac{i}{\sqrt{2}} (\psi_{n,\ell,-|m_\ell|} - (-1)^{m_\ell} \psi_{n,\ell,|m_\ell|}) \propto \sin(|m_\ell|\phi) & \text{for real orbital } (y, yz, \dots) \end{cases}$$

Radial and spherical factors

$$\psi_{n,\ell,m_\ell}(r, \theta, \phi) = R_{n\ell}(r) Y_\ell^{m_\ell}(\theta, \phi)$$

Radial wavefunctions, $R_{n,\ell}(r)$

Radial Wavefunctions, $R_{n\ell}(r)$. Using the dimensionless variable $\rho = \frac{2r}{na_0}$,

$$R_{n\ell}(r) = -\sqrt{\left(\frac{2}{na_0}\right)^3 \frac{(n-\ell-1)!}{2n[(n+\ell)!]^3}} e^{-\rho/2} \rho^\ell L_{n-\ell-1}^{2\ell+1}(\rho),$$

where $a_0 = \frac{4\pi\epsilon_0\hbar^2}{m_e e^2}$ is the **Bohr radius**, and $L_p^k(\rho)$ the solution of the associated Laguerre equation.

Polar and azimuthal wavefunctions, $Y_\ell^{m_\ell}(\theta, \phi) = \Theta_{\ell,m_\ell}(\theta) \Phi_{m_\ell}(\phi)$

$$Y_\ell^{m_\ell}(\theta, \phi) = (-1)^{m_\ell} \sqrt{\frac{(2\ell+1)(\ell-m_\ell)!}{4\pi(\ell+m_\ell)!}} P_\ell^{m_\ell}(\cos\theta) e^{im_\ell\phi}$$

with $P_\ell^{m_\ell}$ the associated Legendre polynomials. The first associated Legendre polynomials are

ℓ	m_ℓ	$P_\ell^{m_\ell}(x)$
0	0	1
1	1	$-(1-x^2)^{\frac{1}{2}}$
	0	x
	-1	$-\frac{1}{2}P_1^1(x)$
2	2	$3(1-x^2)$
	1	$-3x(1-x^2)^{\frac{1}{2}}$
	0	$\frac{1}{2}(3x^2-1)$
	-1	$-\frac{1}{6}P_2^1(x)$
	-2	$\frac{1}{24}P_2^2(x)$

20.5.3 Auxiliary Results on Operators for the Schrödinger Model of the Hydrogen Atom

1. All components of $\hat{\mathbf{L}}$ commute with the kinetic term $\frac{\hat{p}^2}{2m}$ and with $\hat{L}^2$, *proof*

$$[\hat{p}^2, \hat{L}_b] = 0$$

2. All components of $\hat{\mathbf{L}}$ commute with the potential $\hat{V}_{\text{Coulomb}}$, *proof*

$$[\hat{V}_{\text{Coulomb}}, \hat{L}_b] = 0$$

3. Therefore, all components of $\hat{\mathbf{L}}$ commute with the H atom Hamiltonian $\hat{H}_{\text{H-atom}} = \frac{\hat{p}^2}{2m} + \hat{V}_{\text{Coulomb}}$:

$$[\hat{H}_{\text{H-atom}}, \hat{L}_b] = 0$$

4. Simultaneous Commutative Set:

$$[\hat{H}_{\text{H-atom}}, \hat{L}^2] = 0, \quad [\hat{H}_{\text{H-atom}}, \hat{L}_a] = 0, \quad [\hat{L}^2, \hat{L}_a] = 0$$

i.e. the Hamiltonian $\hat{H}_{\text{H-atom}}$, total angular momentum squared $\hat{L}^2$, and any component $\hat{L}_a$ all mutually commute, and thus share a set of common eigenvectors. For the commutation relation $[\hat{L}^2, \hat{L}_a] = 0$, see the section about *spatial angular momentum*.

Hamiltonian & Basic Definitions

The Hamiltonian of the electron in the Hydrogen atom with Coulomb potential is given by

$$\hat{H} = \frac{\hat{p}^2}{2m} + \hat{V}_{\text{Coulomb}},$$

that in space basis becomes

$$\langle \mathbf{r} | \hat{H} = -\frac{\hbar^2}{2m} \nabla^2 \langle \mathbf{r} | - \frac{q^2}{4\pi\epsilon |\mathbf{r}|} \langle \mathbf{r} |.$$

Space Basis Representations

$$\begin{aligned}\langle \mathbf{r} | \Psi \rangle &= \Psi(\mathbf{r}, t) \\ \langle \mathbf{r} | \hat{\mathbf{r}} = \mathbf{r} \langle \mathbf{r} | &\implies \langle \mathbf{r} | \hat{\mathbf{r}} | \Psi \rangle = \mathbf{r} \Psi(\mathbf{r}, t) \\ \langle \mathbf{r} | \hat{\mathbf{p}} = -i\hbar \nabla_{\mathbf{r}} \langle \mathbf{r} | &\implies \langle \mathbf{r} | \hat{\mathbf{p}} | \Psi \rangle = -i\hbar \nabla \Psi(\mathbf{r}, t)\end{aligned}$$

Angular Momentum Operator

$$\hat{\mathbf{L}} = \hat{\mathbf{r}} \times \hat{\mathbf{p}}$$

In Levi-Civita component notation, using Cartesian coordinates

$$\hat{\mathbf{L}} = \hat{\mathbf{e}}_a \hat{L}_a = \hat{\mathbf{e}}_a \varepsilon_{abc} \hat{r}_b \hat{p}_c$$

In the space (position) basis (Cartesian coordinates):

$$\begin{aligned}\langle \mathbf{r} | \hat{\mathbf{L}} | \Psi \rangle &= \langle \mathbf{r} | \hat{\mathbf{e}}_a \hat{L}_a | \Psi \rangle = \\ &= \hat{\mathbf{e}}_a \varepsilon_{abc} r_b (-i\hbar \partial_c \langle \mathbf{r} | \Psi \rangle) = \\ &= -i\hbar \hat{\mathbf{e}}_a \varepsilon_{abc} r_b \partial_c \Psi(\mathbf{r}, t) = \\ &= -i\hbar \mathbf{r} \times \nabla_{\mathbf{r}} \Psi(\mathbf{r}, t) .\end{aligned}$$

$$\langle \mathbf{r} | \hat{\mathbf{p}} | \Psi \rangle$$

$$\langle \mathbf{r} | \hat{\mathbf{p}} | \Psi \rangle = \int \langle \mathbf{r} | \hat{\mathbf{p}} | \mathbf{r}' \rangle \langle \mathbf{r}' | \Psi \rangle d\mathbf{r}' = \int \delta(\mathbf{r} - \mathbf{r}') (-i\hbar \nabla_{\mathbf{r}'} \Psi(\mathbf{r}', t)) d\mathbf{r}' = -i\hbar \nabla_{\mathbf{r}} \Psi(\mathbf{r}, t)$$

$$\langle \mathbf{r} | \hat{p}_b \hat{r}_a | \Psi \rangle$$

$$\langle \mathbf{r} | \hat{p}_b \hat{r}_a | \Psi \rangle = -i\hbar \partial_b \langle \mathbf{r} | \hat{r}_a | \Psi \rangle = -i\hbar \partial_b (r_a \langle \mathbf{r} | \Psi \rangle) = -i\hbar \partial_b (r_a \Psi(\mathbf{r}, t)) = -i\hbar (\delta_{ab} \Psi + r_a \partial_b \Psi)$$

Canonical Commutation Relations (CCR)

Using position basis and Cartesian coordinates, the CCR $[\hat{\mathbf{r}}, \hat{\mathbf{P}}] = i\hbar \mathbb{I}$ reads

$$[r_a, p_b] \Psi = r_a (-i\hbar \partial_b \Psi) - (-i\hbar \partial_b (r_a \Psi)) = i\hbar \delta_{ab} \Psi ,$$

i.e.

$$[r_a, p_b] = i\hbar \delta_{ab}$$

Commutators of Angular Momentum with Position and Momentum

$$\text{Commutator } [r_a, L_b]$$

$$r_a L_b \Psi = r_a (-i\hbar \varepsilon_{bcd} r_c \partial_d \Psi) = -i\hbar \varepsilon_{bcd} r_a r_c \partial_d \Psi$$

$$L_b r_a \Psi = -i\hbar \varepsilon_{bcd} r_c \partial_d (r_a \Psi) = -i\hbar \varepsilon_{bcd} r_c \delta_{ad} \Psi - i\hbar \varepsilon_{bcd} r_c r_a \partial_d \Psi$$

Subtracting the two expressions gives:

$$[r_a, L_b] \Psi = i\hbar \varepsilon_{bcd} r_c \delta_{ad} \Psi = i\hbar \varepsilon_{bca} r_c \Psi,$$

and thus

$$[r_a, L_b] = -i\hbar (\mathbf{r} \times)_{ab} \Psi \quad (\text{in general } \neq 0)$$

(Recall: $(\mathbf{a} \times \mathbf{b})_i = \varepsilon_{ijk} a_j b_k = \{\mathbf{a}_\times\}_{ik} \cdot \{\mathbf{b}\}_k$, i.e. $\{\mathbf{a}_\times\}_{ik} = \varepsilon_{ijk} a_j$)

Commutator $[p_a, L_b]$

$$p_a L_b \Psi = -i\hbar \partial_a (-i\hbar \varepsilon_{bcd} r_c \partial_d \Psi) = -\hbar^2 \varepsilon_{bcd} \partial_a (r_c \partial_d \Psi) = -\hbar^2 \varepsilon_{bad} \partial_d \Psi - \hbar^2 \varepsilon_{bcd} r_c \partial_a \partial_d \Psi$$

$$L_b p_a \Psi = -i\hbar \varepsilon_{bcd} r_c \partial_d (-i\hbar \partial_a \Psi) = -\hbar^2 \varepsilon_{bcd} r_c \partial_d \partial_a \Psi$$

Subtracting the two expressions gives:

$$[p_a, L_b] \Psi = -\hbar^2 \varepsilon_{bad} \partial_d \Psi = \hbar^2 \varepsilon_{abd} \partial_d \Psi = i\hbar \varepsilon_{abc} p_c \Psi$$

Cartesian coordinates.

- $[p_x, L_x] \equiv 0$
- $[p_x, L_y] = \hbar^2 \varepsilon_{xyz} \partial_z \Psi = i\hbar p_z$
- $[p_x, L_z] = \hbar^2 \varepsilon_{xzy} \partial_y \Psi = -\hbar^2 \partial_y \Psi = -i\hbar p_y$

Commutator $[\hat{p}^2, L_b]$

$$[p^2, L_b] = \sum_a p_a p_a L_b - L_b \sum_a p_a p_a = \sum_a (p_a [p_a, L_b] + [p_a, L_b] p_a)$$

Using $[p_a, L_b] = i\hbar \varepsilon_{abc} p_c$:

$$\sum_a (p_a (i\hbar \varepsilon_{abc} p_c) + (i\hbar \varepsilon_{abc} p_c) p_a) = i\hbar \varepsilon_{abc} (p_a p_c + p_c p_a)$$

Since $p_a p_c = p_c p_a$ (symmetric in a, c) and ε_{abc} is antisymmetric in a, c , the sum vanishes identically:

$$[p^2, L_b] = 0 \quad \text{for all components } b$$

Example. For $b = x$:

$$[p^2, L_x] = -i\hbar^3 [1 \cdot \partial_z \partial_y \Psi - 1 \cdot \partial_y \partial_z \Psi] = 0$$

Commutator of Coulomb Potential $\hat{V}$ with Angular Momentum $\hat{L}_b$

In space basis, the Coulomb potential reads:

$$\langle \mathbf{r} | \hat{V} | \Psi \rangle = -\frac{q^2}{4\pi\epsilon_0} \frac{1}{r} \Psi(\mathbf{r}, t) = -k \frac{1}{|\mathbf{r}|} \Psi(\mathbf{r}, t)$$

Let's evaluate whether $\hat{V}$ and $\hat{L}_b$ commute:

1. Action of $\hat{V}\hat{L}_b$:

$$\langle \mathbf{r} | \hat{V} \hat{L}_b | \Psi \rangle = -k \frac{1}{\sqrt{x_a x_a}} (-i\hbar \epsilon_{bcd} r_c \partial_d \Psi)$$

2. Action of $\hat{L}_b \hat{V}$, with $|\mathbf{r}| = \sqrt{x_a x_a}$:

$$\langle \mathbf{r} | \hat{L}_b \hat{V} | \Psi \rangle = -i\hbar \epsilon_{bcd} r_c \partial_d \left(-k \frac{\Psi}{\sqrt{x_a x_a}} \right) = i\hbar k \epsilon_{bcd} r_c \left[-\frac{\partial_d |\mathbf{r}|}{|\mathbf{r}|^2} \Psi + \frac{1}{|\mathbf{r}|} \partial_d \Psi \right]$$

Since $\partial_d |\mathbf{r}| = \partial_d \sqrt{x_a x_a} = \frac{x_d}{|\mathbf{r}|}$:

$$\langle \mathbf{r} | \hat{L}_b \hat{V} | \Psi \rangle = i\hbar k \epsilon_{bcd} r_c \left[-\frac{r_d}{|\mathbf{r}|^3} \Psi + \frac{1}{|\mathbf{r}|} \partial_d \Psi \right]$$

3. Commutator $[\hat{V}, \hat{L}_b]$:

$$\langle \mathbf{r} | [\hat{V}, \hat{L}_b] | \Psi \rangle = -i\hbar k \frac{\epsilon_{bcd} r_c r_d}{|\mathbf{r}|^3} \Psi = 0$$

Why? The product $r_c r_d$ is **symmetric** under swapping $c \leftrightarrow d$, while ϵ_{bcd} is **antisymmetric** under swapping $c \leftrightarrow d$. Summing over c, d gives zero!

Example for $b = x$. $\epsilon_{xcd} r_c r_d = \epsilon_{xyz} yz + \epsilon_{xzy} zy = yz - zy = 0$.

Part V

Solid State Physics

INTRODUCTION TO SOLID STATE PHYSICS

Contents.

- *Solid State Physics*
- *Semiconductors*

References

- D.Tong, University of Cambridge, Solid State Physics
 - Particles in a Magnetic Field
 - Band Structure: lattices, Bloch's theorem, tight-binding and nearly-free electron models,...
 - Electron dynamics in Solids
 - Phonons
- C.Kittel, Introduction to Solid State Physics
 - Semiconductor Crystals (Ch.8, p.185)
 - Fermi Surfaces and Metals (Ch.9, p.221)
 - Surface and Interface Physics (Ch.17, p.487)
 - * ..., p - n junctions, semiconductor lasers, LED,...
- D.A.Neamen, Semiconductor Physics and Devices
 - Part I, Semiconductor Material Properties
 - * Introduction to the QM of Solids (Ch.3, p.58)
 - * The Semiconductors in Equilibrium (Ch.4, p.106)
 - * Carrier Transport Phenomena (Ch.5, p.156)
 - * Non-equilibrium Excess Carriers in Semiconductors (Ch.6, p.192)
 - Part II, Fundamental Semiconductor Devices
 - * p - n Junction (Ch.7, p.241)
 - * p - n junction diode (Ch.8, p.276)
 - * MOSFET (Ch.10,11)
 - * ...
 - Part III, Specialized Semiconductor Devices

- * Optical Devices (Ch.14, p.618)
 - Solar cells
 - Photodetectors
 - Photoluminescence and electroluminescence
 - LED
 - Laser diodes
- [LibreTexts Engineering, Material Science, Supplemental Modules: Semiconductors](#)

NOTES ABOUT SOLID STATE PHYSICS

22.1 Crystals

Primary lattice vectors, $\mathbf{a}_j$, defines periodicity (and symmetry) of the lattice,...

$$f(\mathbf{r}) = f\left(\mathbf{r} + \sum_j n_j \mathbf{a}_j\right),$$

with $n_j \in \mathbb{Z}$.

22.1.1 Bloch's theorem

...

INTRODUCTION TO SEMICONDUCTORS

Following mainly *D.A.Naemen, Semiconductor Physics and Devices*

- *Introduction to the QM of Solids*
- *Semiconductors in Equilibrium*
- *Carrier Transport Phenomena*
- *Non-equilibrium Excess Carriers in Semiconductors*
- *p-n Junction*
- *p-n Junction Diode*

23.1 Introduction to the QM of Solids

23.1.1 Energy bands

First, the energy level splitting is discussed. Then, two models (Kronig-Penny, and nearly free electron models) are introduced to describe the concept of allowed and forbidden energy bands, starting from Schrodinger equation.

Energy level splitting

This section deal with energy level splitting in a two-element system.

Non-interacting systems

Let $|L\rangle$, and $|R\rangle$ two localized eigen-states of non-interacting identical subsystems, with the same energy E_0 ,

$$\begin{aligned}\hat{H}_0|L\rangle &= E_0|L\rangle \\ \hat{H}_0|R\rangle &= E_0|R\rangle\end{aligned}$$

As these states are localized in space, they're (approximately?) orthogonal, $\langle L|R\rangle = 0$.

Interacting subsystems - Perturbation potential

When the two systems are brought close together, they start interacting weakly, so that the Hamiltonian becomes

$$\hat{H} = \hat{H}_0 + \hat{V} .$$

The matrix elements of the Hamiltonian in the base $\{|L\rangle, |R\rangle\}$ (where are all the other eigen-functions? We're not interested in them, right now, but only on the pair of eigenfunctions with the same energy when the syb-systems are not interacting) read

$$\begin{aligned}\langle L|\hat{H}|L\rangle &=: E'_0 \\ \langle R|\hat{H}|R\rangle &=: E'_0 \\ \langle L|\hat{H}|R\rangle &= \langle R|\hat{H}|L\rangle^* =: -t\end{aligned}$$

Energy levels - Eigenvalues of the Hamiltonian operator

If the new eigen-functions of the interacting system can be written as a linear combination of the non-interacting functions, $|\Psi_{1,2}\rangle = \ell_{1,2}|L\rangle + r_{1,2}|R\rangle$, the eigen-problem becomes

$$\hat{H}|\Psi_{1,2}\rangle = E_{1,2}|\Psi_{1,2}\rangle .$$

Projecting onto $\langle L|$, and $\langle R|$,

$$\begin{cases} \ell E'_0 - rt = E\ell \\ -\ell t + rE'_0 = Er \end{cases} \quad \rightarrow \quad \begin{cases} \ell(E'_0 - E) - rt = 0 \\ -\ell t + r(E'_0 - E) = 0 \end{cases}$$

This system has non trivial solutions, if the determinant of the linear system is zero,

$$0 = \begin{vmatrix} E'_0 - E & -t \\ -t & E'_0 - E \end{vmatrix} = (E'_0 - E) - t^2 .$$

The energy levels of the interacting states are slightly shifted w.r.t. the non-interacting level,

$$E = E'_0 \mp t .$$

Eigen-states

The eigen-states are

- the **bonding state**, for $E = E'_0 - t$, $\ell = -r$, and with normalization

$$|\Psi\rangle_- = \frac{1}{\sqrt{2}} (|L\rangle + |R\rangle)$$

- the **anti-bonding state**, for $E = E'_0 + t$, $\ell = r$, and with normalization

$$|\Psi\rangle_+ = \frac{1}{\sqrt{2}} (|L\rangle - |R\rangle)$$

Kronig-Penny model

Kronig-Penny potential

Kronig-Penny model is a 1-dimensional model with periodic square wave potential,

$$V(x) = \begin{cases} V_0 & , \quad x \in \text{regions II} = [na + (n-1)b, n(a+b)] \\ 0 & , \quad x \in \text{regions I} = [(n-1)(a+b), na + (n-1)b] \end{cases}$$

General expression of Bloch states

Following *Bloch's theorem*, the eigen-states of an electron (in position base) in a infinite periodic lattice can be written as

$$\Psi(\mathbf{r}) = u(\mathbf{r})e^{i\mathbf{k}\cdot\mathbf{r}} ,$$

with $u(\mathbf{r}) = u\left(\mathbf{r} + \sum_j n_j \mathbf{a}_j\right)$ with the same periodicity as the lattice, $n_j \in \mathbb{Z}$, or in 1-dimensional problems,

$$\Psi(x) = u(x)e^{ikx} .$$

Eigenvalue problem for the Hamiltonian operator

The eigenvalue problem for the Hamiltonian operator reads

$$E\Psi = \hat{H}\Psi = \left[\frac{\hat{p}^2}{2m} + V(x) \right] \Psi ,$$

with $\hat{p}$ the momentum operator, whose expression in position base is given by (13.1), and in 1-dimensional problems $\hat{p}\Psi = -i\hbar\partial_x\Psi$. The operator acting twice gives $\hat{p}^2\Psi = -\hbar^2\partial_{xx}\Psi$.

Introducing the expression of the eigenfunctions from Bloch's theorem¹, it follows

$$Eu(x) = -\frac{\hbar^2}{2m} (u''(x) + 2iku'(x) - k^2u(x)) + V(x)u(x) ,$$

or

$$\begin{aligned} 0 &= u'' + 2iku'(x) - \left(k^2 - \frac{2mE}{\hbar^2}\right)u(x) && \text{regions I} \\ 0 &= u'' + 2iku'(x) - \left(k^2 - \frac{2m(E - V_0)}{\hbar^2}\right)u(x) && \text{regions II} \end{aligned}$$

Solution of the ODEs.

Continuity of $\Psi(x)$, and $\partial_x\Psi(x)$. for matching solutions.**

Energy levels.

Nearly free electron model

¹ The derivatives read $\Psi' = (u' + iku) e^{ikx}$, and $\Psi'' = (u'' + 2iku' - k^2u) e^{ikx}$.

23.2 Semiconductors in Equilibrium

23.2.1 Structure of a semi-semiconductor

Symbol	Concentration of
p	Holes in the valence bands of Si -atom lattice
n	Free electrons in the conduction band
N_A	Acceptor atoms, usually class III
N_D	Donor atoms, usually class V
N_A^-	Ionized acceptor atoms, having collected e^- (usually from neighboring Si atoms)
N_D^+	Ionized donor atoms, having released e^-

Usually, N_A , N_D , N_A^- , N_D^+ only depends on the space coordinate, as a consequence of doping. The concentration of free electrons and holes may be time dependent instead (...).

Intrinsic semiconductor

Lattice of Si atoms. Some free electrons have left the valence bands of the atoms and entered the conduction band. For a new free electron, a new hole is left behind in the lattice. If the medium is **electrically neutral** the numbers of free electrons and holes are locally equal, and thus their concentrations

$$n_i(\mathbf{r}) = p_i(\mathbf{r}) .$$

Doped semiconductors

p-type

The semiconductor is doped with atoms of class III, replacing some of the Si atoms in the lattice. These atoms introduce a hole in the lattice and are prone to collect e^- from neighboring Si atoms.

Under *full-ionization condition*, all the doping atoms are ionized, and thus $N_A^- = N_A$. In a **electrically neutral** region with no donor atoms, the concentration of the holes is equal to the sum of the concentration of the ionized acceptor atoms and the free electrons,

$$p = n + N_A^- .$$

In a standard *p*-type semiconductor, $N_A \gg n_i$, and thus $p \sim N_A^-$.

n-type

The semiconductor is doped with atoms of class V, replacing some of the Si atoms in the lattice. These atoms introduce a “extra” loosely bound electron in the lattice and are prone to release it in the conduction band of the semiconductor.

Under *full-ionization condition*, all the doping atoms are ionized, and thus $N_D^+ = N_D$. In a **electrically neutral** region with no acceptor atoms, the concentration of the free electrons is equal to the sum of the concentration of the ionized donor atoms and holes,

$$n = p + N_D^+ .$$

In a standard *n*-type semiconductor, $N_D \gg p_i$, and thus $n \sim N_D^+$.

Full ionization

todo

23.2.2 Electric charge density

$$\rho(\mathbf{r}, t) = -qn(\mathbf{r}, t) + qp(\mathbf{r}, t) + qN_D^+(\mathbf{r}, t) - qN_A^-(\mathbf{r}, t), \quad (23.1)$$

being

- q the elementary charge, the charge of the electron (here the opposite to get a positive numerical value of q in Coulomb)
- $n(\mathbf{r}, t)$ the number volume density of the mobile electrons
- $p(\mathbf{r}, t)$ the number volume density of the holes in the valence bands of the lattice
- $N_D^+(\mathbf{r}, t)$ the number volume density of the positive donor ions
- $N_A^-(\mathbf{r}, t)$ the number volume density of the negative acceptor ions

Usually, $N_D^+(\mathbf{r})$, $N_A^-(\mathbf{r})$, corresponding to the density of the donor and acceptor atoms in the lattice, that have fixed positions.

Remark. todo Discuss the values of N_D^+ and N_D in terms of energy levels at different temperatures. Add this discussion to [full-ionization section](#) or in another section and then point to it?

23.3 Carrier Transport Phenomena

In this section:

- *Currents in semiconductors*: hole and electron number current, and electric current density
- *Einstein relation*
- *Balance equations*: number and charge density dynamical equations

23.3.1 Currents in semiconductors

Two main processes:

1. **diffusion** due to non-uniform density
2. **drift** due to an electric field $\mathbf{e}(\mathbf{r})$

The overall current has the contribution of the motion of both electrons (n , for negative charges) and holes (p , for positive “charges”),

$$\mathbf{j}(\mathbf{r}, t) = \mathbf{j}_n(\mathbf{r}, t) + \mathbf{j}_p(\mathbf{r}, t),$$

and both these contributions have their diffusion and drift part,

$$\begin{aligned} \mathbf{j}_n &= \mathbf{j}_{n,diff} + \mathbf{j}_{n,drift} \\ \mathbf{j}_p &= \mathbf{j}_{p,diff} + \mathbf{j}_{p,drift} \end{aligned}$$

For the contribution of the electrons, the diffusion current follows a Fick's law for the number density n (multiplied by the constant charge $-q$ of the elementary charge), while the drift coefficient is written as the product of the negative charge density and their drift velocity,

$$\begin{aligned}\mathbf{j}_n &= \mathbf{j}_{n,diff} + \mathbf{j}_{n,drift} = \\ &= -D_n(-q)\nabla n + \underbrace{(-q)n}_{\rho_n} \mathbf{v}_n = \\ &= D_n q \nabla n + q \mu_n n \mathbf{e} ,\end{aligned}\tag{23.2}$$

being the average drift velocity, $\mathbf{v}_n = -\mu_n \mathbf{e}$, a function on the local electric field through the electron mobility μ_n . Here the minus sign follows the definition of the electron mobility with positive value, and from the opposite direction of the local electric field $\mathbf{e}$ and the drift velocity of negative charges.

For the contribution of the holes,

$$\mathbf{j}_p = -D_p q \nabla p + q \mu_p p \mathbf{e} .\tag{23.3}$$

	$\mu_n [\dots]$	$\mu_p [\dots]$
Silicon	1350	480
Gallium Arsenide	8500	400
Germanium	3900	1900

Remark. While charges in free-space have acceleration proportional to the electric field, in solids they have drift velocity proportional to the electric field, due to collisions with the lattice. This mechanism can be described in terms of conductivity σ (or its inverse, resistivity ρ_R) of the medium, similarly to Ohm's law

$$\mathbf{j} = \sigma \mathbf{e} , \quad \mathbf{e} = \rho_R \mathbf{j} .$$

Using the expression of the drift current, as the sum of the n and p contributions in (23.2) and (23.3) respectively,

$$\mathbf{j}_{drift} = q(\mu_n n + \mu_p p) \mathbf{e} ,$$

it immediately follows the formula for the resistivity (and the conductivity)

$$\rho_R = \frac{1}{\sigma} = \frac{1}{q(\mu_n n + \mu_p p)} .\tag{23.4}$$

todo See discussion about this relation in 5.1.4. Velocity Saturation

23.3.2 Einstein relation

...

$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{kT}{q} .\tag{23.5}$$

23.3.3 Balance equations

The density of free electrons and holes are governed by the following PDEs

$$\begin{aligned}\partial_t n - \nabla \cdot \left(\frac{\mathbf{j}_n}{q} \right) &= (G_n - R_n) \\ \partial_t p + \nabla \cdot \left(\frac{\mathbf{j}_p}{q} \right) &= (G_p - R_p)\end{aligned}\tag{23.6}$$

where the expression of current densities $\mathbf{j}_n, \mathbf{j}_p$ are given in (23.2), (23.3) respectively, and the terms $G_{n,p}, R_{n,p}$ represent source and sink terms representing **generation** or **recombination** of free charges and holes.

As for a new free electron there's a new hole, then $G_n = G_p$. As recombination occurs between the same number of free electrons and holes, $R_n = R_p$.

Multiplying the first and the second equation in (23.6) by $-q$ and q respectively, the balance equation for the *free* charge density $\rho_f = -qn + qp$,

$$\partial_t \rho_f + \nabla \cdot \mathbf{j} = 0.$$

23.4 Non-equilibrium Excess Carriers in Semiconductors

23.5 *p-n* Junctions

23.5.1 Introduction to the pn Junction and its Operational Regimes

When a *p*-type and an *n*-type semiconductor are brought into intimate contact, the immense carrier concentration gradients at the interface drive a transient diffusion process. Mobile electrons and holes annihilate each other at the metallurgical interface, leaving behind uncompensated, fixed dopant ions. This region, stripped of mobile carriers, is known as the **depletion region**. The fixed charges generate an internal electric field that opposes further diffusion, establishing a balance.

Depending on the external voltage V_a applied across the device, the junction operates in one of three primary regimes:

- **Zero Bias (Thermal Equilibrium, $V_a = 0$):** No external voltage is applied. The internal electric field perfectly balances the carrier diffusion gradients. The net current density for both electrons and holes is identically zero ($J_n = 0, J_p = 0$).
- **Forward Bias ($V_a > 0$):** A positive potential is applied to the *p*-side relative to the *n*-side. This external potential opposes the built-in field, lowering the electrostatic barrier and narrowing the depletion width. Diffusion forces win over drift, resulting in an exponential injection of minority carriers across the junction.
- **Reverse Bias ($V_a < 0$):** A negative potential is applied to the *p*-side relative to the *n*-side. The external voltage reinforces the built-in field, widening the depletion region and increasing the potential barrier. Diffusion drops to zero, and the current is limited to a minute reverse saturation current driven by thermal generation.

23.5.2 The Depletion Approximation and Shockley Transport

To derive closed-form analytical expressions for the electrostatics and current-voltage relations, we employ the **Depletion Approximation** (also called the *Abrupt Junction Approximation*). This framework assumes that the transition between doping zones is perfectly sharp and that the depletion region is completely devoid of mobile carriers ($\rho = \text{constant}$ inside the zone, $\rho = 0$ outside).

Concentration	p -bulk	p -interface	n -interface	n -bulk
N_A^-	$\sim N_A$	$\sim N_A$		
N_D^+			$\sim N_D$	$\sim N_D$
p	$\sim N_A^-$			
n				$\sim N_D^+$
ρ		$-N_A^-q$	N_D^+q	

Zero Bias (Thermal Equilibrium)

By applying the depletion approximation, the net space-charge density $\rho(x)$ is treated as piecewise constant:

$$\rho(x) = \begin{cases} 0 & x < x_p \\ -qN_A & x_p \leq x < 0 \\ qN_D & 0 \leq x \leq x_n \\ 0 & x > x_n \end{cases}$$

Integrating Gauss' law, $\partial_x e(x) = \frac{\rho(x)}{\epsilon}$, with the assumption of electrical neutrality, $0 = q(N_A x_p + N_D x_n)$, and zero field outside the interface,

$$e(x) = \begin{cases} 0 & x < x_p \\ -\frac{qN_A}{\epsilon} (x - x_p) & x_p \leq x < 0 \\ \frac{qN_D}{\epsilon} (x - x_n) & 0 \leq x \leq x_n \\ 0 & x > x_n \end{cases}$$

Integrating the relation between the electric field and the electric potential, $e(x) = -\partial_x \phi(x)$, with the reference $\phi(x) = 0$ in the p -bulk, $x < x_p$,

$$\phi(x) = \begin{cases} 0 & x < x_p \\ \frac{qN_A}{2\epsilon} (x - x_p)^2 & x_p \leq x < 0 \\ \frac{qN_D}{2\epsilon} (x - x_n)^2 + \frac{qN_A}{2\epsilon} x_p^2 - \frac{qN_D}{2\epsilon} x_n^2 & 0 \leq x \leq x_n \\ \frac{q}{2\epsilon} (N_A x_p^2 - N_D x_n^2) =: V_{bi} & x > x_n \end{cases}$$

Global charge neutrality requires $-N_A x_p = N_D x_n$, yielding the equilibrium depletion width W_0 :

$$W_0 = x_p + x_n = \sqrt{\frac{2\epsilon_s V_{bi}}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right)}$$

Details

For the continuity of the potential, for $0 \leq x \leq x_n$,

$$\begin{aligned}\phi(x) &= -\frac{qN_D}{\varepsilon} \left[\frac{x^2}{2} - x_n x \right] + \phi(0) = \\ &= -\frac{qN_D}{\varepsilon} \left[\frac{x^2}{2} - x_n x \right] + \frac{qN_A}{2\varepsilon} x_p^2 = \\ &= -\frac{qN_D}{2\varepsilon} (x - x_n)^2 + \frac{qN_A}{2\varepsilon} x_p^2 + \frac{qN_D}{2\varepsilon} x_n^2\end{aligned}$$

Thus the potential difference across the depletion region V_{bi} reads,

$$\phi(x_n) = \frac{q}{2\varepsilon} (N_A x_p^2 + N_D x_n^2) =: V_{bi}.$$

The width of the depletion region w can be written as

$$\begin{aligned}w &= x_n - x_p = \\ &= x_n \left(1 + \frac{N_D}{N_A} \right) = x_n N_D \left(\frac{1}{N_D} + \frac{1}{N_A} \right) = \\ &= -x_p \left(\frac{N_A}{N_D} + 1 \right) = -x_p N_A \left(\frac{1}{N_D} + \frac{1}{N_A} \right),\end{aligned}$$

then the potential can be written as

$$\begin{aligned}V_{bi} &= \frac{q}{2\varepsilon} \left(\frac{N_A^2 x_p^2}{N_A} + \frac{N_D^2 x_n^2}{N_D} \right) = \\ &= \frac{q}{2\varepsilon} N_A^2 x_p^2 \left(\frac{1}{N_A} + \frac{1}{N_D} \right) = \\ &= \frac{q}{2\varepsilon} w^2 \left(\frac{1}{N_A} + \frac{1}{N_D} \right)^{-1}.\end{aligned}$$

Thus the width of the depletion region can be written as a function of the built-in potential as

$$w = \sqrt{\frac{2\varepsilon V_{bi}}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right)}.$$

todo Deal with some statistical mechanics to show this

The total voltage drop across the region is the **built-in potential** V_{bi} , determined by the bulk doping concentrations and the Law of Mass Action ($n_0 p_0 = n_i^2$):

$$V_{bi} = V_t \ln \left(\frac{N_A N_D}{n_i^2} \right)$$

where $V_t = \frac{k_B T}{q}$ is the thermal voltage.

Forward Bias Under Shockley Assumptions

To find the explicit I - V relationship, we introduce the **Shockley Ideal Conditions**:

1. Low-level injection ($n \ll N_A$ in the p -bulk; $p \ll N_D$ in the n -bulk).
2. No recombination or generation occurs inside the depletion region.
3. The electric field in the neutral bulk regions is negligible; transport there is purely diffusion-driven.

Width of the depletion region. The applied forward voltage V_a lowers the barrier, modifying the depletion width to:

$$W(V_a) = \sqrt{\frac{2\epsilon_s(V_{bi} - V_a)}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right)}$$

Minority charge injection. Lowering the barrier causes minority charge injection at the boundaries of the depletion region. Compared to the equilibrium condition with no external voltage,

$$\begin{aligned} \Delta n(x_p) &= n_{p0} \left[\exp\left(\frac{V_a}{V_t}\right) - 1 \right] \\ \Delta p(x_n) &= p_{n0} \left[\exp\left(\frac{V_a}{V_t}\right) - 1 \right] \end{aligned}$$

Charge transport dynamics. If Shockley conditions hold, no recombination in the depletion region occurs; recombination occurs in the bulk of p - and n - sections: in bulk regions, the electric field $\vec{e}$ is negligible, and thus the current is driven by diffusion only.

Governing equation of the holes in the n -bulk immediately follows from equation (23.6), with the assumption of no drift current in (23.3),

$$\begin{aligned} \partial_t \Delta p_n &= -\nabla \cdot \left(\frac{j_{p,diff}}{q} \right) + \Delta(G - R) = \\ &= -\nabla \cdot (-D \Delta p_n \nabla n) + \Delta(G - R), \end{aligned}$$

or in 1-dimensional problems with constant coefficients

$$\partial_t \Delta p_n = D \partial_{xx} \Delta p_n + \Delta(G - R)$$

Assuming *steady-state diffusion equation* (**todo justify this limit**) governs the diffusion of the minority charge carriers in the bulk, and using the **law of mass action** to write the source term (see box below), the governing equation becomes

$$D_p \Delta p_n'' = \frac{1}{\tau_p} \Delta p_n,$$

with $\tau_p = \frac{1}{k_r n_{n0}}$.

Law of mass action for $G - R$

In the n -bulk region,

- $n_{n0} \gg p_{n0}$,
- $\Delta n = \Delta p$ if the electrical charge remains zero
- $\Delta p_n \gg p_{n0}$
- $\Delta n_n \ll n_{n0}$

Thus the source term becomes

$$\begin{aligned}
 R - G &= k_r p n - k_r p_0 n_0 = \\
 &= k_r (p_{n0} + \Delta p_n) (n_{n0} + \Delta n_n) - k_r p_{n0} n_{n0} = \\
 &= k_r (p_{n0} \Delta n_n + n_{n0} \Delta p_n + \Delta n_n \Delta p_n) = \\
 &\simeq k_r n_{n0} \Delta p_n
 \end{aligned}$$

as:

- $G \sim G_0 = R_0$ at equilibrium. **todo** Find some time/space to justify all these sentences about equilibrium
- the conditions at the beginning of the box give:
 - $n_{n0} \Delta p_n \gg p_{n0} \Delta n_n$ (because the deltas are equal, and $n_{n0} \gg p_{n0}$)
 - $n_{n0} \Delta p_n \gg \Delta p_n \Delta n_n$ (because the deltas are equal, and $n_{n0} \gg \Delta n_n$)

The differential problem supplied with proper boundary conditions becomes

$$\begin{cases} \frac{\Delta p_n(x)}{\tau_p} = D_p \Delta p_n''(x) \\ p(x_n) = p_{n0} \left[\exp\left(\frac{V_a}{V_t}\right) - 1 \right] \\ p(x \rightarrow +\infty) = 0 \end{cases}$$

and its solution reads

$$\Delta p_n(x) = p_{n0} \left[\exp\left(\frac{V_a}{V_t}\right) - 1 \right] \exp\left(-\frac{x - x_n}{L_p}\right),$$

with $L_p = \sqrt{\tau_p D_p}$ for the holes in the n -bulk, $x > x_n$

$$\Delta n_p(x) = n_{p0} \left[\exp\left(\frac{V_a}{V_t}\right) - 1 \right] \exp\left(\frac{x - x_p}{L_n}\right),$$

with $L_n = \sqrt{\tau_n D_n}$ for the electrons in the p -bulk, $x < x_p$.¹²

Current under steady conditions.

- Depletion region $x \in [x_n, x_p]$. Under steady conditions, no charge accumulation occurs along the p - n junction and thus the current must be constant. There's no recombination or generation inside the depletion region $x \in [x_p, x_n]$, thus the following conditions hold for holes and free electrons

$$\text{const} = AJ_p(x) = AJ_p(x_p) = AJ_p(x_n) \quad \forall x \in [x_p, x_n]$$

$$\text{const} = AJ_n(x) = AJ_n(x_p) = AJ_n(x_n),$$

and thus

$$\text{const} = AJ(x) \quad \forall x \in [x_p, x_n].$$

Evaluating the current at the boundary of the depletion region (in the model, just outside that, so that the electric field is zero and the current is driven only by diffusion)

$$\begin{aligned}
 I &= AJ(x) = AJ_n(x) + AJ_p(x) = \\
 &= AJ_n(x_p) + AJ_p(x_n) = \\
 &= -A(-q)D_n n'_p(x_p) - AqD_p p'_n(x_n) = \\
 &= AqD_n \frac{1}{L_n} n_{p0} \left[\exp\left(\frac{V_a}{V_t}\right) - 1 \right] + AqD_p \frac{1}{L_p} p_{n0} \left[\exp\left(\frac{V_a}{V_t}\right) - 1 \right] = \\
 &= qA \left(\frac{D_n n_{p0}}{L_n} + \frac{D_p p_{n0}}{L_p} \right) \left[\exp\left(\frac{V_a}{V_t}\right) - 1 \right].
 \end{aligned}$$

¹ The boundary conditions of the diffusion of n in the p -bulk are $n(x \rightarrow -\infty) = 0$, $n(x_p) = n_{p0}[\dots]$.

² Usually $n_0 \ll \Delta n \ll N_A$.

This equation readily gives the $V(I)$ relation for an ideal p - n junction,

$$I(V) = I_0 \left[\exp \left(\frac{V}{V_t} \right) - 1 \right], \quad (23.7)$$

with $I_0 = qA(\dots) = qAn_i^2 \left(\frac{D_n}{L_n N_A} + \frac{D_p}{L_p N_D} \right)$.

- n -bulk, $x \in [x_n, +\infty)$. In this region, the electric field is assumed to be negligible, and thus only diffusion drives the current of the charge carriers. For the holes,

$$\begin{aligned} J_p(x) &= -D_p p'_n(x) = \\ &= \frac{D_p}{L_p} p_{n0} \exp \left(\frac{V_a}{V_t} \right) \exp \left(-\frac{x - x_n}{L_p} \right) = \\ &= J_p(x_n) \exp \left(-\frac{x - x_n}{L_p} \right). \end{aligned}$$

and thus $J_n(x) = J - J_p(x)$.

- p -bulk, $x \in (-\infty, x_p] \dots$
-

23.5.3 Advanced Description: Quasi-Fermi Levels and Non-Equilibrium Transport

todo Uncomment

23.6 p - n Diodes

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